



Towards Practical Meshlet Compression

BASTIAN KUTH

MAX OBERBERGER

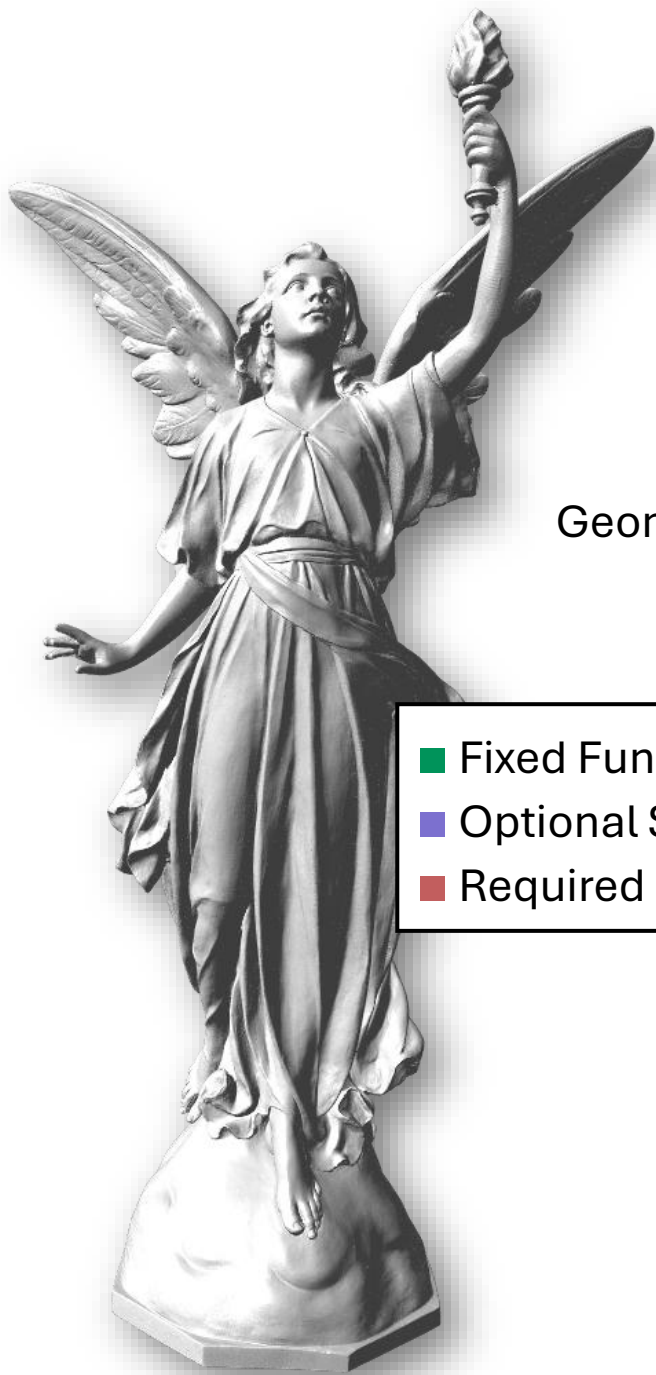
FELIX KAWALA

SANDER REITTER

SEBASTIAN MICHEL

MATTHÄUS CHAJDAS

QUIRIN MEYER

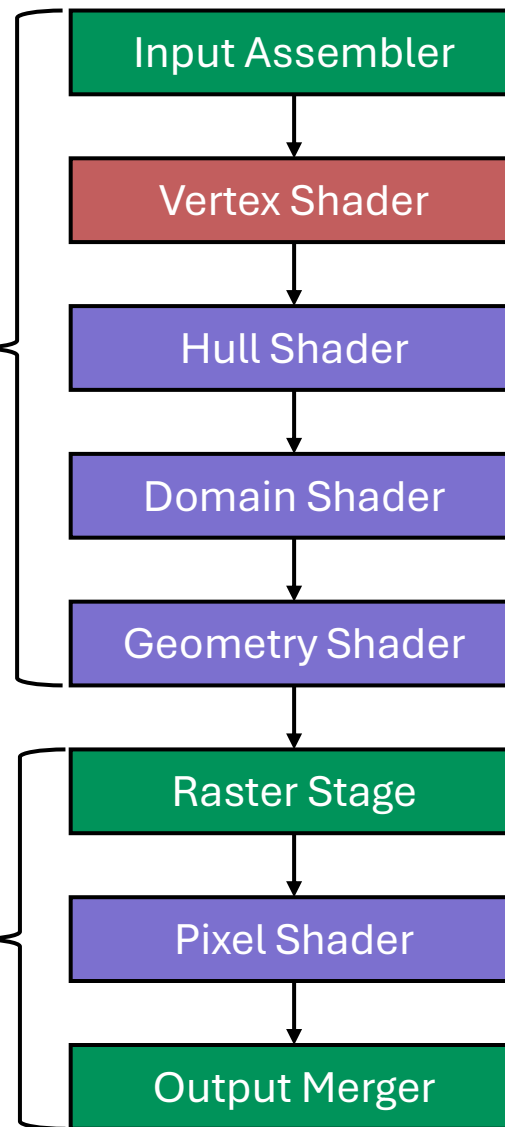


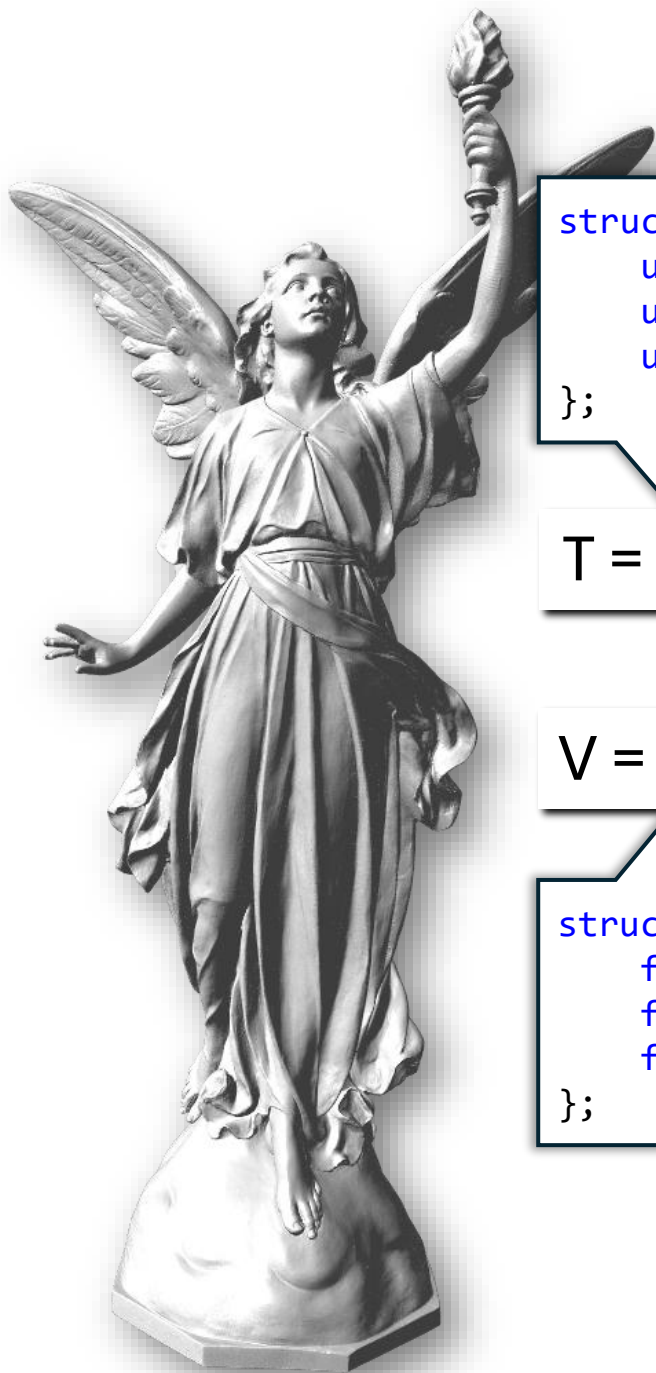
Geometry Stage

- Fixed Function Unit
- Optional Stage
- Required Stage

Pixel Stage

Vertex Pipeline





```
struct Triangle {  
    uint32_t v0;  
    uint32_t v1;  
    uint32_t v2;  
};
```

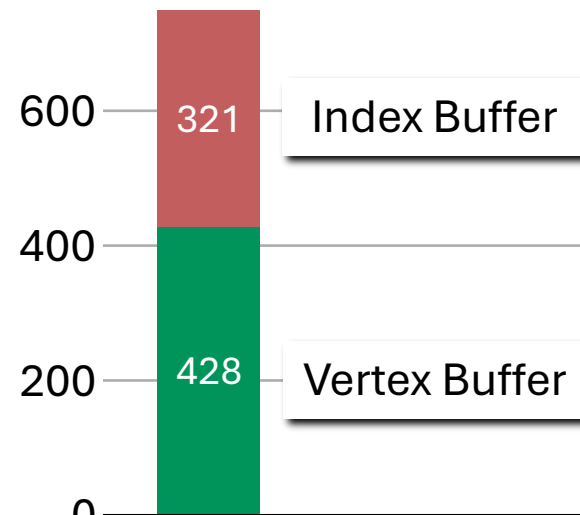
$T = 28,055,728$

$V = 14,028,307$

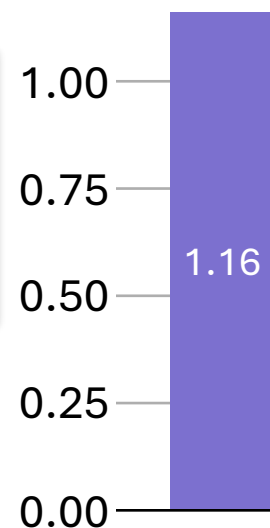
```
struct Vertex {  
    float3 position;  
    float3 normal;  
    float2 texCoord;  
};
```

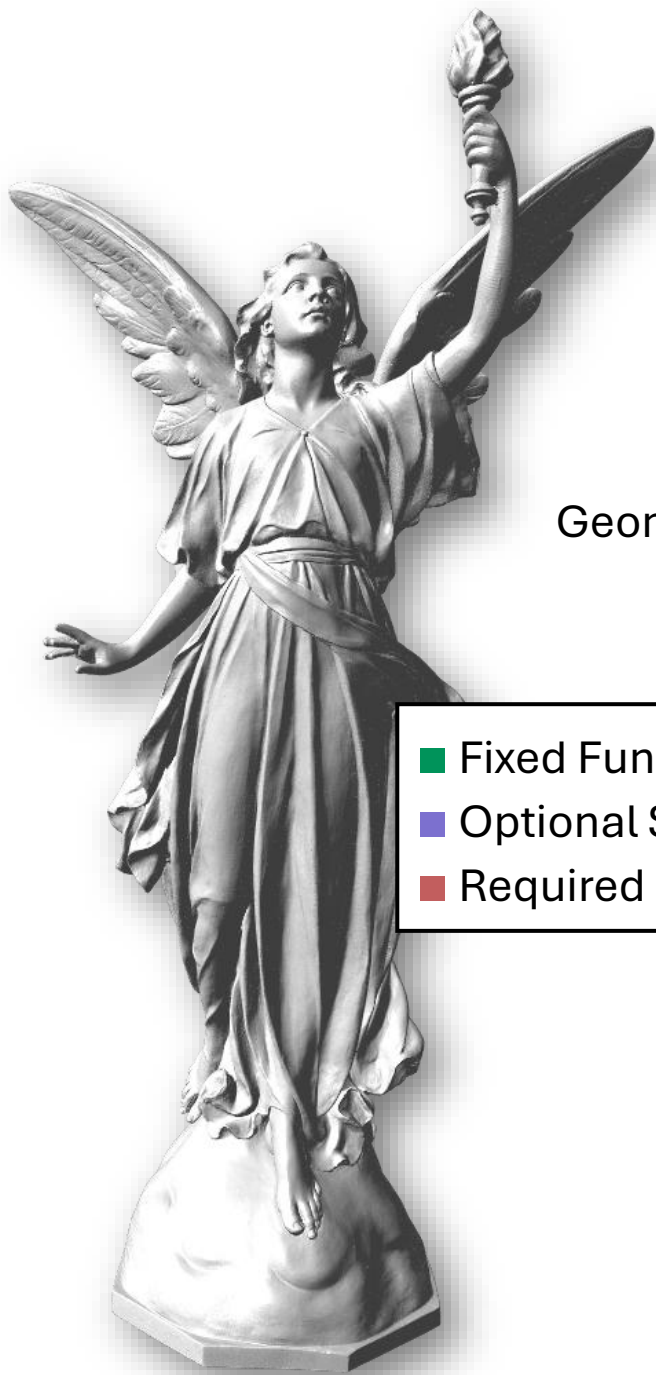
Vertex
Pipeline

Buffers Size in MiB



Render Time in ms on RX 7900 XTX



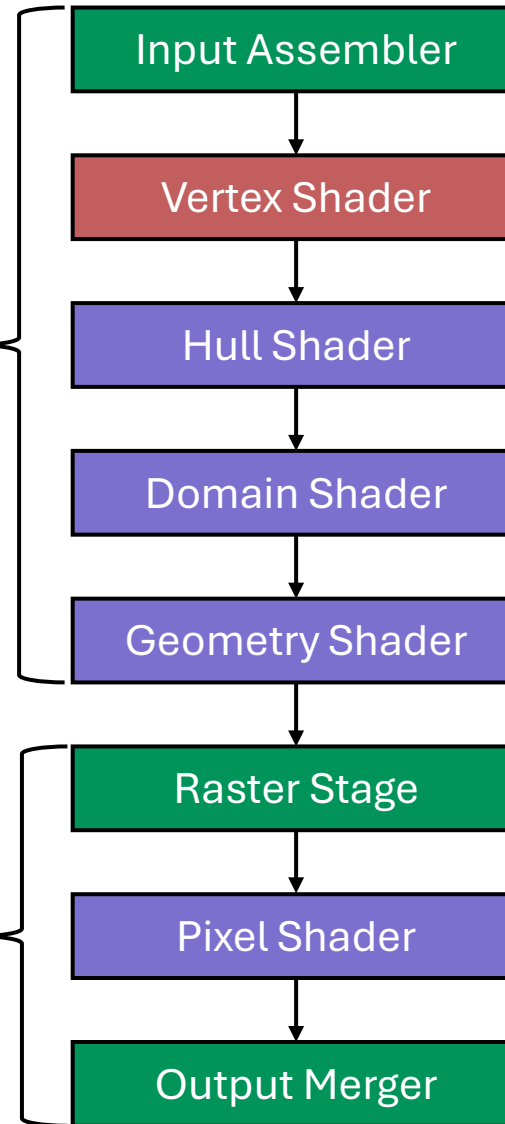


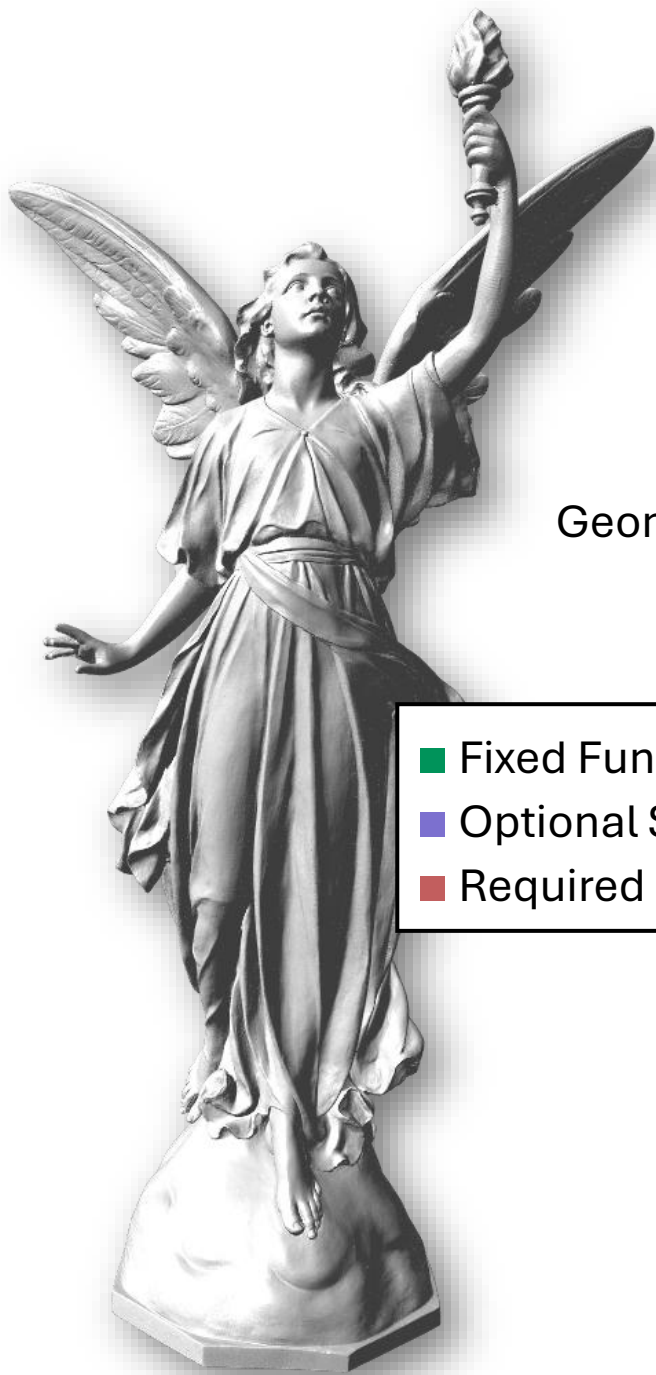
Geometry Stage

- Fixed Function Unit
- Optional Stage
- Required Stage

Pixel Stage

Vertex Pipeline





Geometry Stage

Pixel Stage

Vertex
Pipeline

Mesh-
Shading

Input Assembler

```
[NumThreads(128, 1, 1)]  
[OutputTopology("triangle")]  
void MeshShader(  
    uint gtid : SV_GroupThreadID,  
    uint2 gid : SV_GroupID,  
    out indices uint3 tris[256],  
    out vertices TransformedVertex verts[128]  
) {  
    ...  
    SetMeshOutputCounts(vertexCount, primitiveCount);  
}
```

- Fixed Function Unit
- Optional Stage
- Required Stage

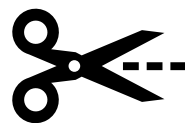
Geometry Shader

Mesh Shader

Raster Stage

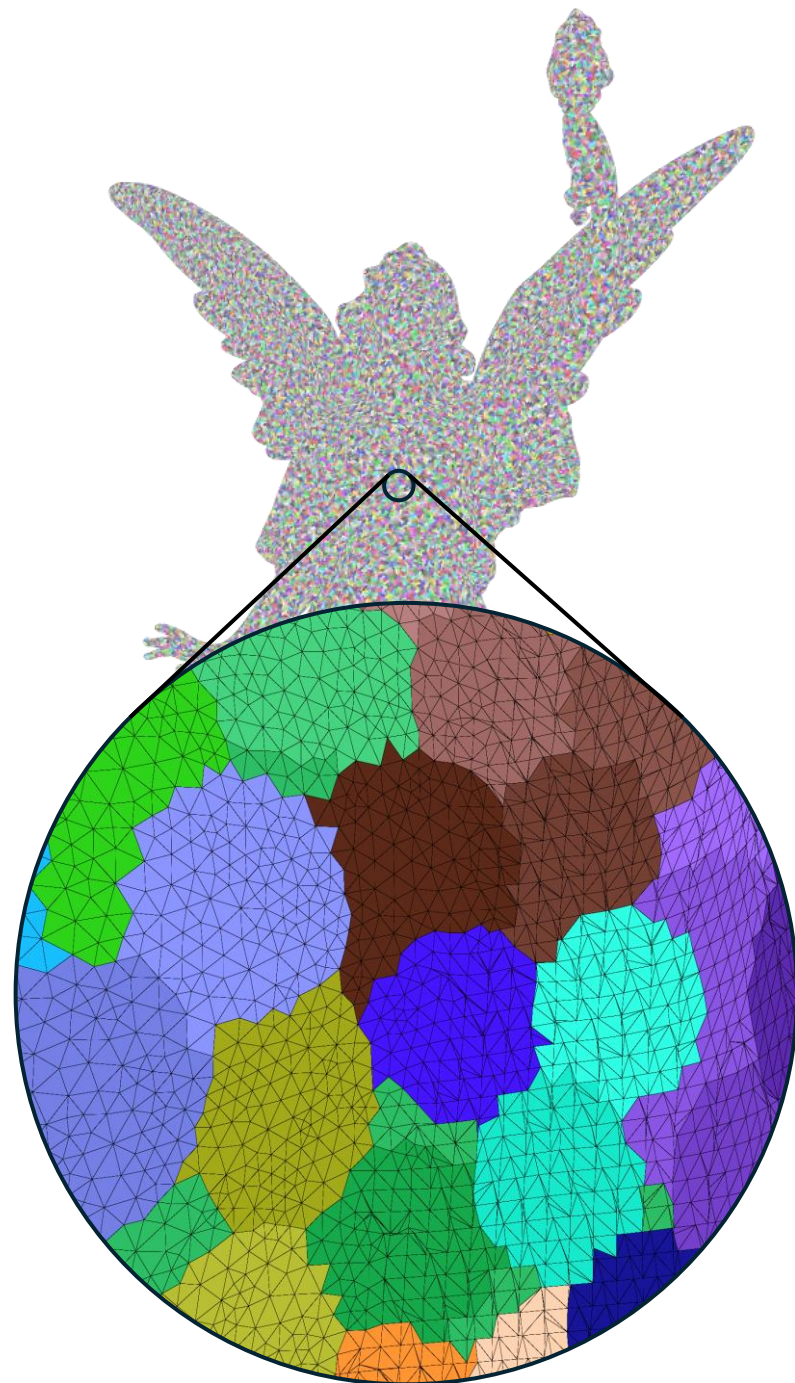
Pixel Shader

Output Merger



$V = 128$

$T = 256$



V = 128
T = 256

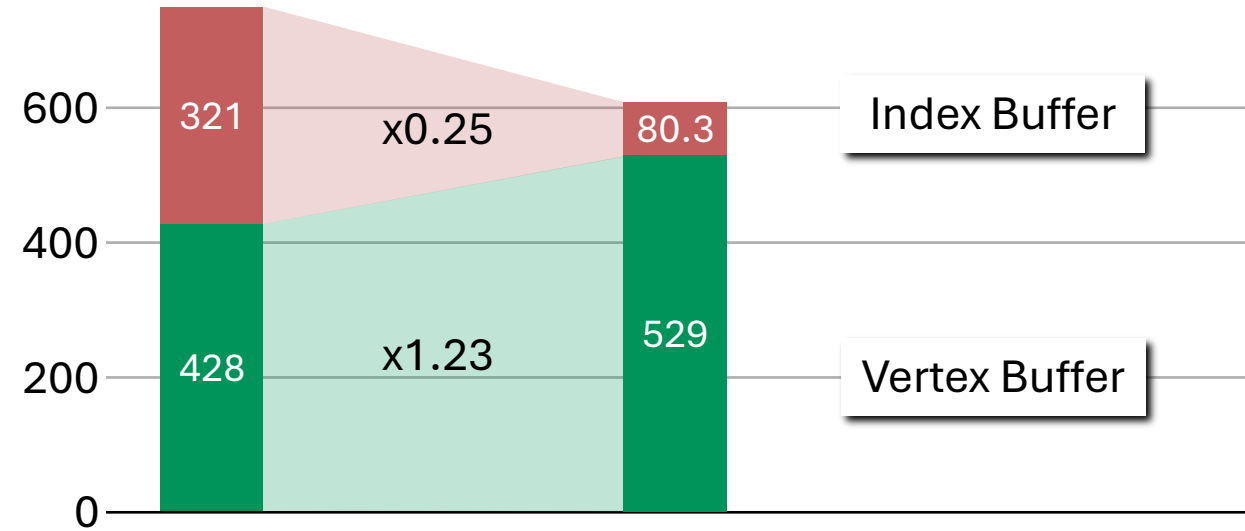
```
struct Triangle {  
    uint32_t v0;  
    uint32_t v1;  
    uint32_t v2;  
};
```

```
struct Triangle {  
    uint8_t v0;  
    uint8_t v1;  
    uint8_t v2;  
};
```

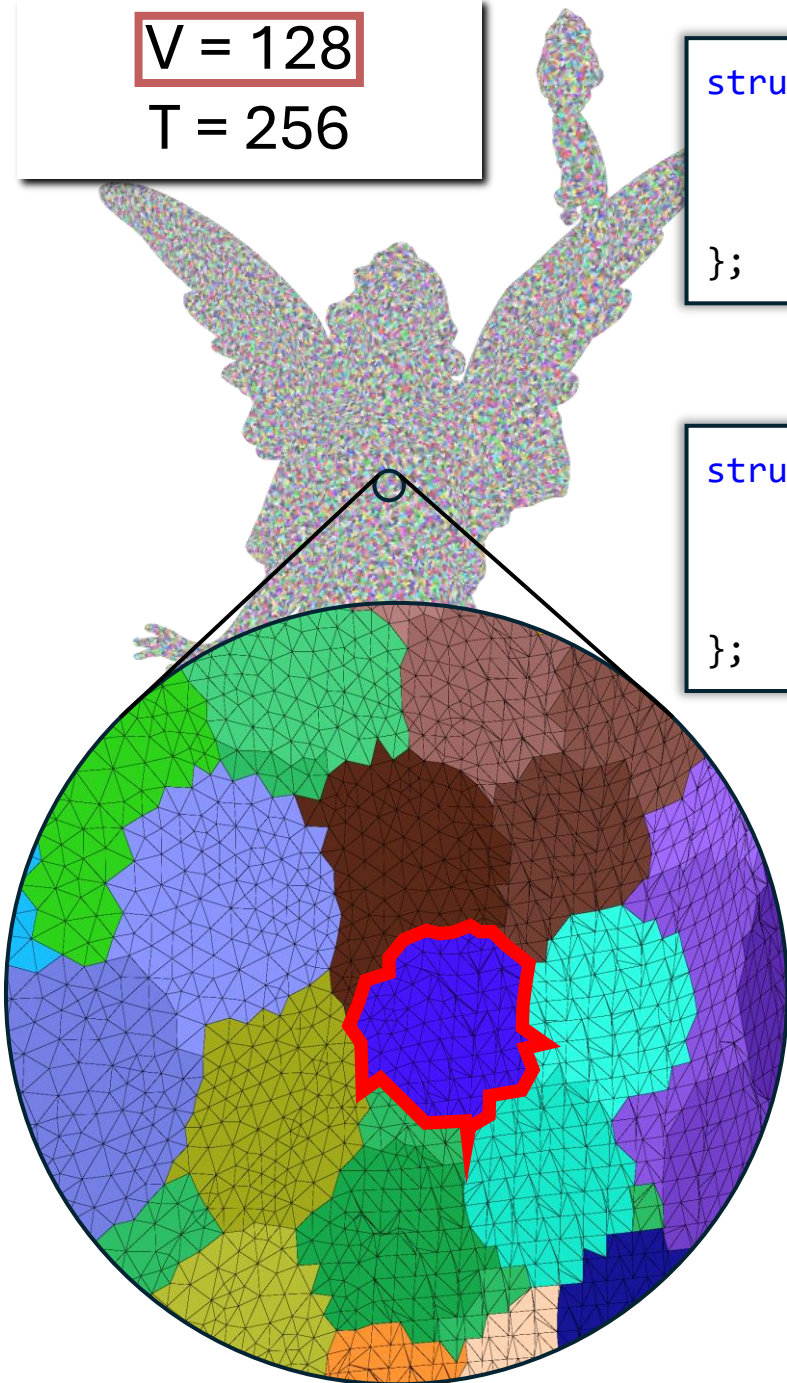
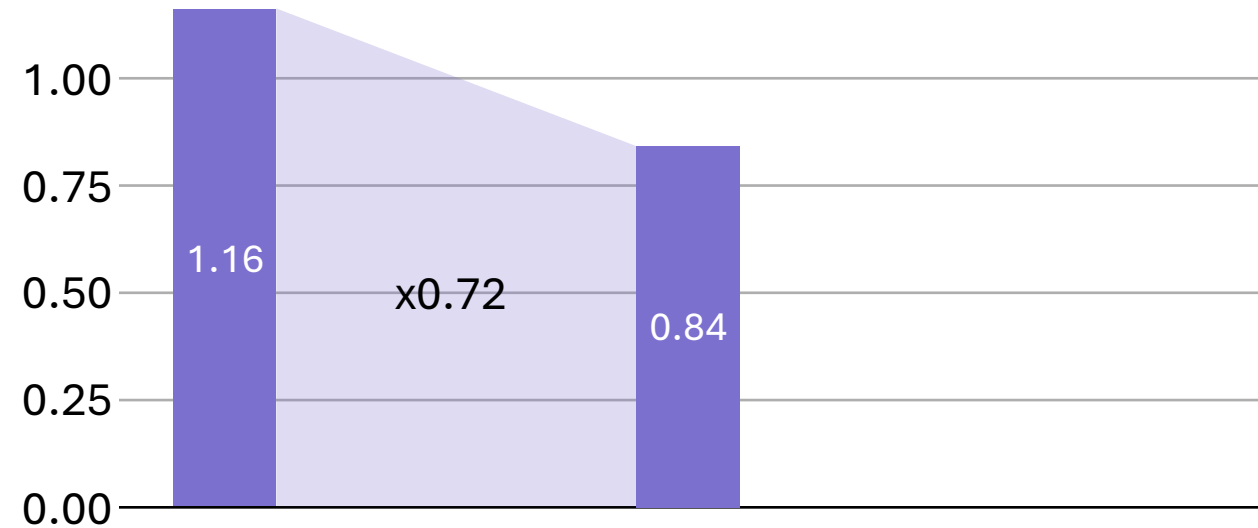
Vertex
Pipeline

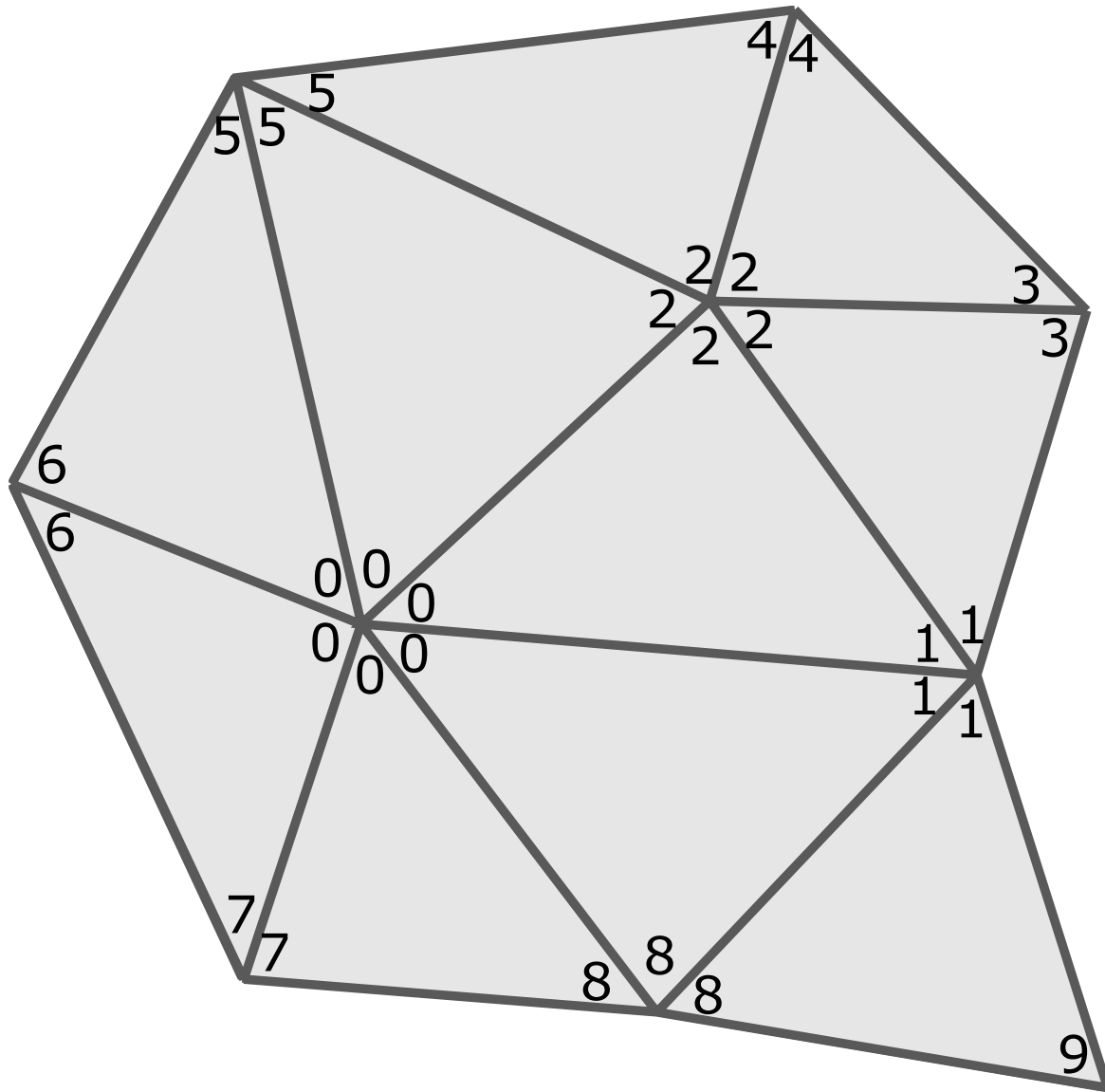
Mesh-
Shading

Buffers Size in MiB



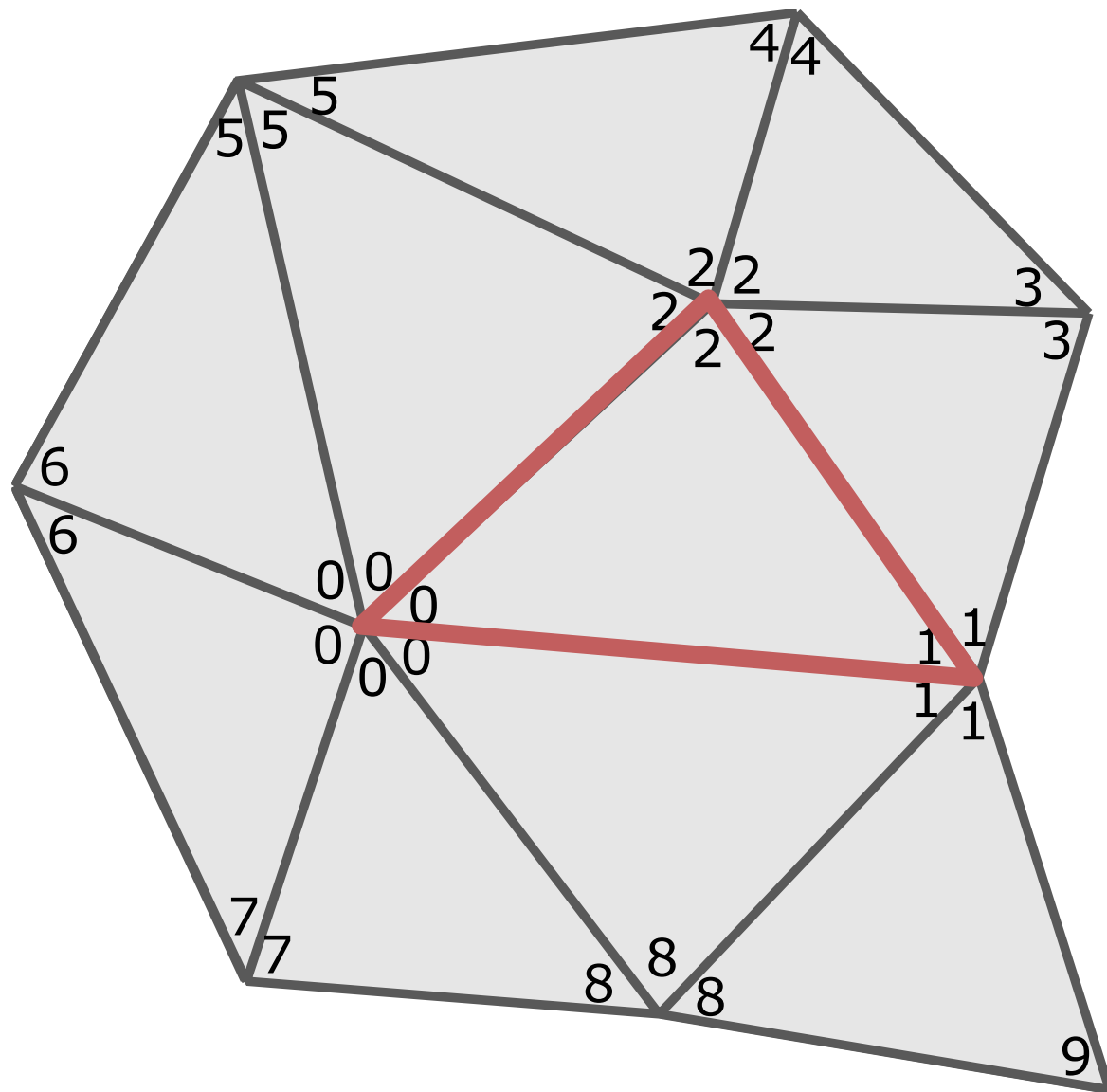
Render Time in ms on RX 7900XTX





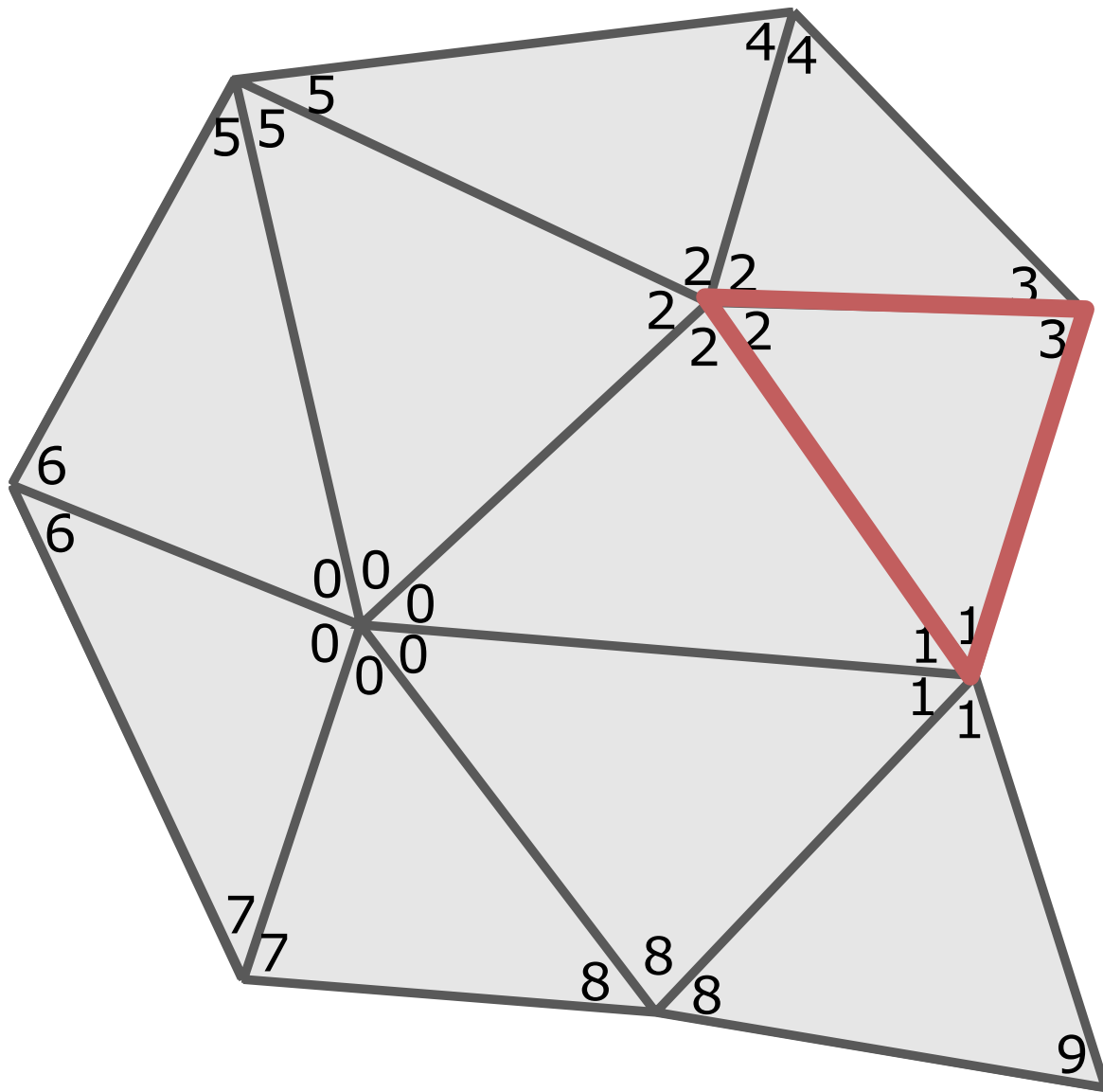
Basic Index Buffer

0	2	2	2	2	0	0	0	0	1
1	1	3	4	5	5	6	7	8	8
2	3	4	5	0	6	7	8	1	9



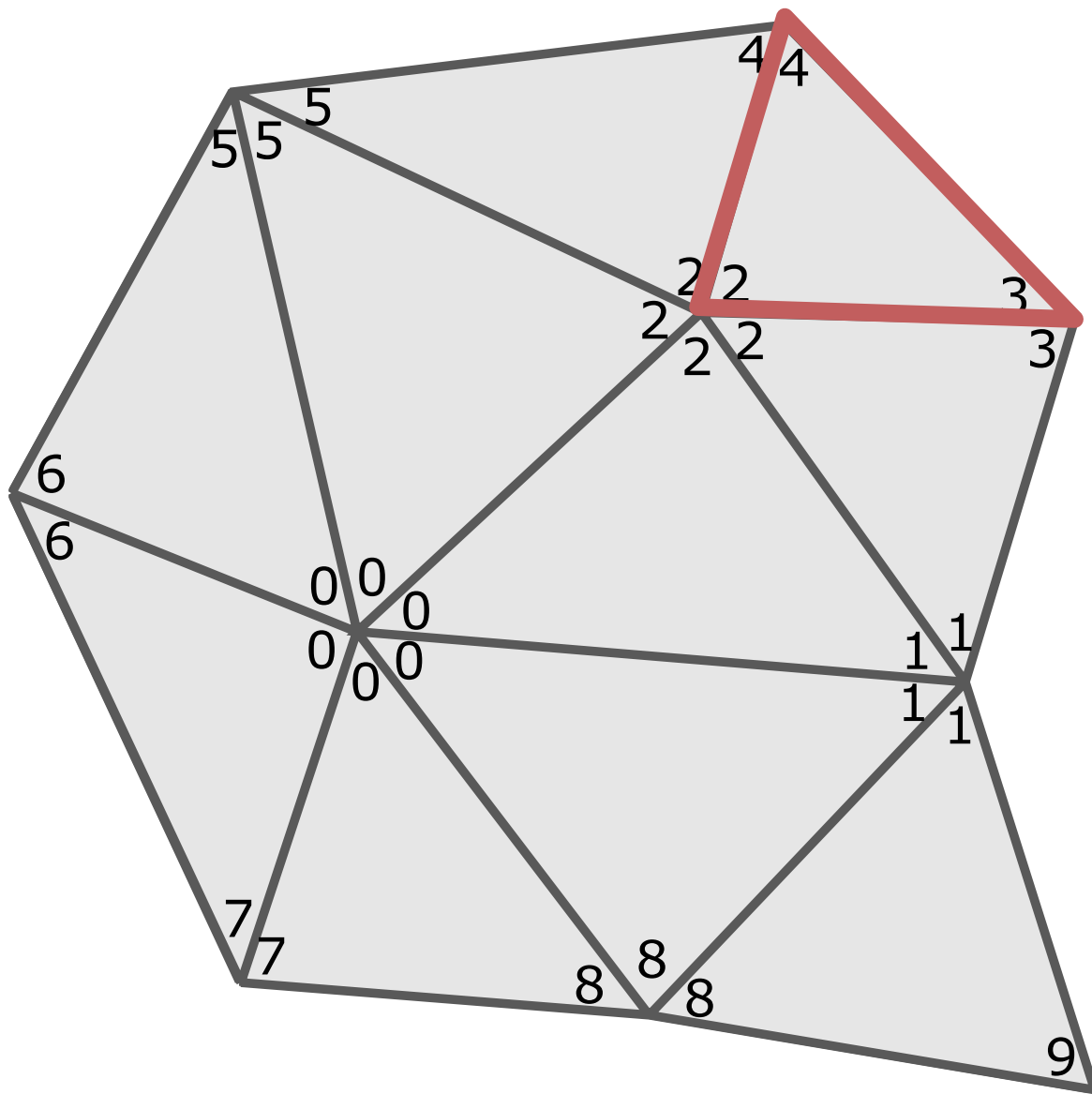
Basic Index Buffer

0	2	2	2	2	0	0	0	0	1
1	1	3	4	5	5	6	7	8	8
2	3	4	5	0	6	7	8	1	9



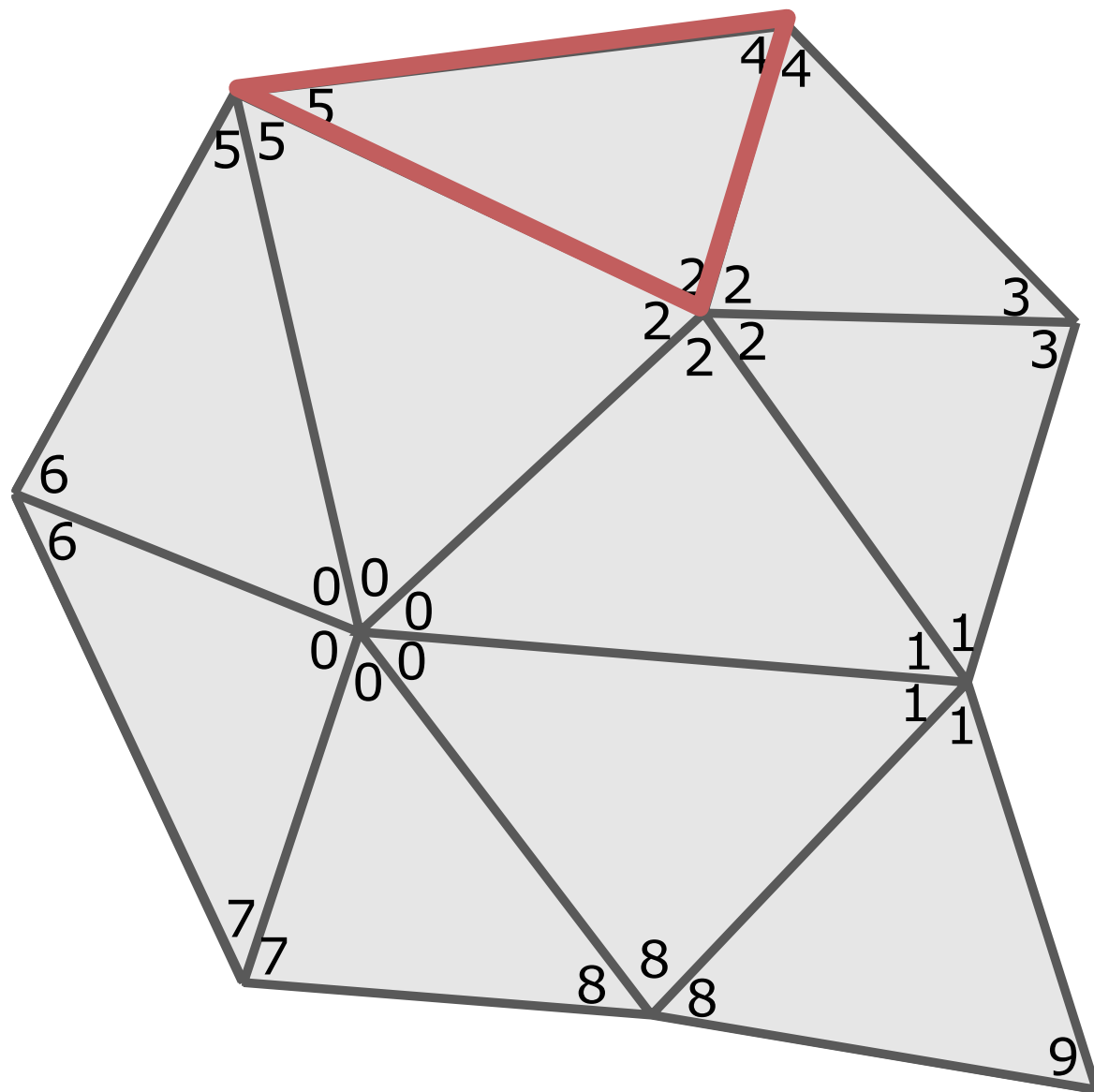
Basic Index Buffer

0	2	2	2	2	0	0	0	0	1
1	1	3	4	5	5	6	7	8	8
2	3	4	5	0	6	7	8	1	9



Basic Index Buffer

0	2	2	2	2	0	0	0	0	1
1	1	3	4	5	5	6	7	8	8
2	3	4	5	0	6	7	8	1	9



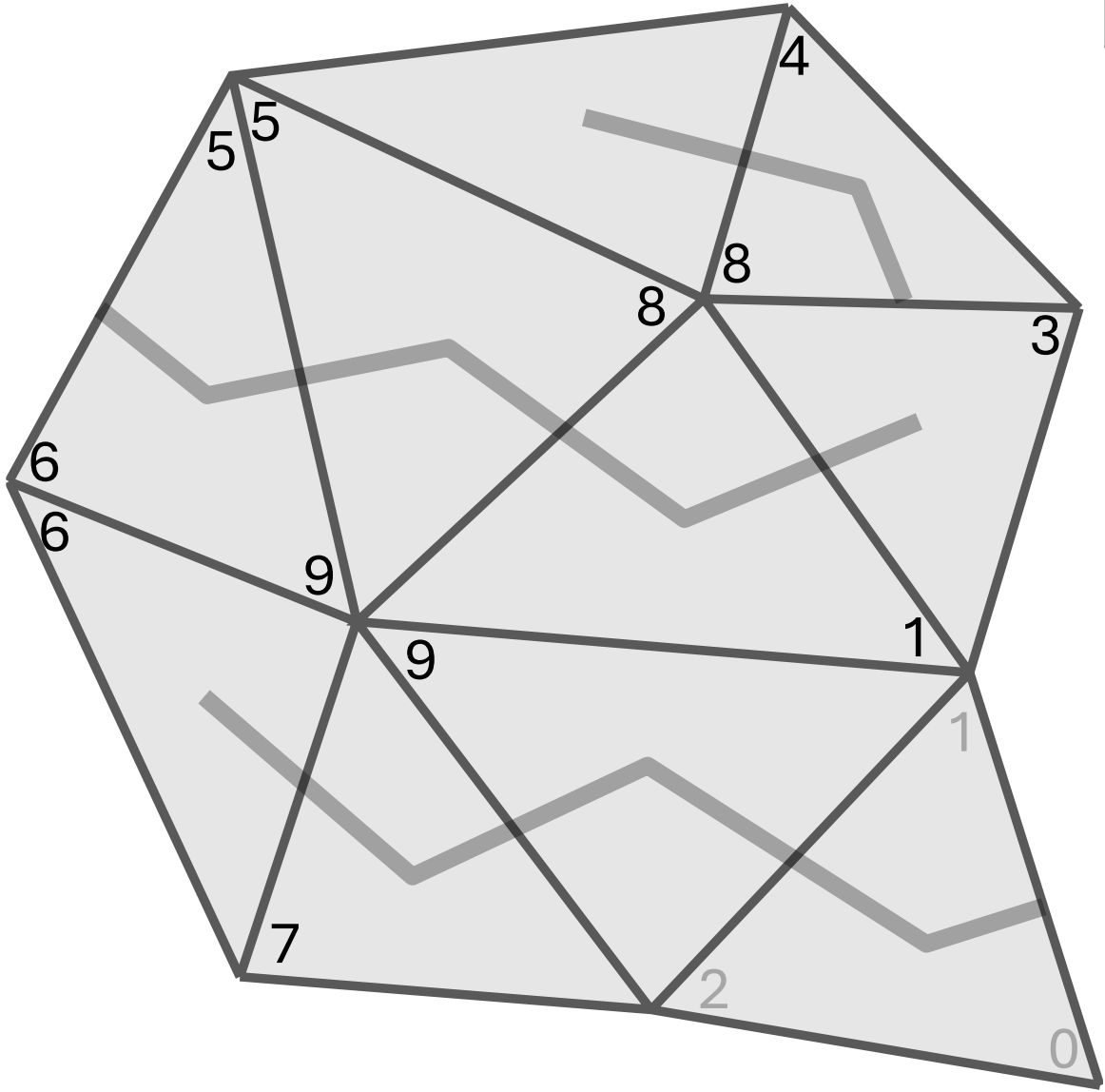
Basic Index Buffer

0	2	2	2	2	0	0	0	0	1
1	1	3	4	5	5	6	7	8	8
2	3	4	5	0	6	7	8	1	9

Basic $3 \times 10 \times 8$ = 240bit

Alternating Triangle Strip (ATS)

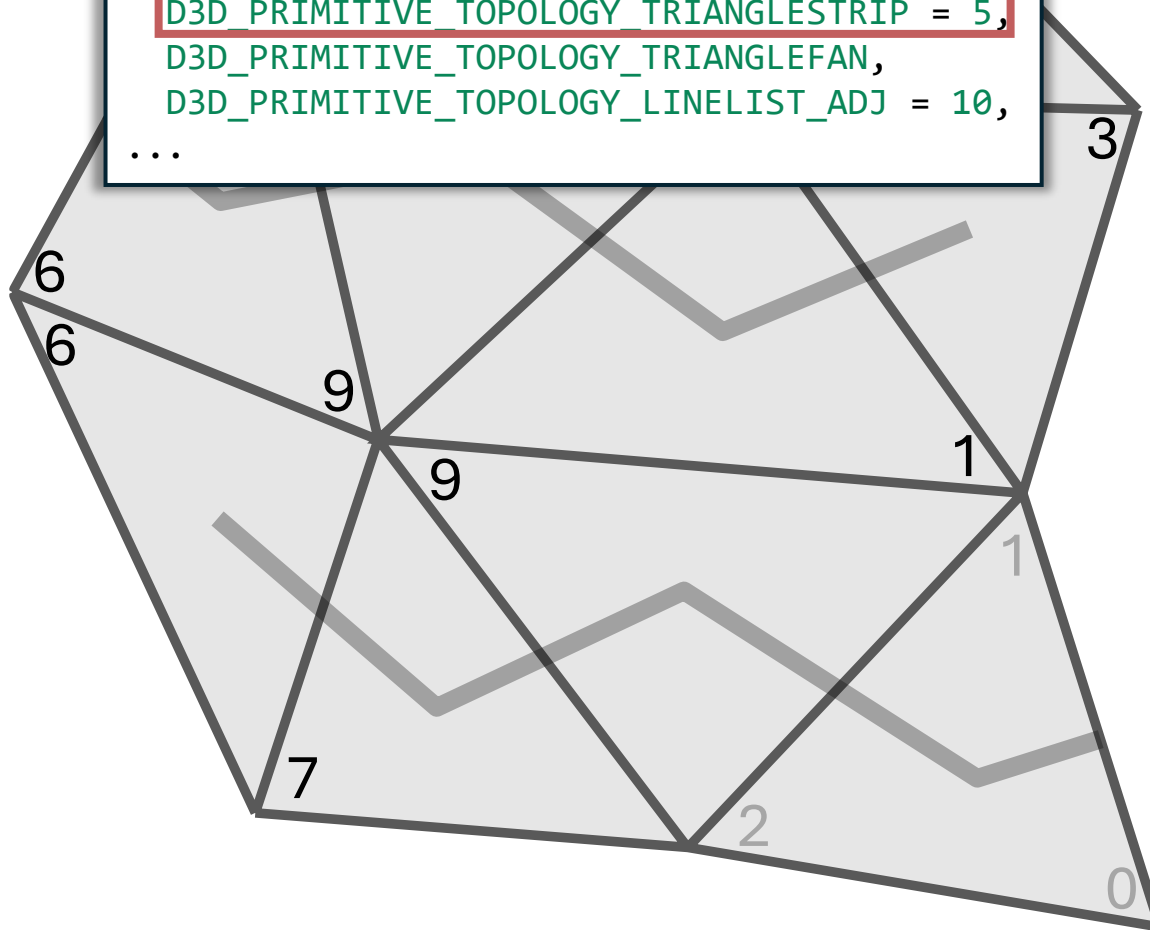
0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



```
typedef enum D3D_PRIMITIVE_TOPOLOGY {
    D3D_PRIMITIVE_TOPOLOGY_UNDEFINED = 0,
    D3D_PRIMITIVE_TOPOLOGY_POINTLIST = 1,
    D3D_PRIMITIVE_TOPOLOGY_LINELIST = 2,
    D3D_PRIMITIVE_TOPOLOGY_LINESTRIP = 3,
    D3D_PRIMITIVE_TOPOLOGY_TRIANGLELIST = 4,
    D3D_PRIMITIVE_TOPOLOGY_TRIANGLESTRIP = 5,
    D3D_PRIMITIVE_TOPOLOGY_TRIANGLEFAN,
    D3D_PRIMITIVE_TOPOLOGY_LINELIST_ADJ = 10,
    ...
}
```

Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

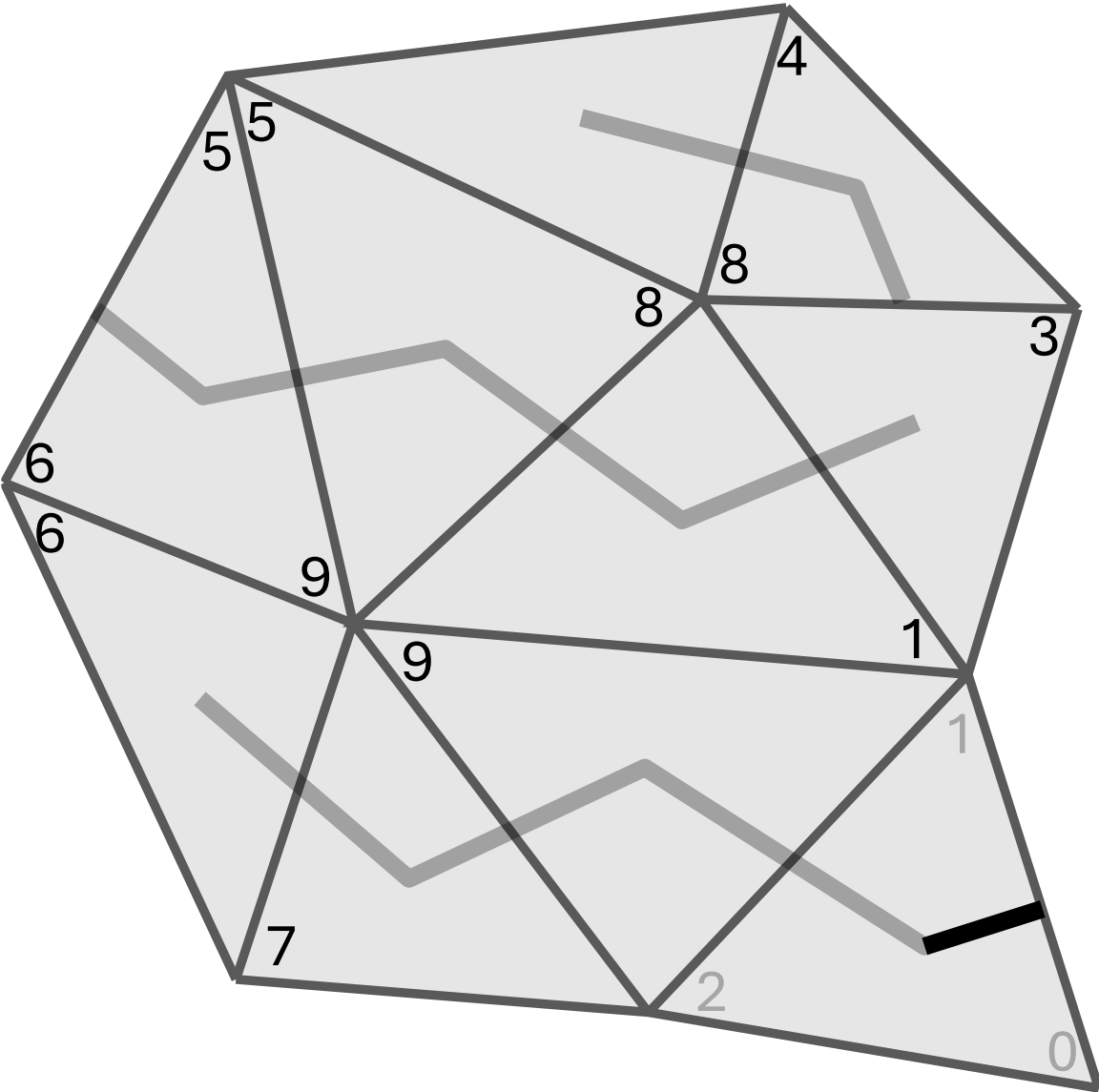


Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0
1
2

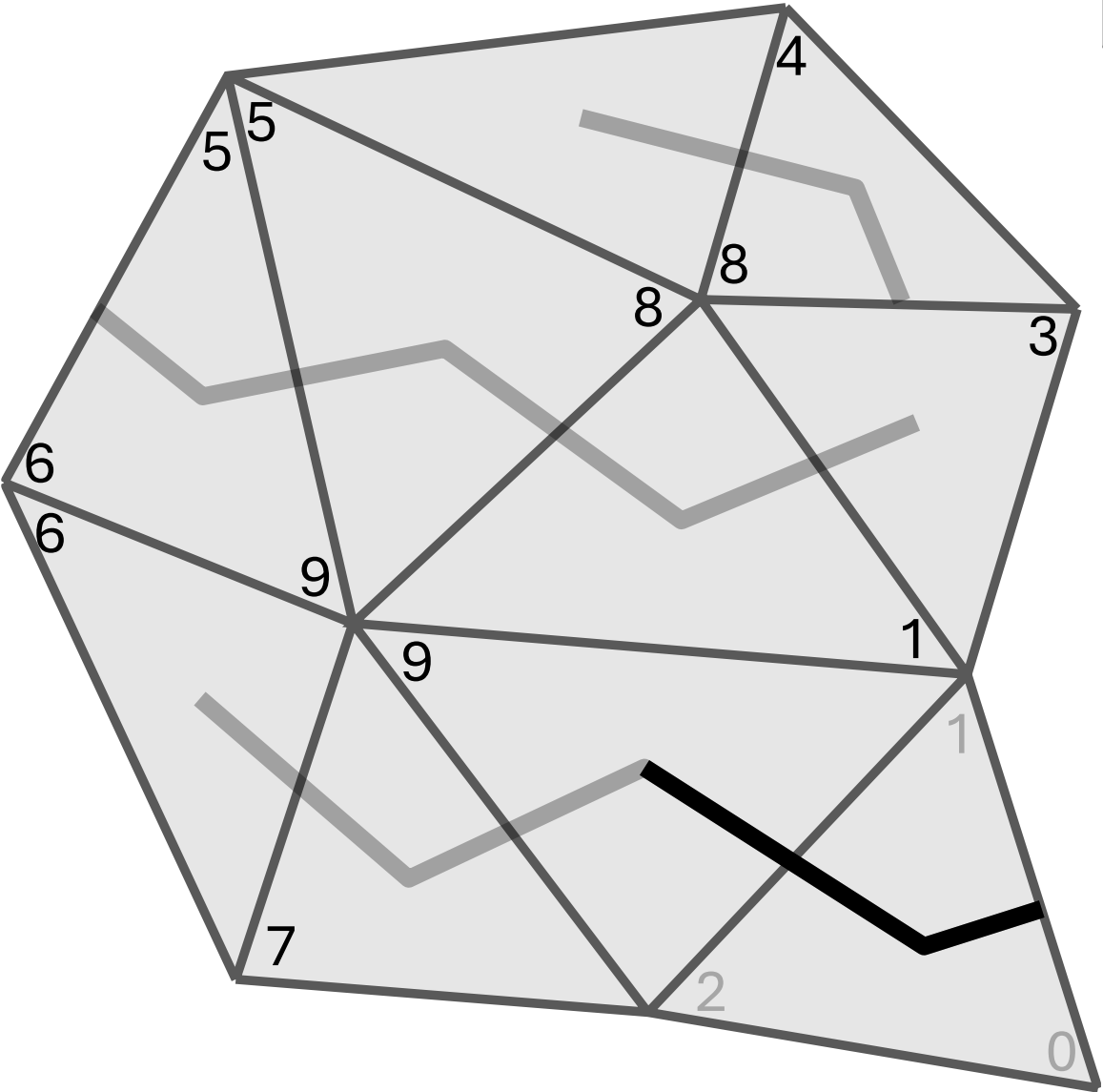


Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



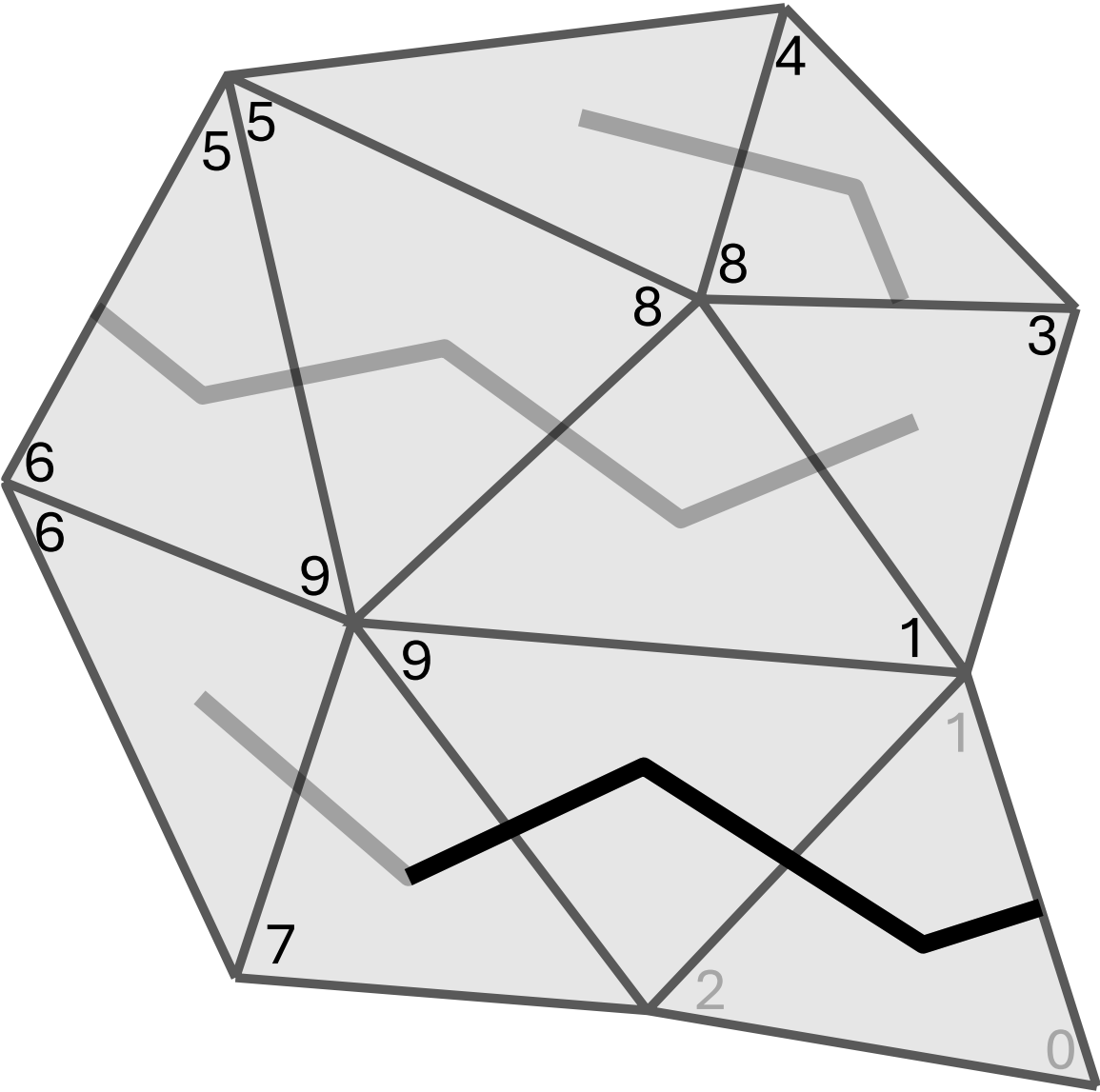
0	9
1	2
2	1



Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

0	9	2
1	2	9
2	1	7

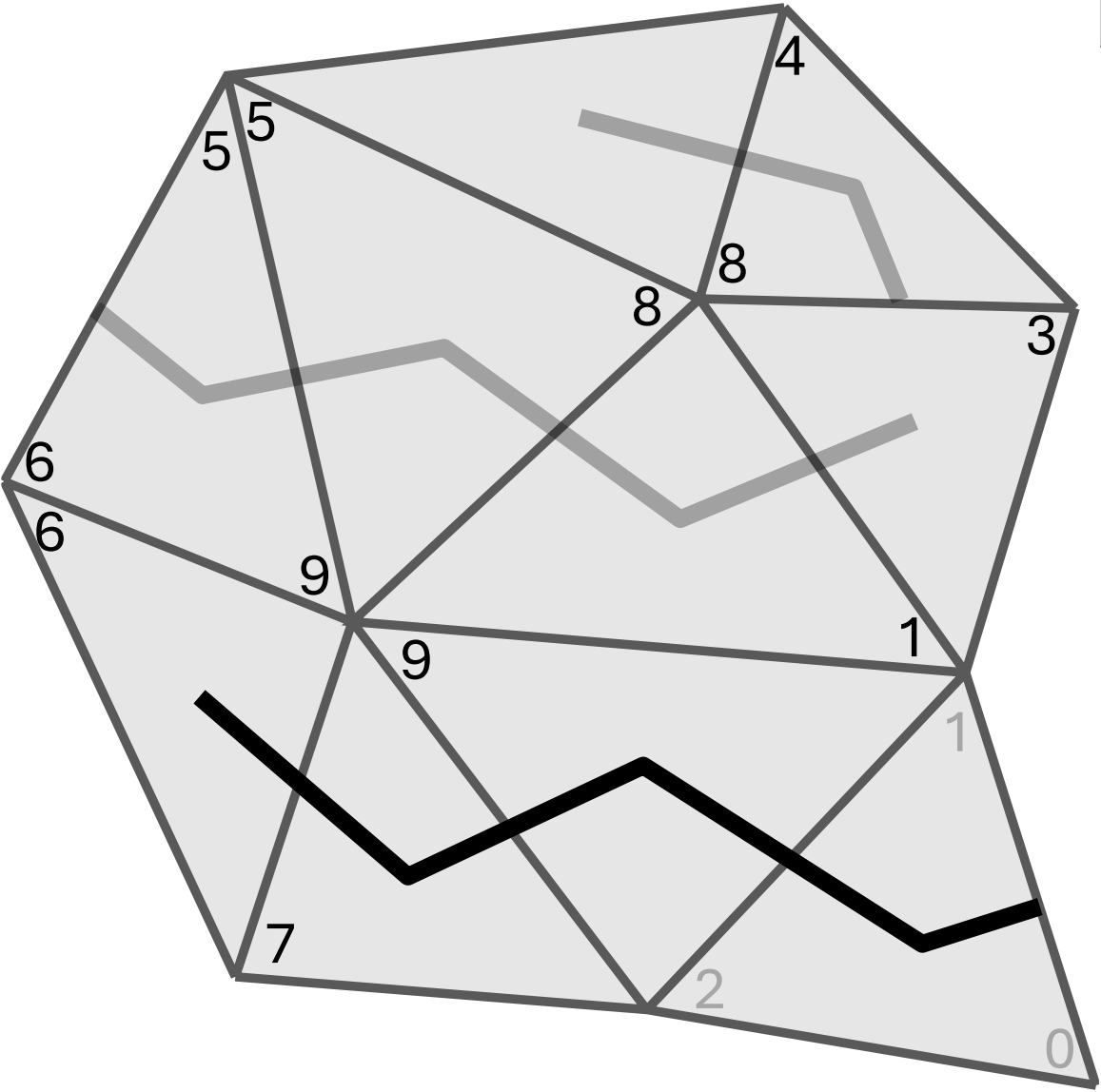


Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6
1	2	9	7
2	1	7	9

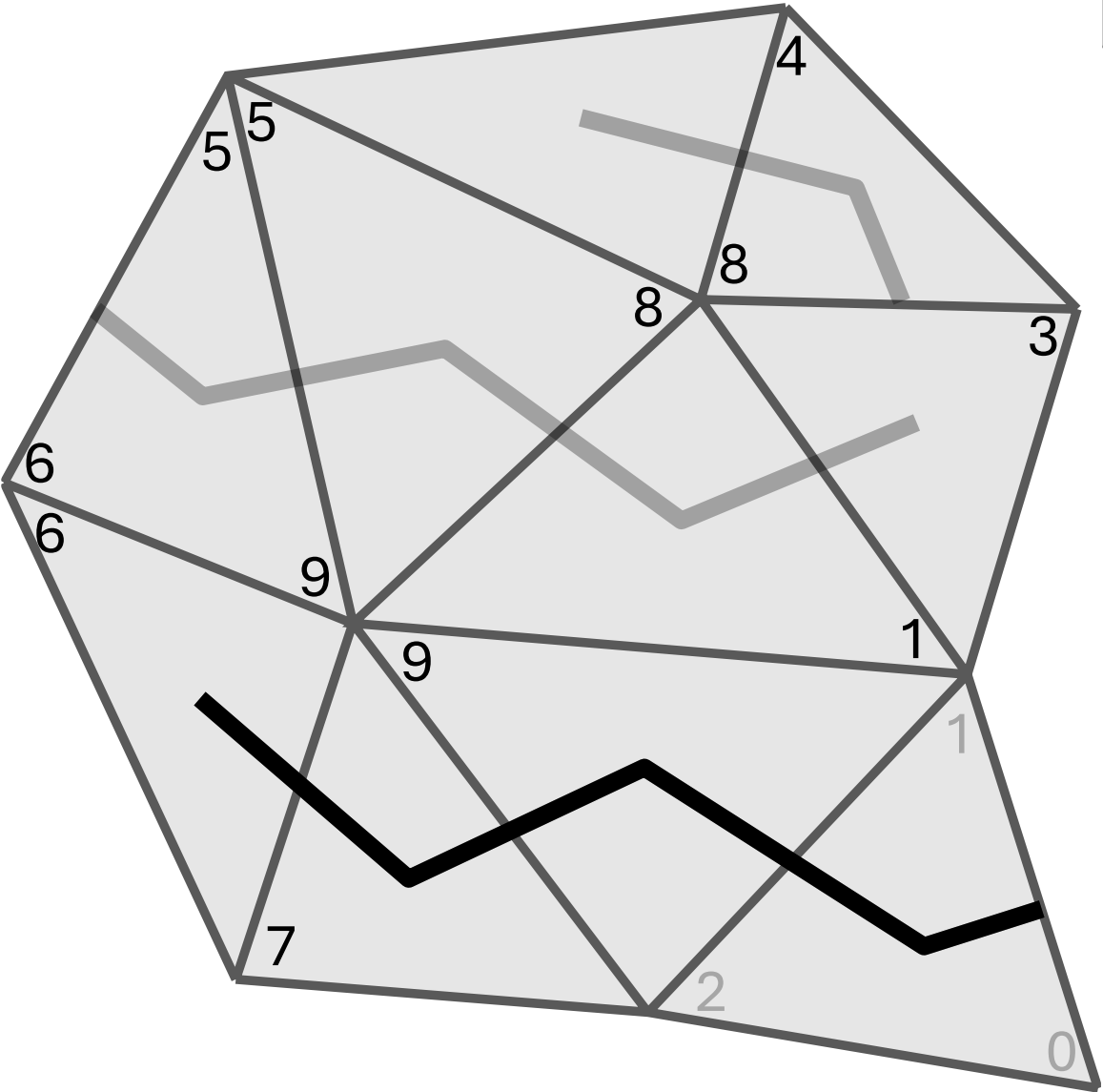


Alternating Triangle Strip (ATS)

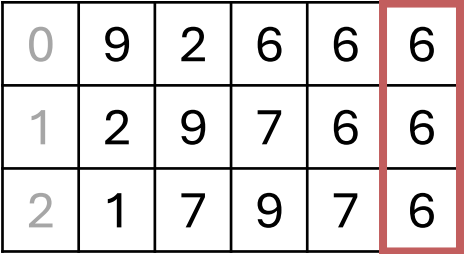
0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6	6
1	2	9	7	6
2	1	7	9	7



0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

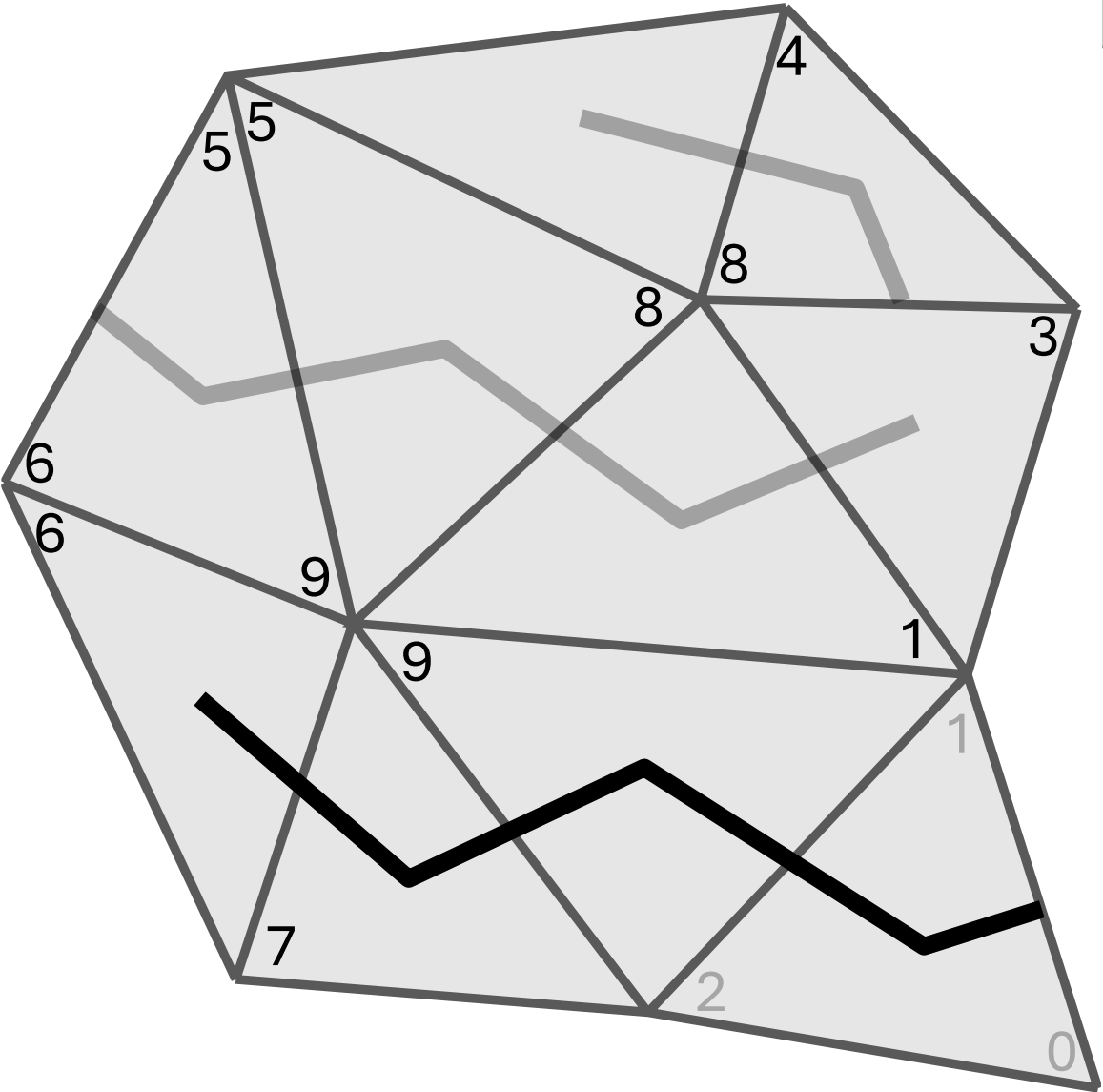


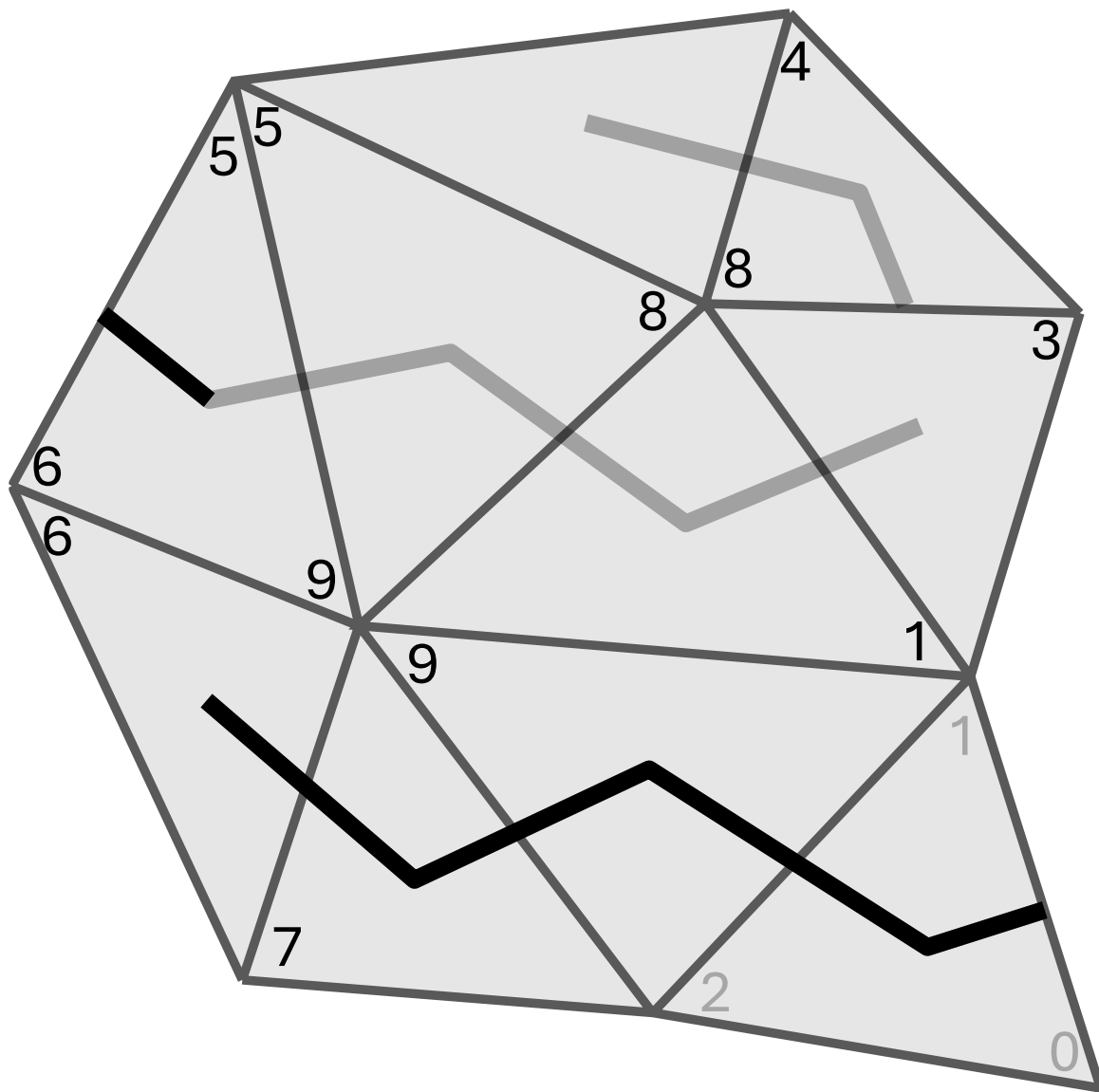
Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6	6	6	6
1	2	9	7	6	6	6
2	1	7	9	7	6	5



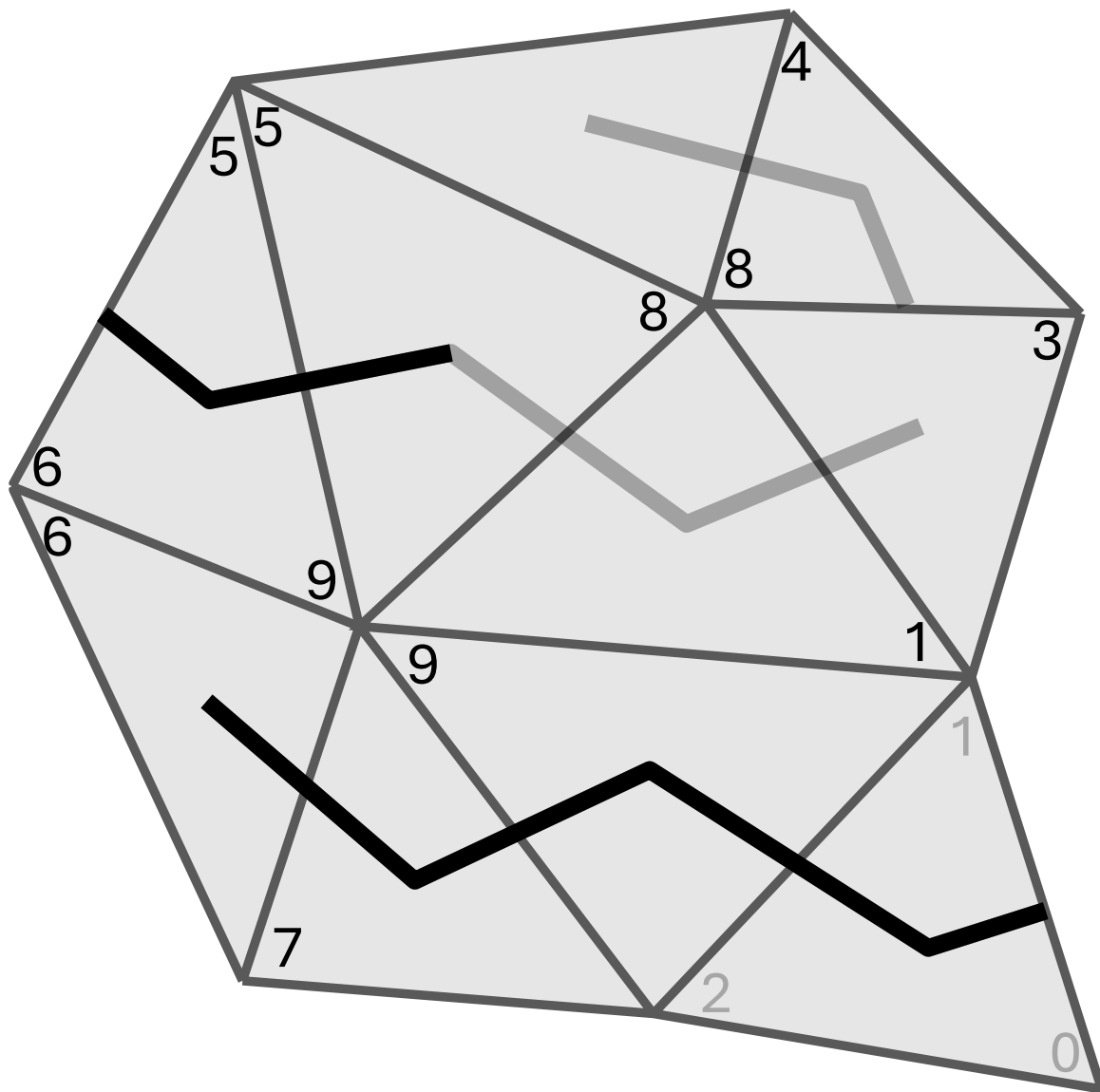


Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6	6	6	6	9
1	2	9	7	6	6	6	5
2	1	7	9	7	6	5	6

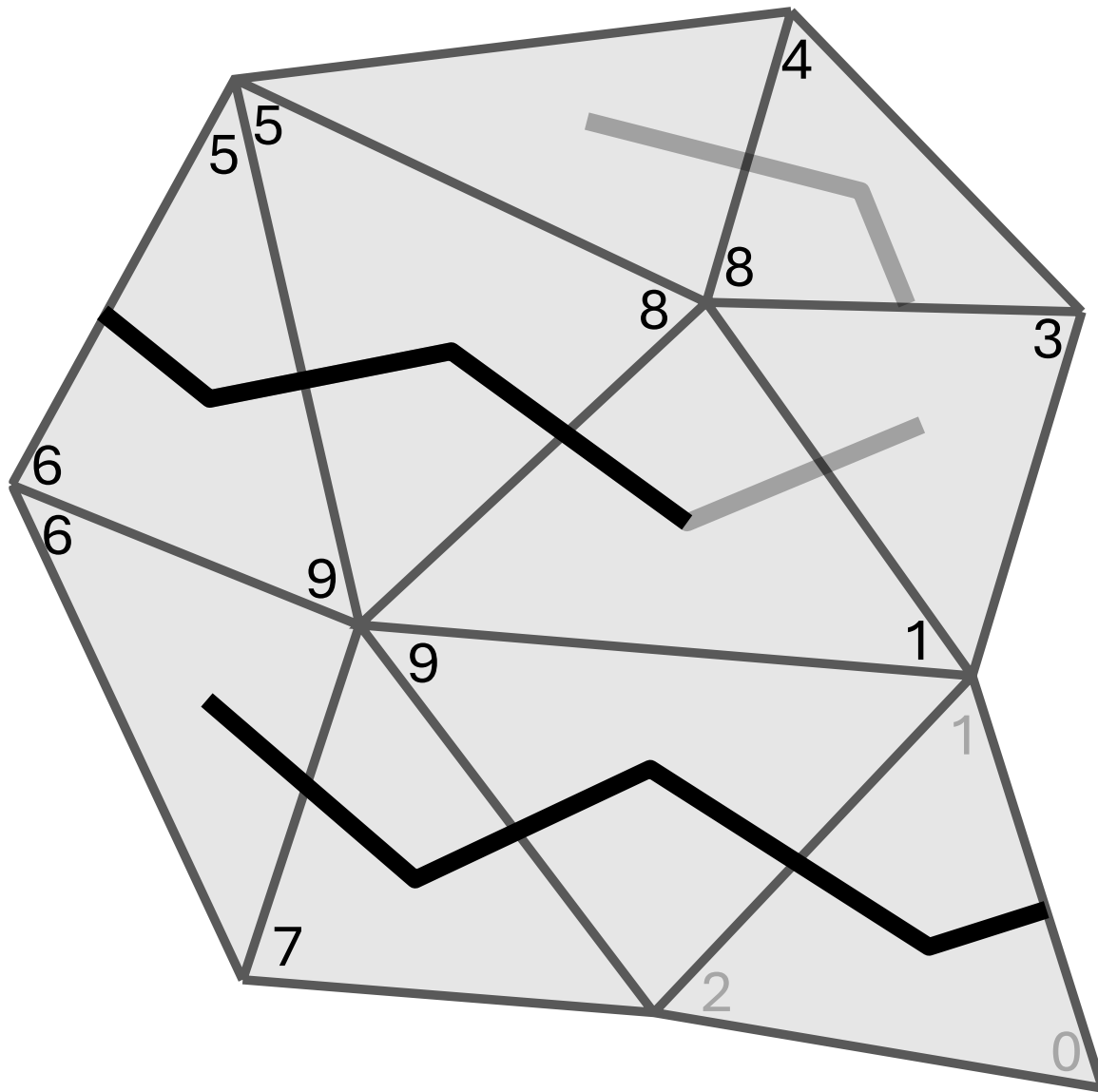


Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6	6	6	6	9	5
1	2	9	7	6	6	6	5	9
2	1	7	9	7	6	5	6	8

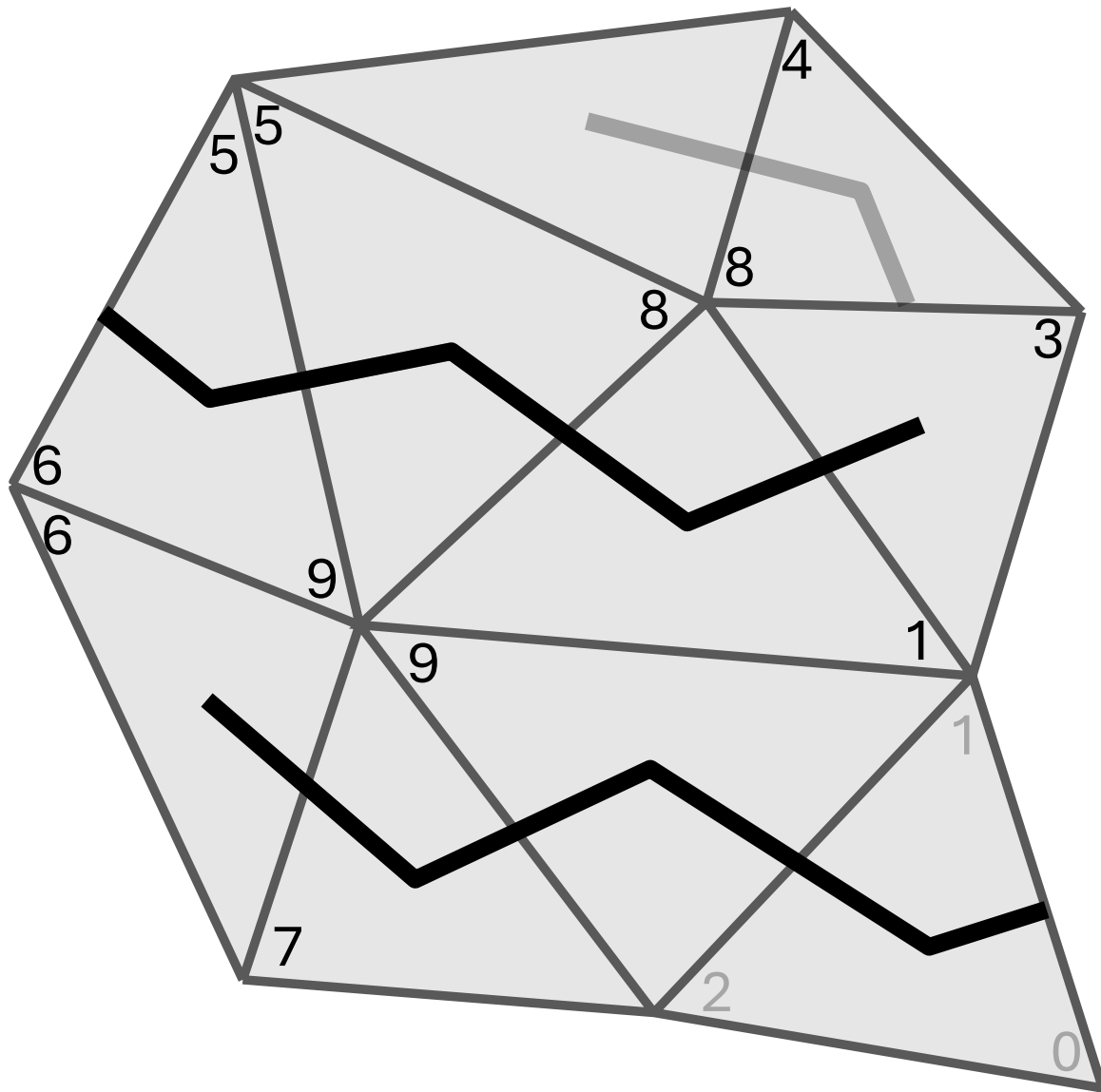


Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6	6	6	6	9	5	1
1	2	9	7	6	6	6	5	9	8
2	1	7	9	7	6	5	6	8	9



Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



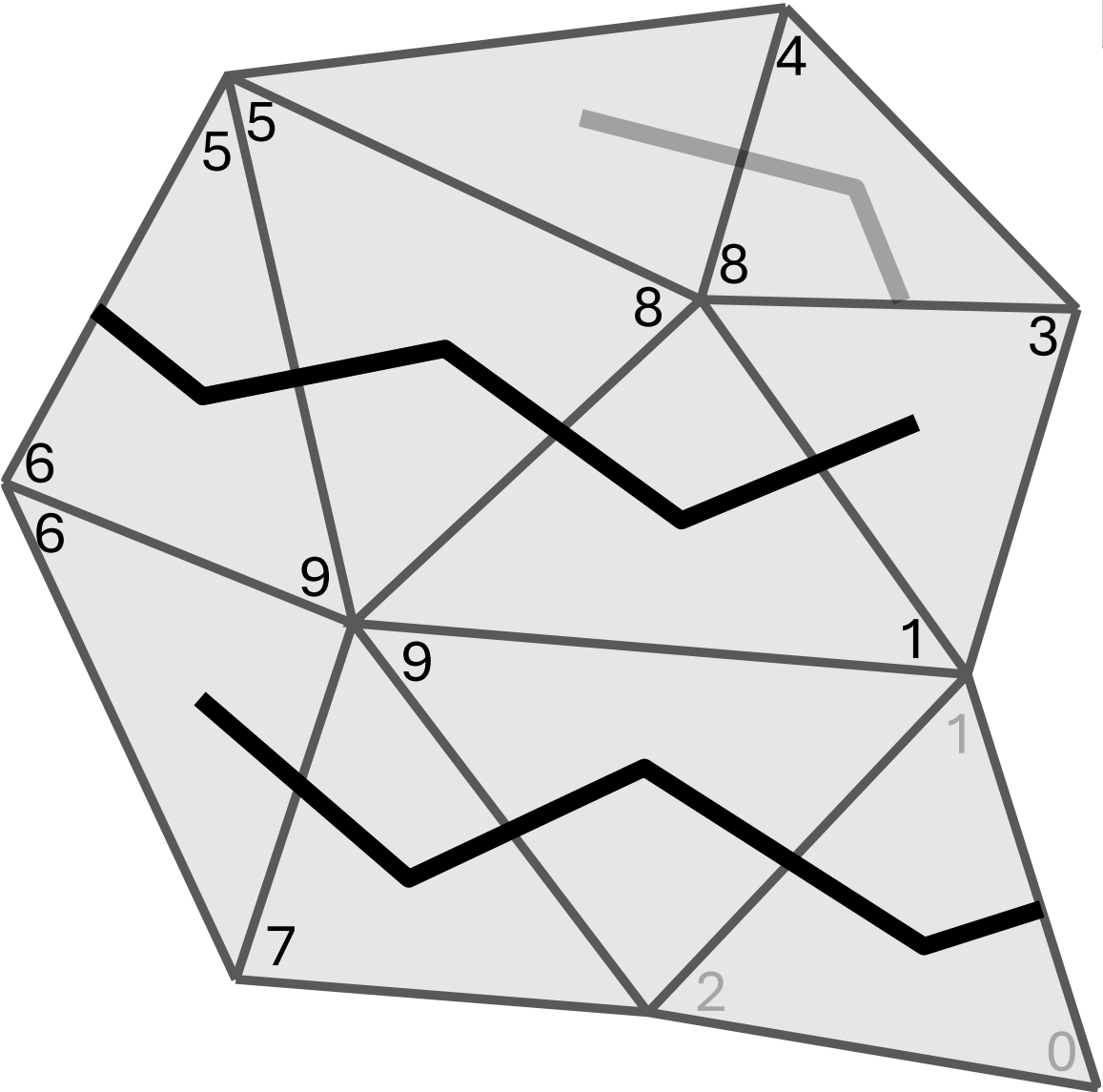
0	9	2	6	6	6	6	9	5	1	8
1	2	9	7	6	6	6	5	9	8	1
2	1	7	9	7	6	5	6	8	9	3

Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6	6	6	6	9	5	1	8	3
1	2	9	7	6	6	6	5	9	8	1	3
2	1	7	9	7	6	5	6	8	9	3	1

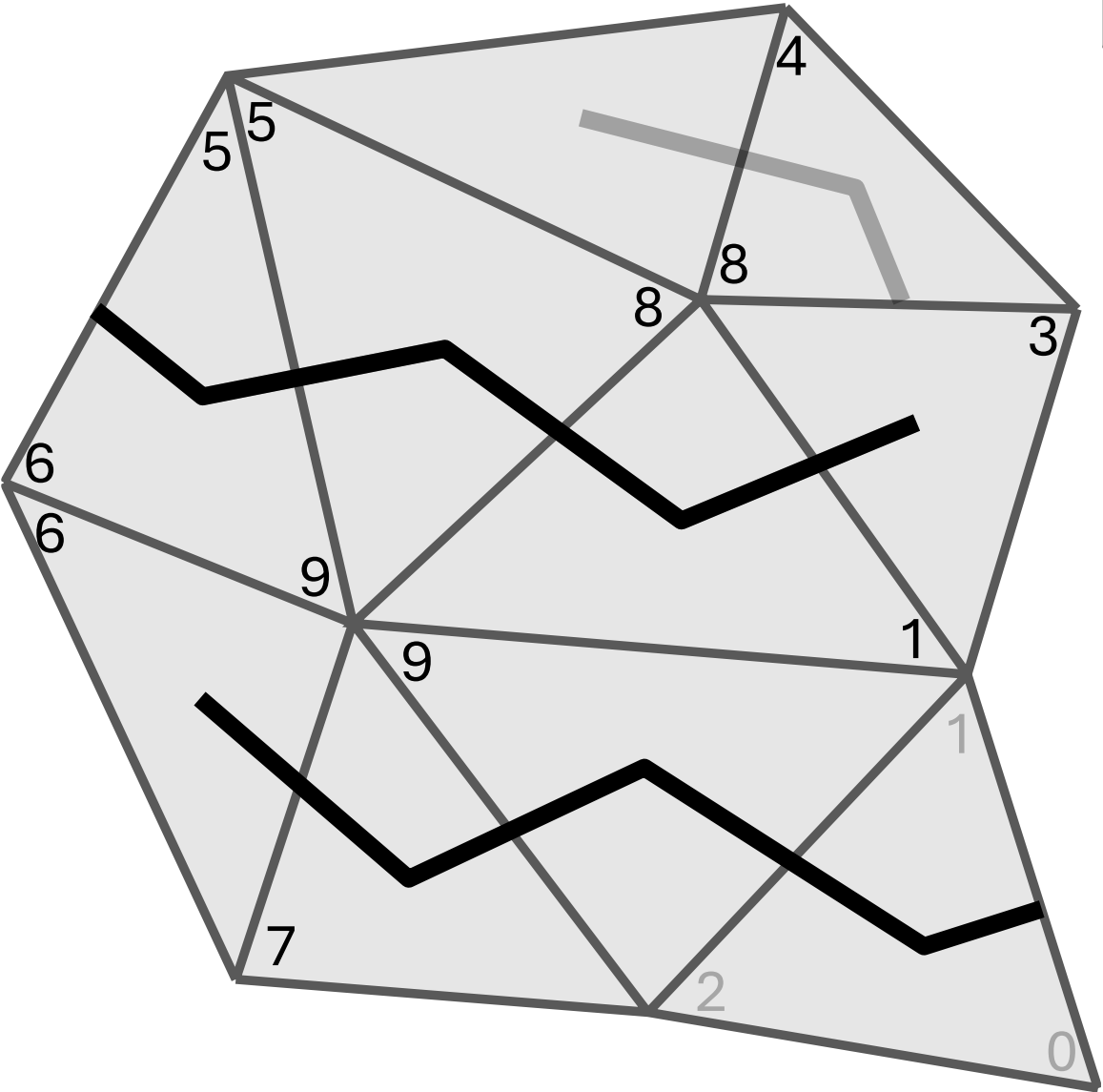


Alternating Triangle Strip (ATS)

0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

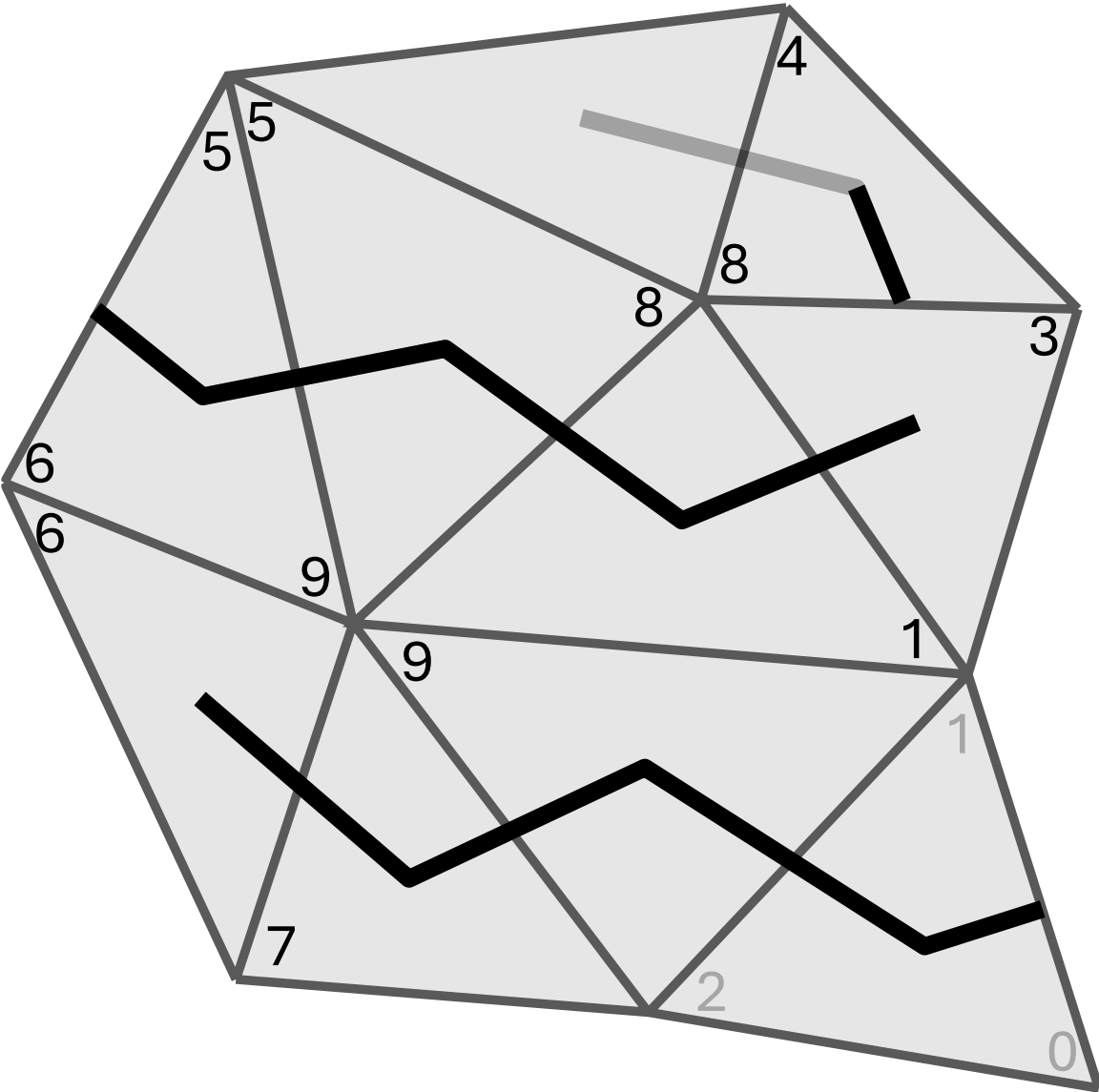


0	9	2	6	6	6	6	9	5	1	8	3	3
1	2	9	7	6	6	6	5	9	8	1	3	3
2	1	7	9	7	6	5	6	8	9	3	1	8



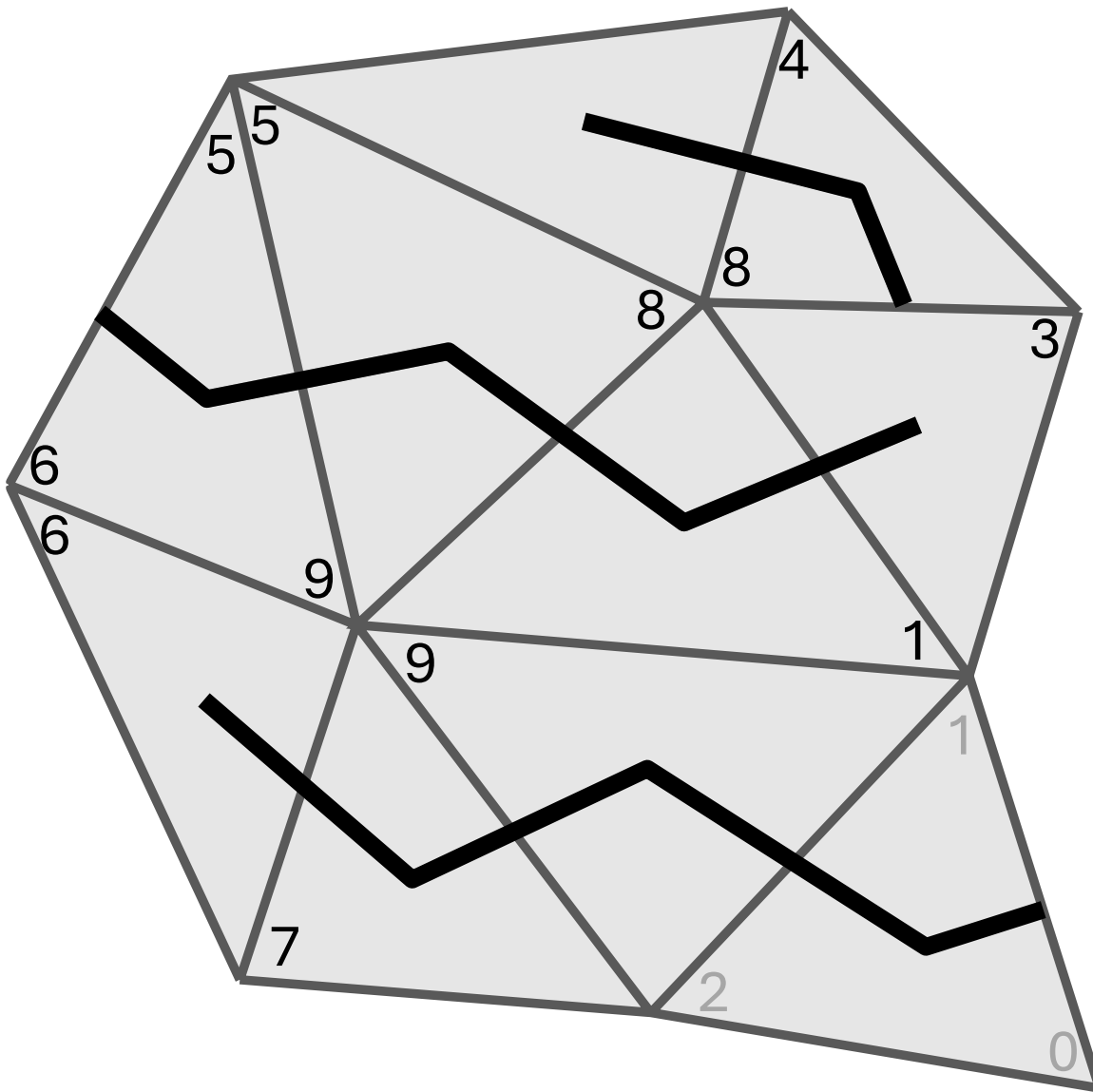
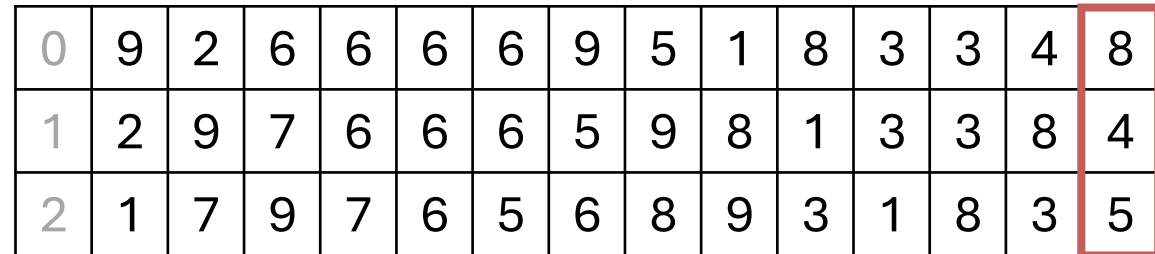
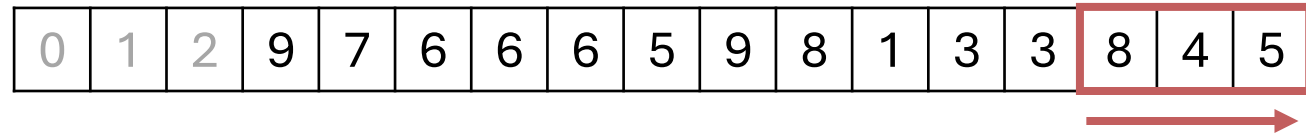
Alternating Triangle Strip (ATS)

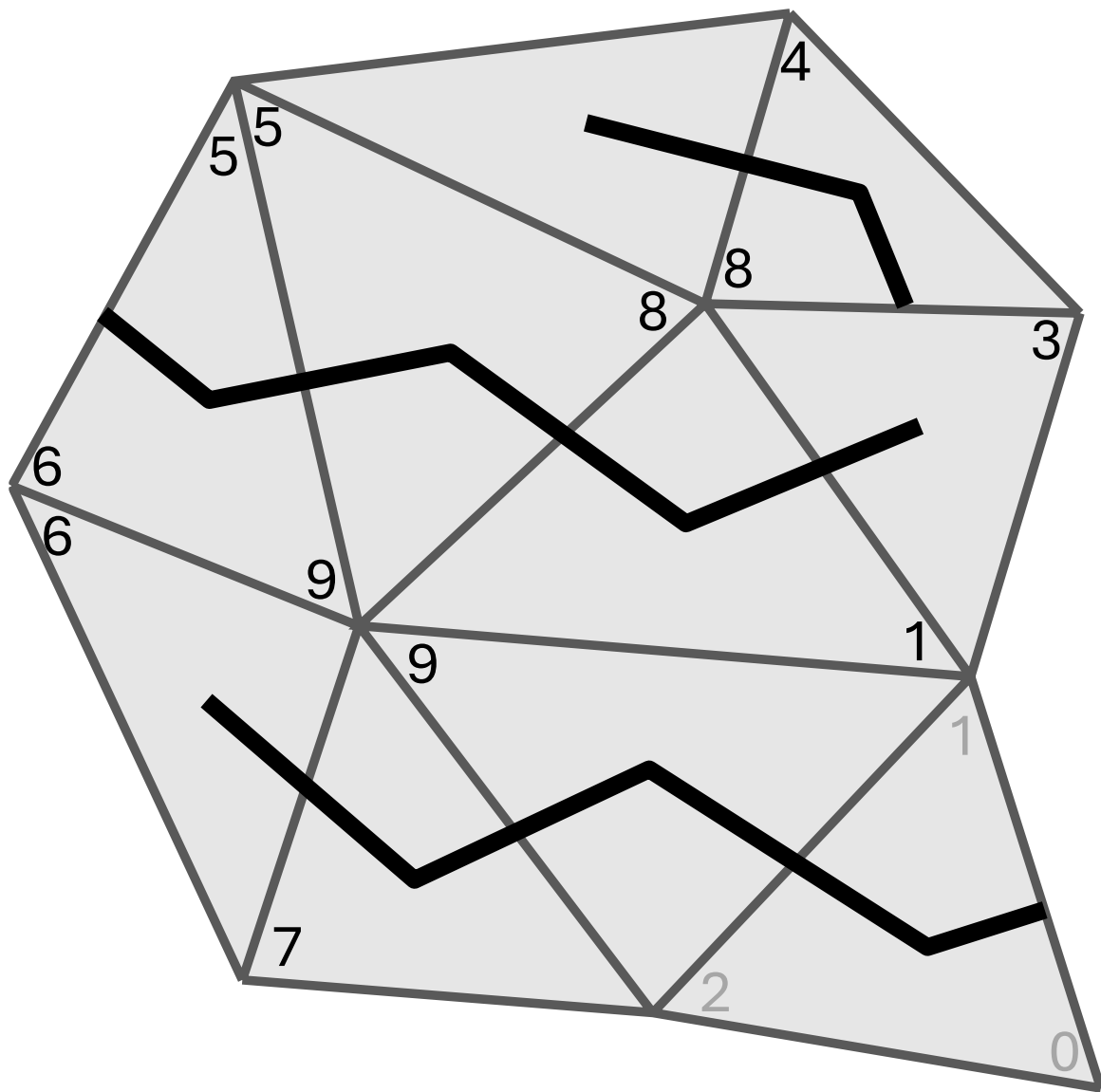
0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



0	9	2	6	6	6	6	9	5	1	8	3	3	4
1	2	9	7	6	6	6	5	9	8	1	3	3	8
2	1	7	9	7	6	5	6	8	9	3	1	8	3

Alternating Triangle Strip (ATS)



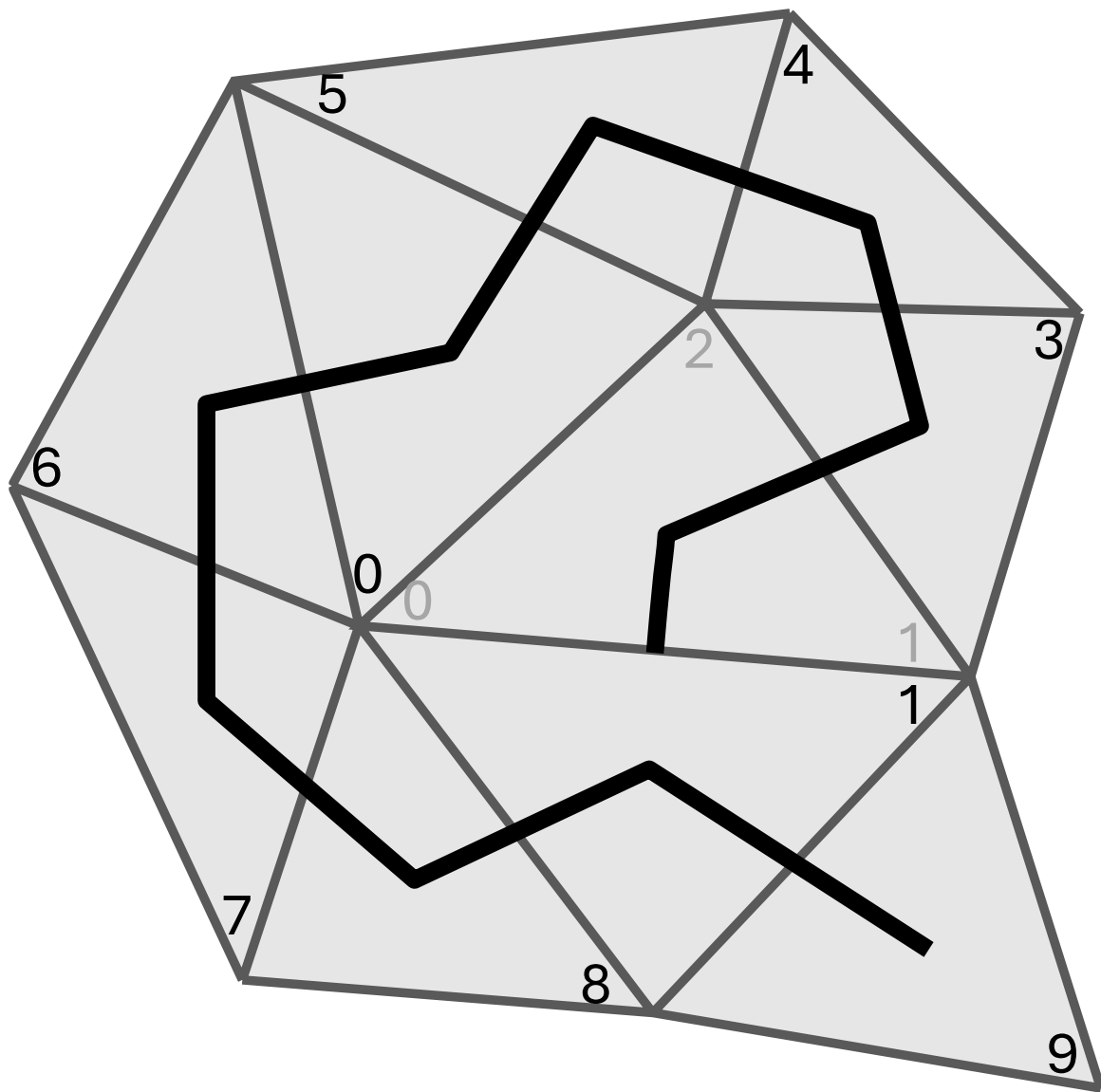


Alternating Triangle Strip (ATS)

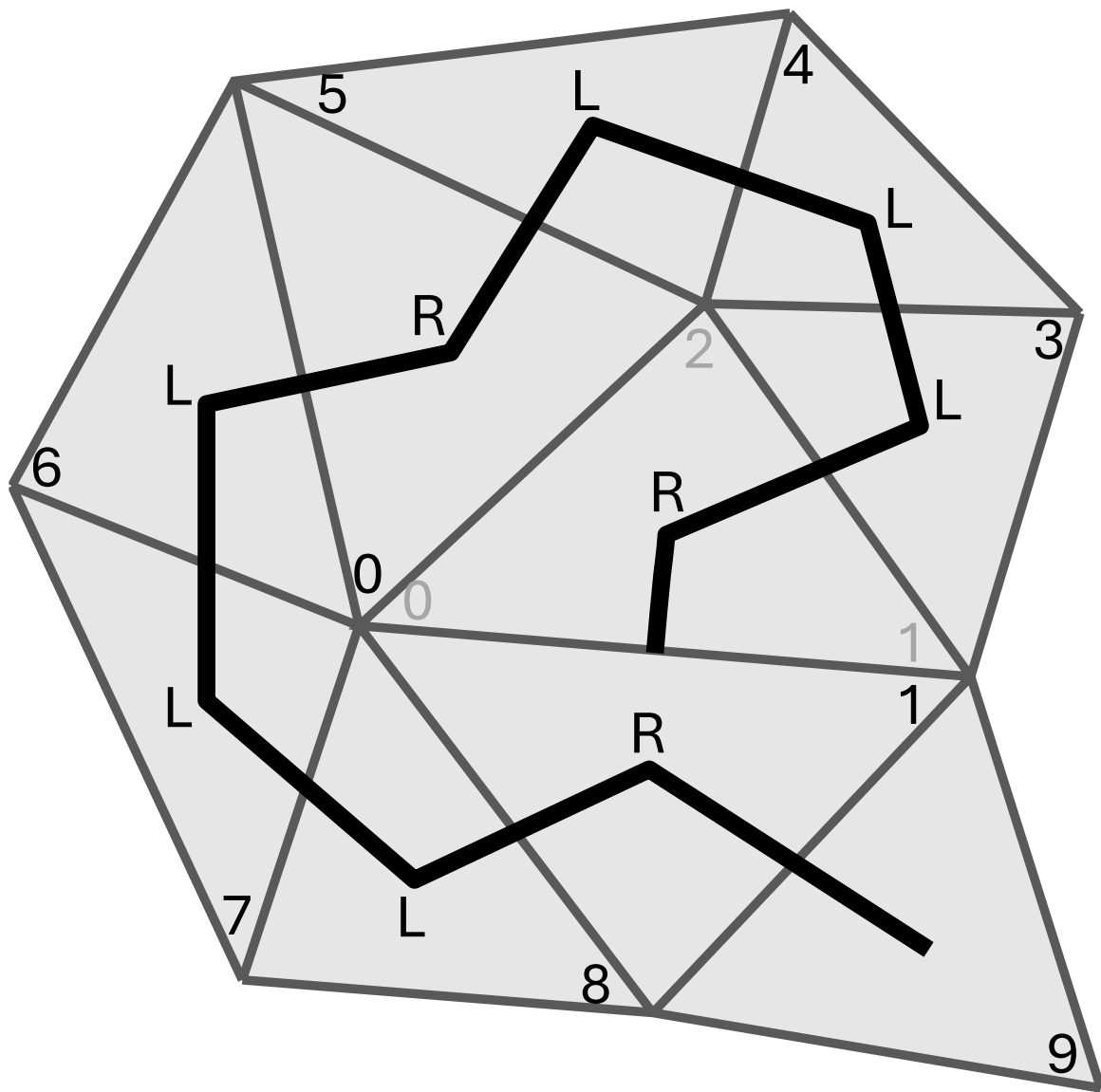
0	1	2	9	7	6	6	6	5	9	8	1	3	3	8	4	5
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

0	9	2	6	6	6	6	9	5	1	8	3	3	4	8
1	2	9	7	6	6	6	5	9	8	1	3	3	8	4
2	1	7	9	7	6	5	6	8	9	3	1	8	3	5

Basic	$3 \times 10 \times 8$	$= 240\text{bit}$
ATS	14×8	$= 112\text{bit}$

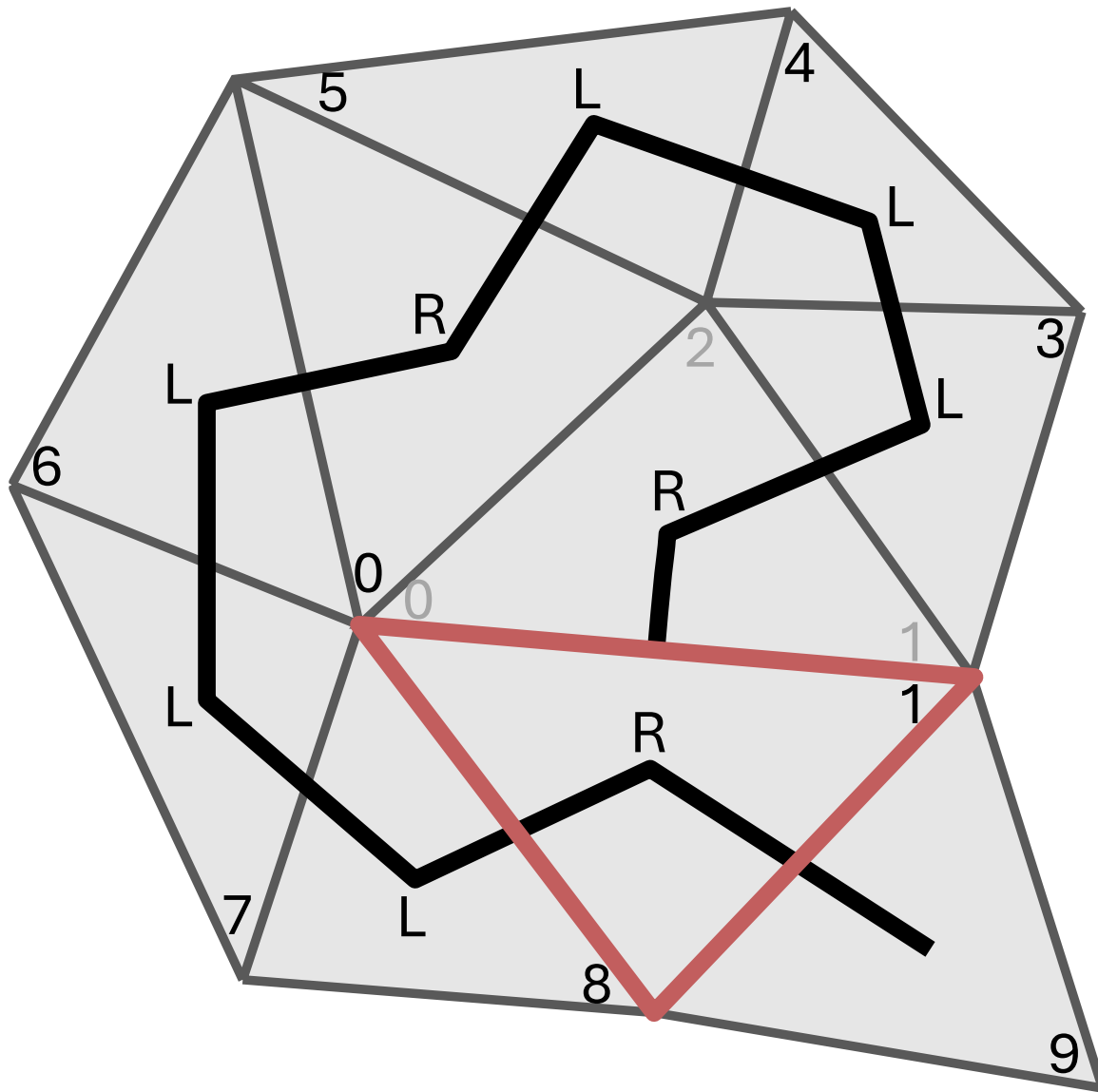


Generalized Triangle Strip (GTS)											
0	1	2	3	4	5	0	6	7	8	1	9
			R	L	L	L	R	L	L	L	R



Generalized Triangle Strip (GTS)

0	1	2	3	4	5	0	6	7	8	1	9
			R	L	L	L	R	L	L	L	R

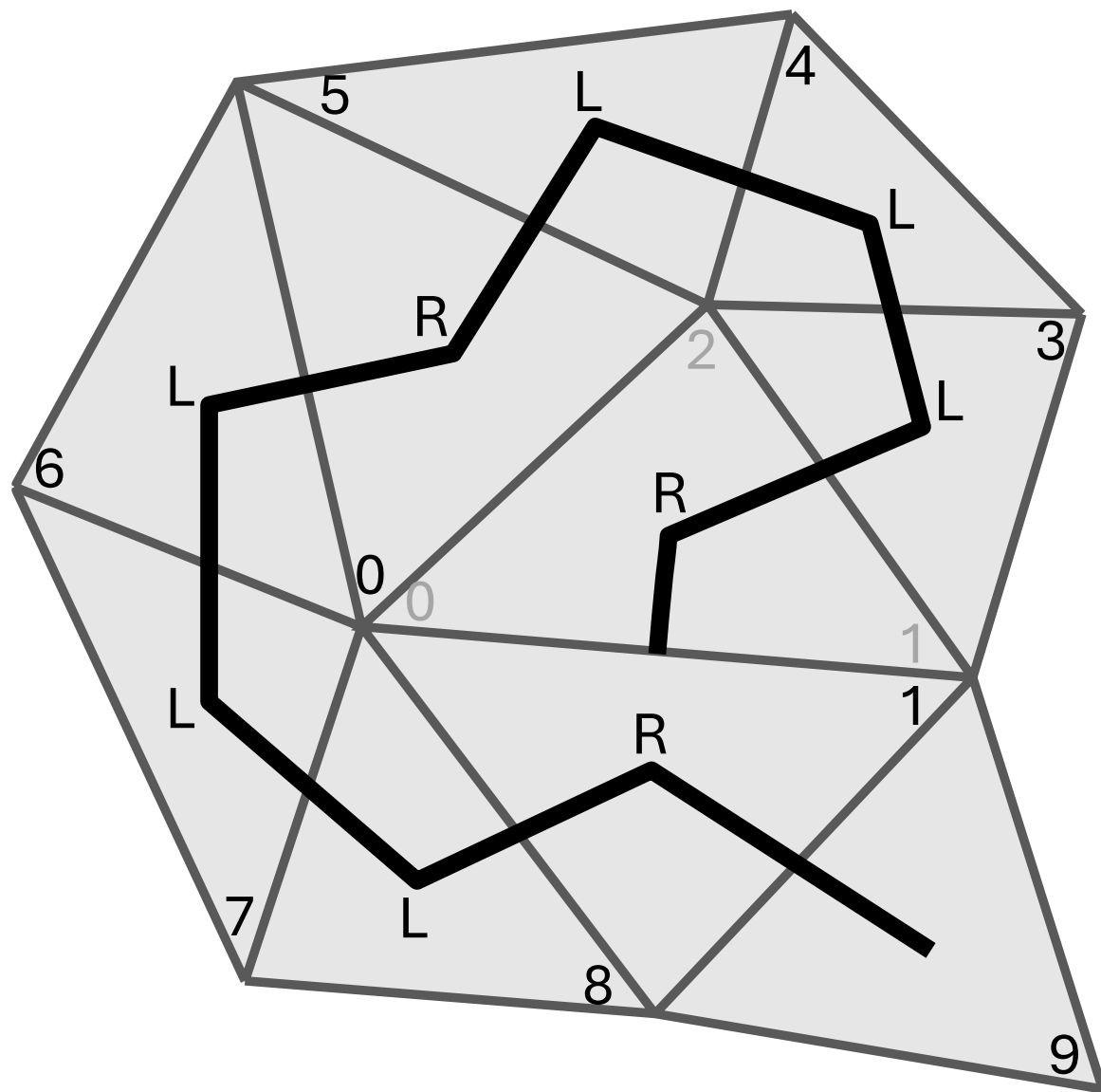


Generalized Triangle Strip (GTS)

0	1	2	3	4	5	0	6	7	8	1	9
			R	L	L	L	R	L	L	L	R

```
int firstbithigh(in int value);
```

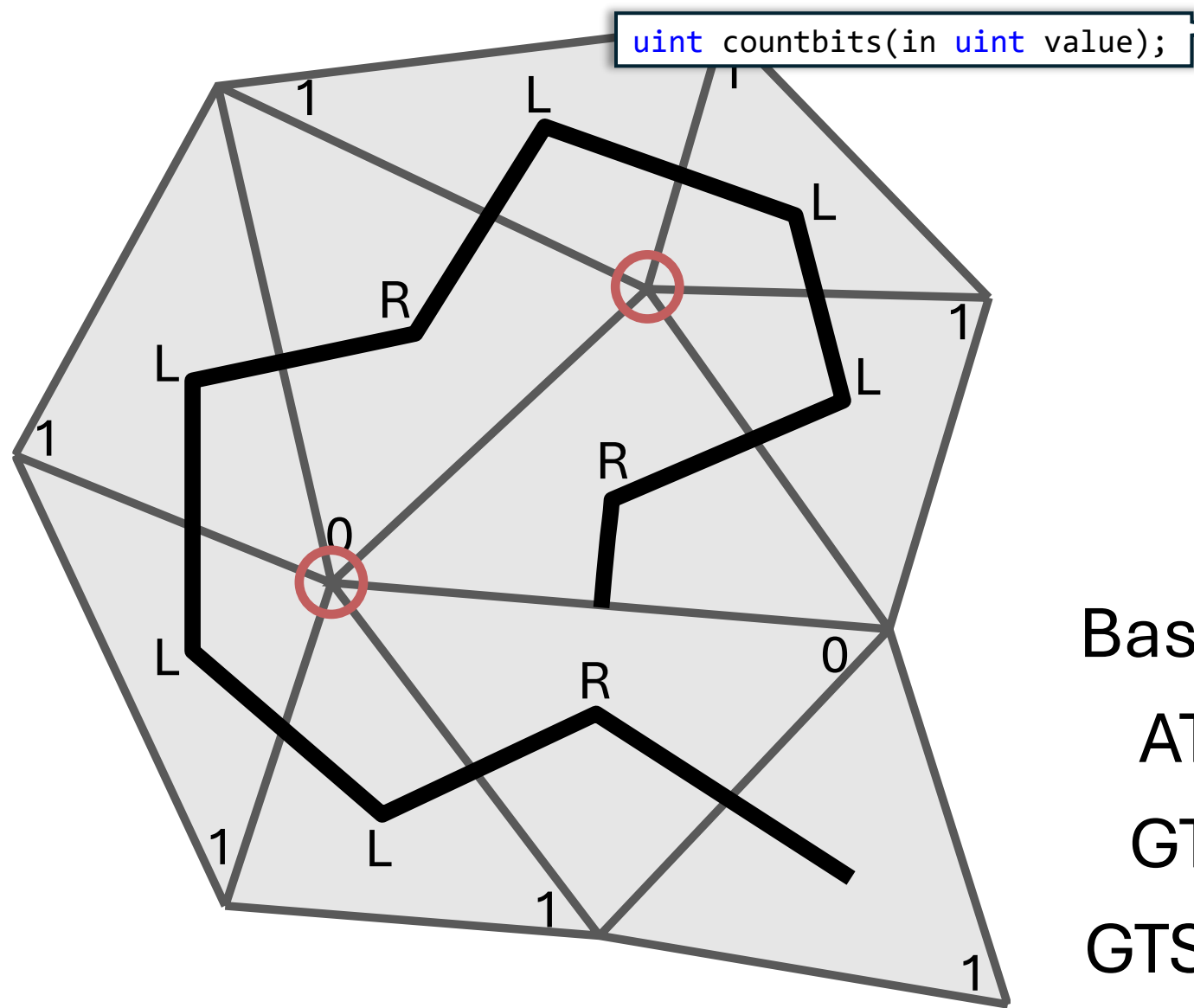
Basic	$3 \times 10 \times 8$	= 240bit
ATS	14×8	= 112bit
GTS	9×9	= 81bit



Generalized Triangle Strip (GTS)

0	1	2	3	4	5	0	6	7	8	1	9
			R	L	L	L	R	L	L	L	R

Basic	$3 \times 10 \times 8$	= 240bit
ATS	14×8	= 112bit
GTS	9×9	= 81bit

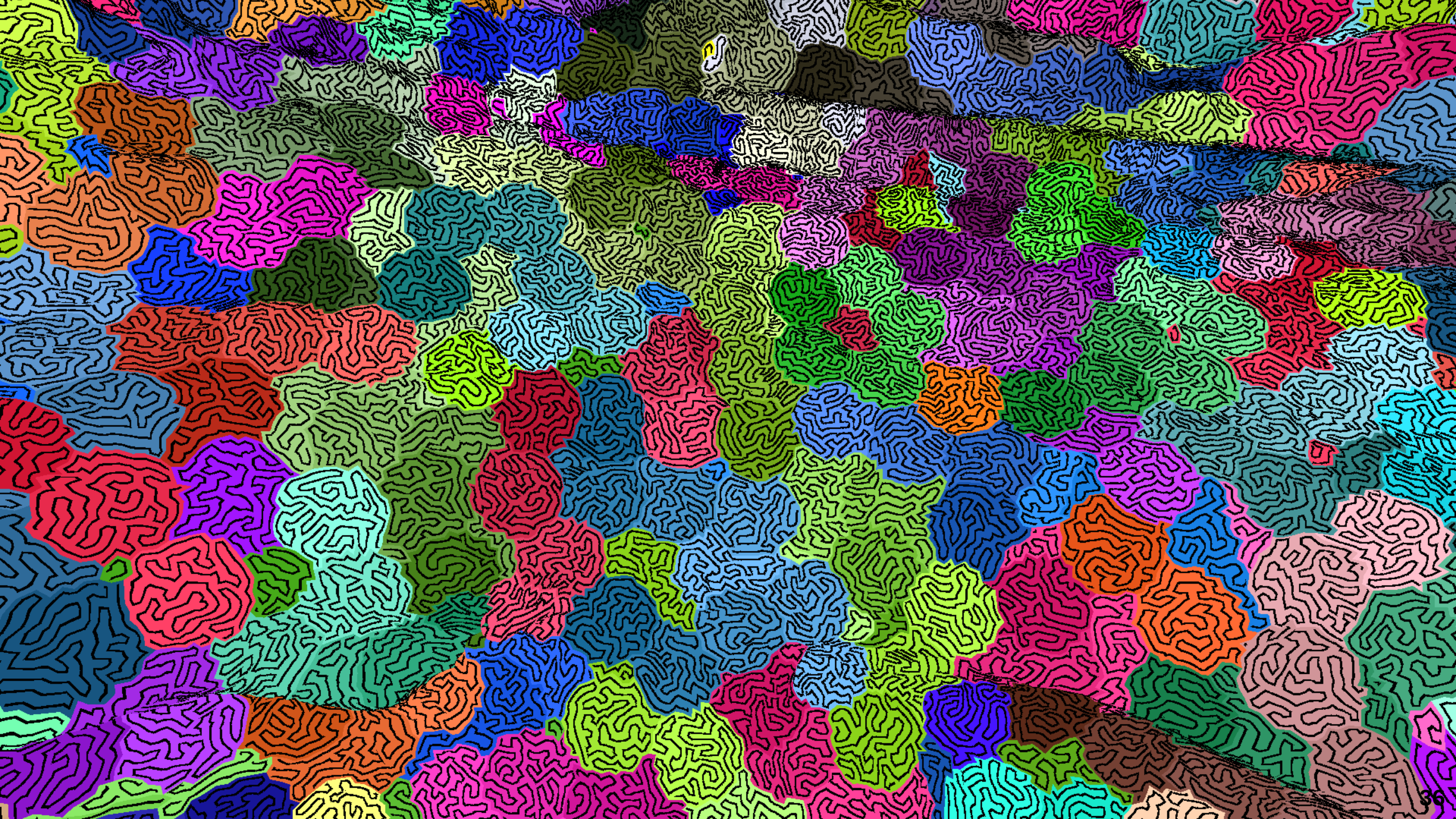


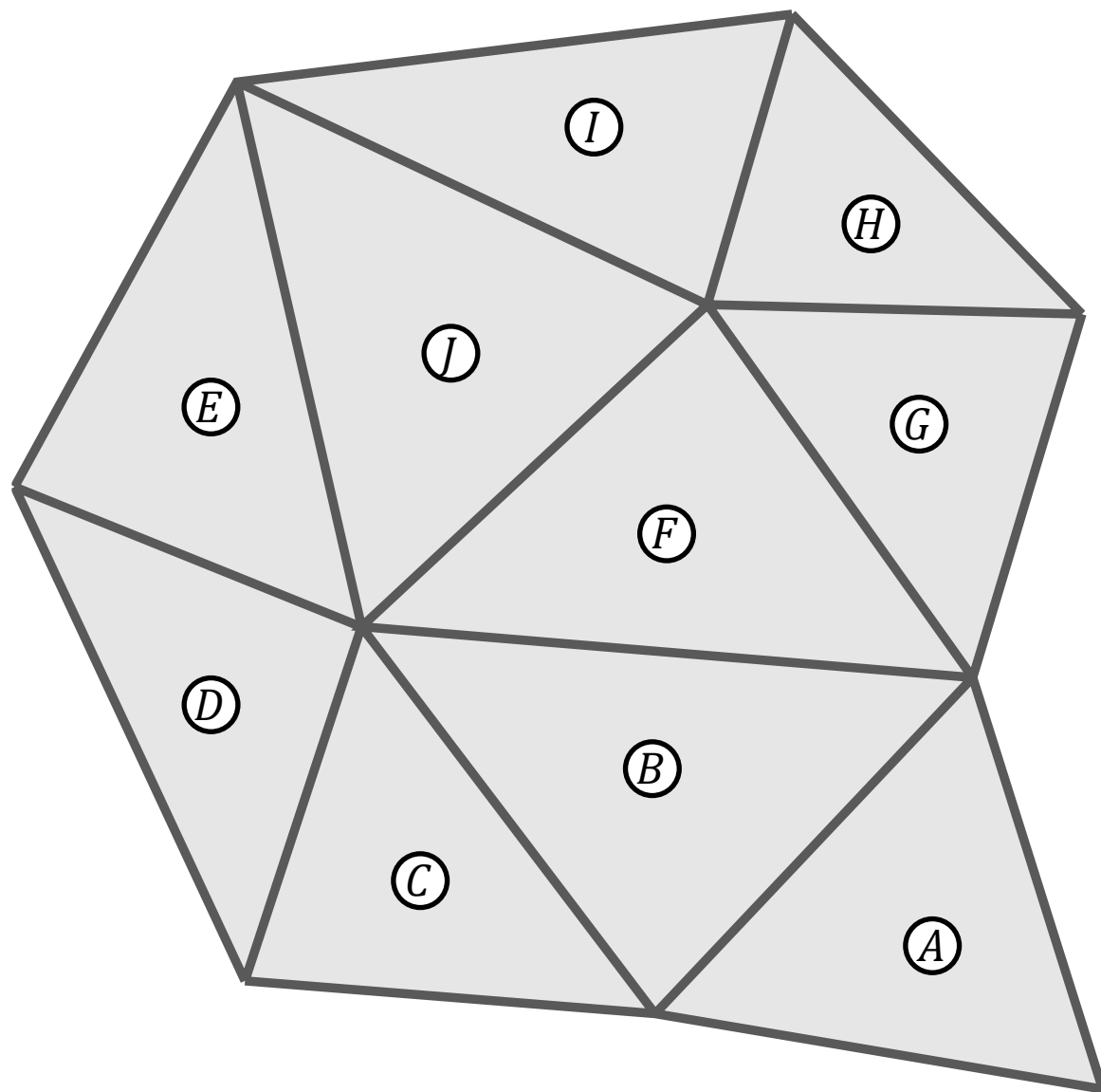
Generalized Triangle Strip with Reuse

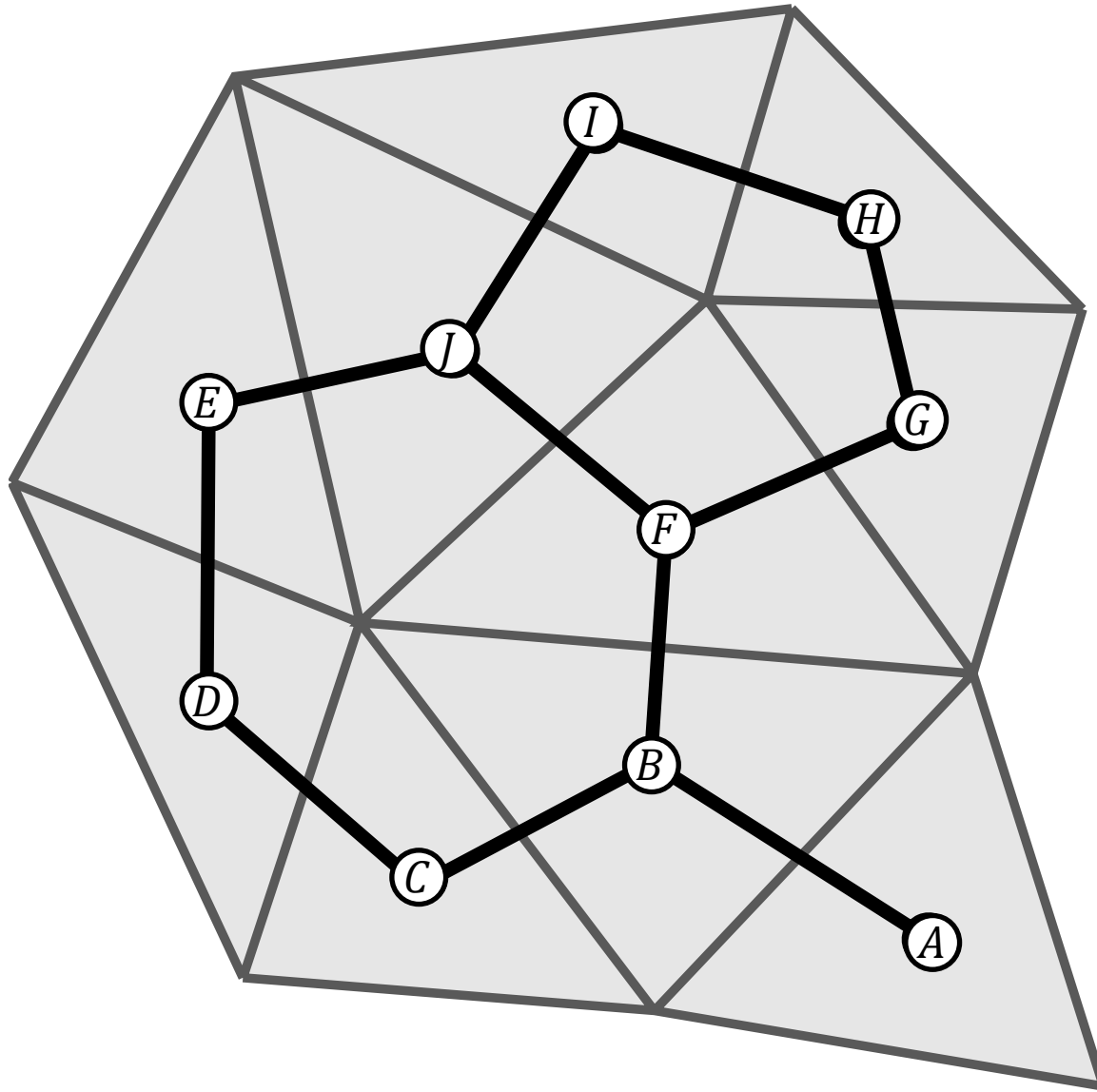
1	1	1	1	1	1	0	1	1	1	0	1
			R	L	L	L	R	L	L	L	R

Arrows from the 0s in the first row point to a box containing 0 and 1.

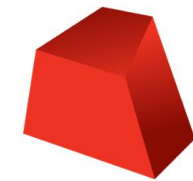
Basic	$3 \times 10 \times 8$	= 240bit
ATS	14×8	= 112bit
GTS	9×9	= 81bit
GTSR	$2 \times 9 + 2 \times 8$	= 34bit





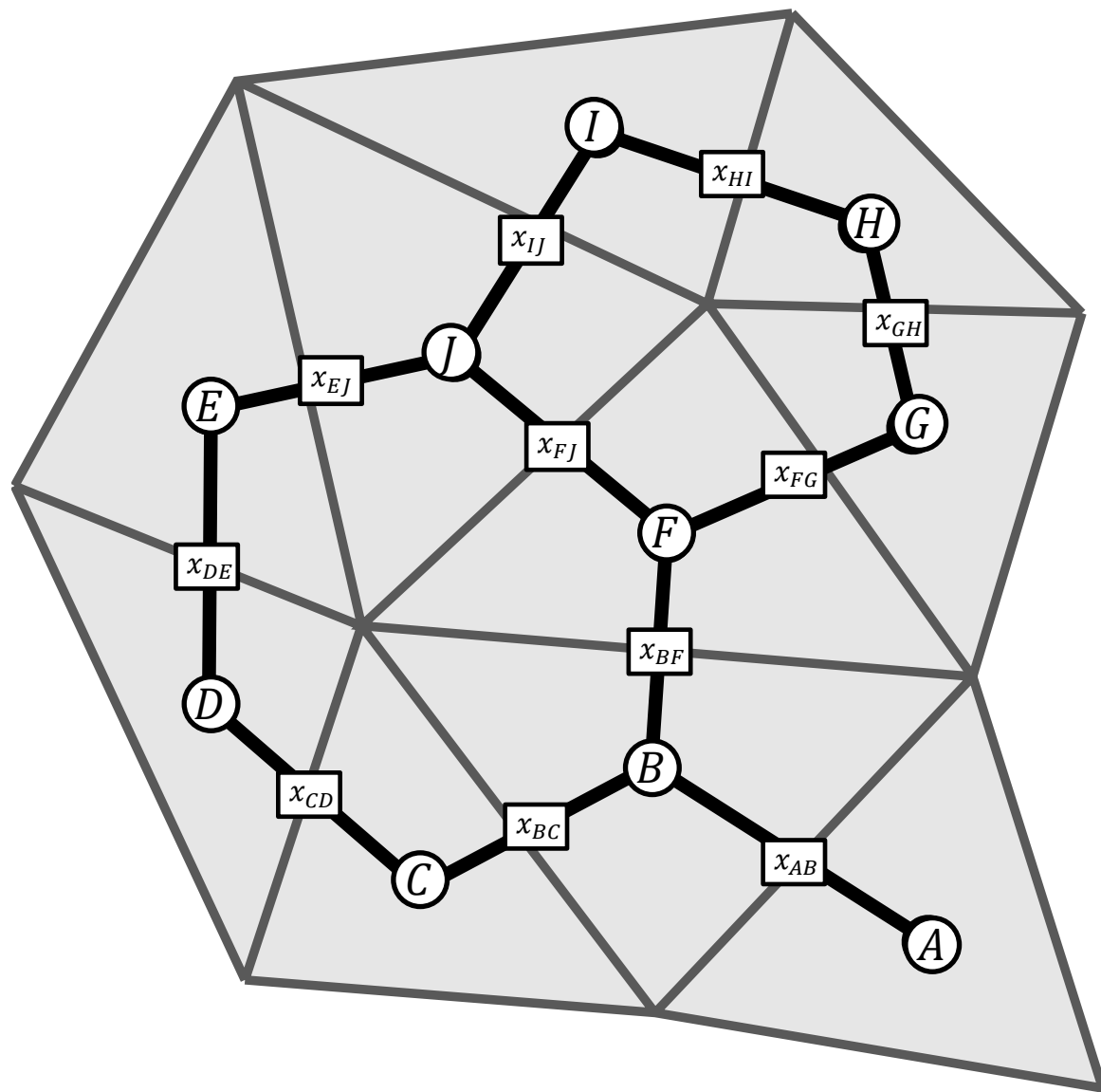


Find a vector $\mathbf{x}$
 that maximizes $\mathbf{c}^T \mathbf{x}$
 subject to $A\mathbf{x} \leq \mathbf{b}$
 and $\mathbf{x} \geq \mathbf{0}$.

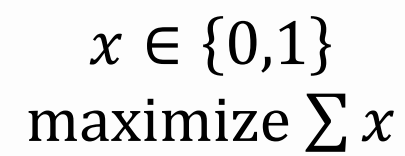


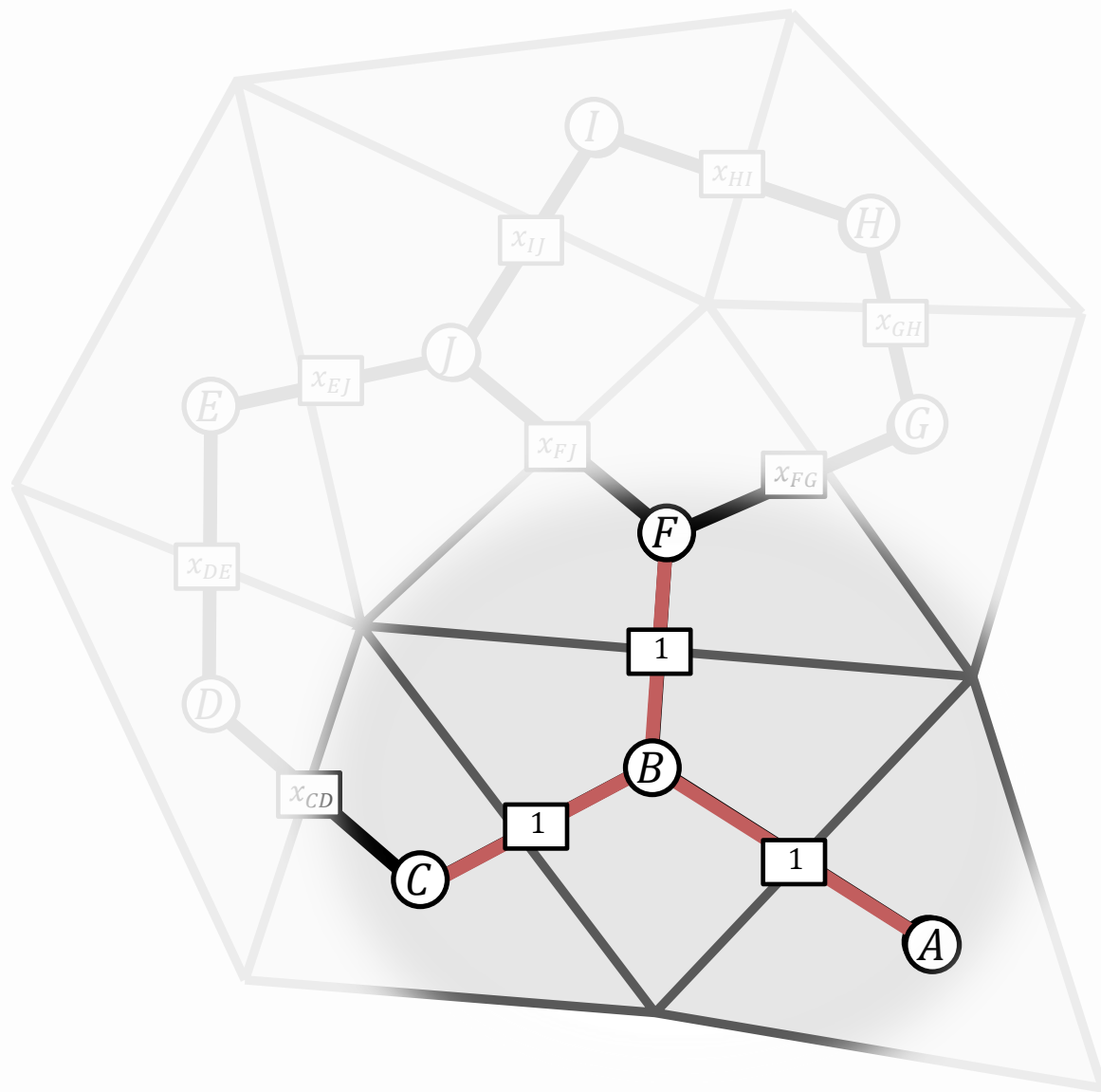
GUROBI
 OPTIMIZATION





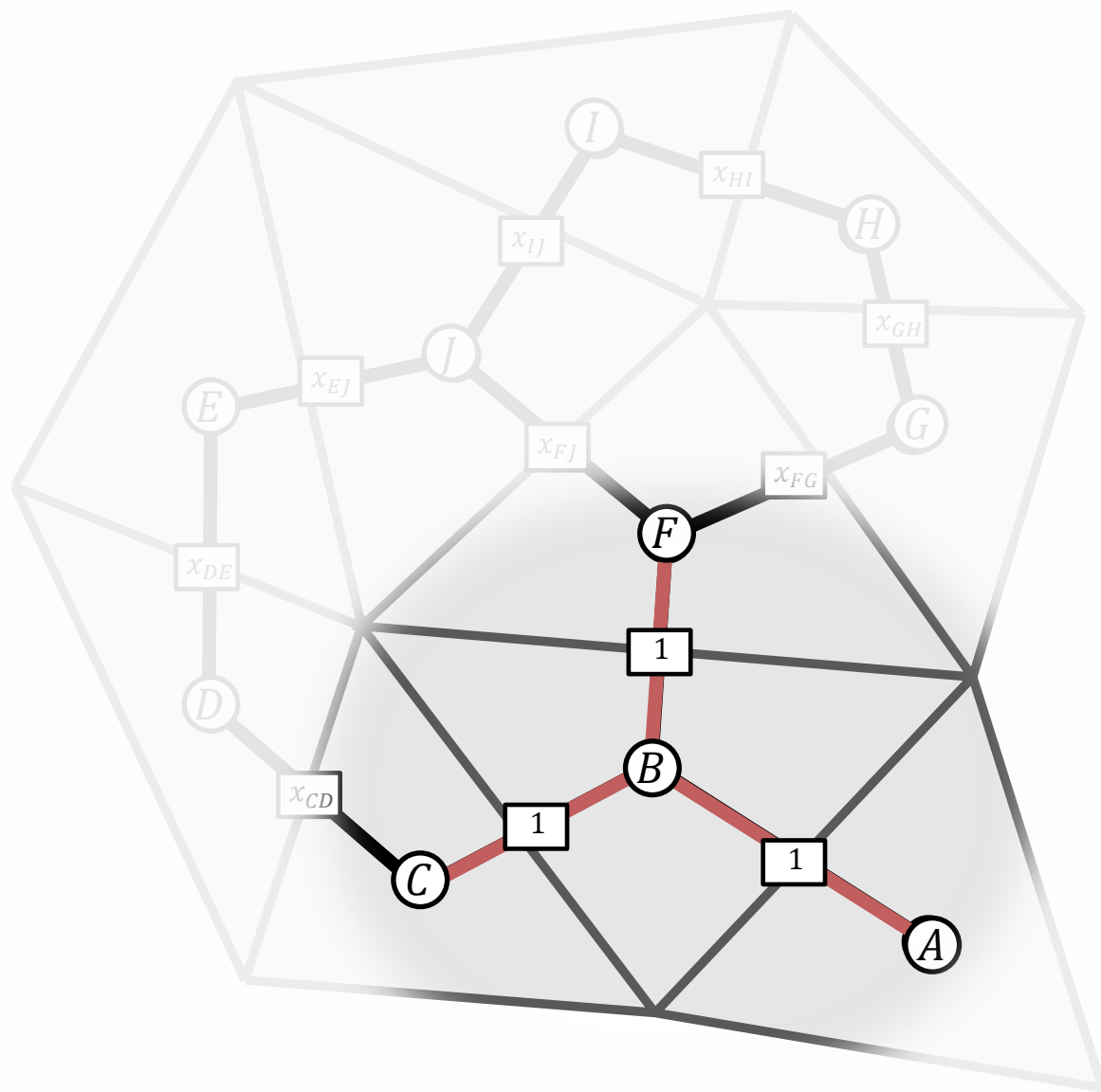
$x \in \{0,1\}$
 maximize $\sum x$





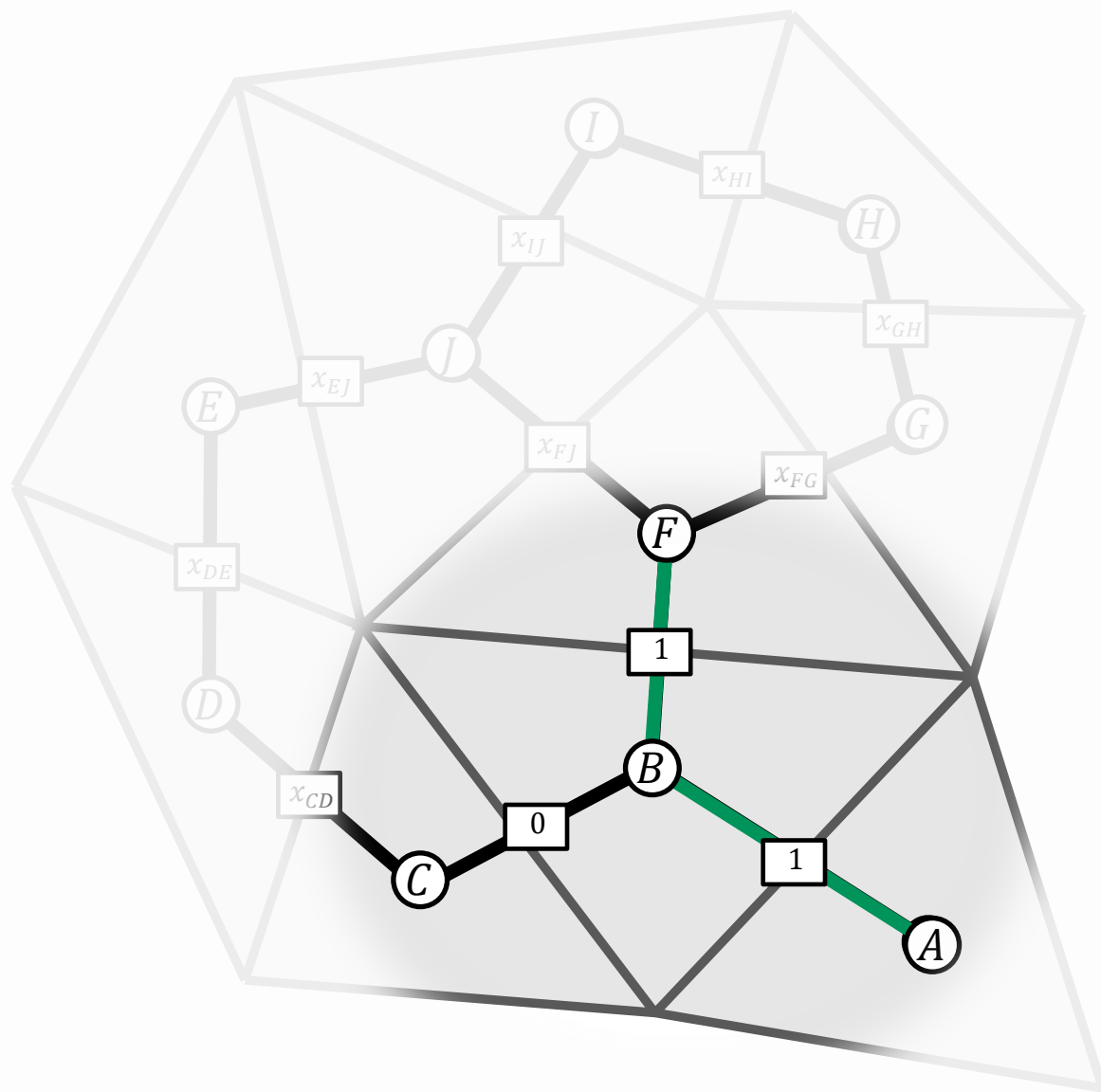
$x \in \{0,1\}$
 maximize $\sum x$

$\Rightarrow$ prevent forks



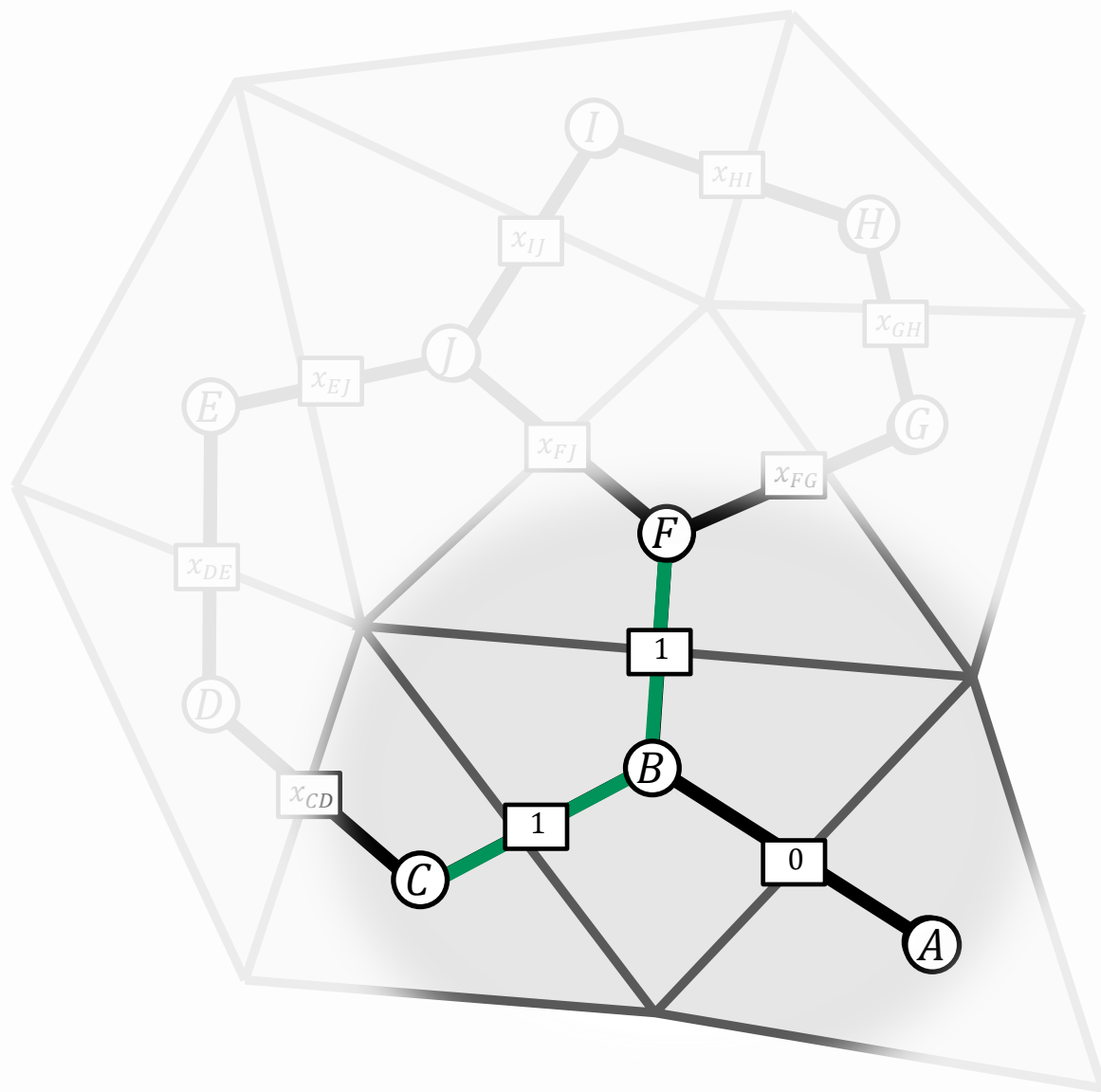
$x \in \{0,1\}$
 maximize $\sum x$

prevent forks:
 $x_{AB} + x_{BC} + x_{BF} \leq 2$



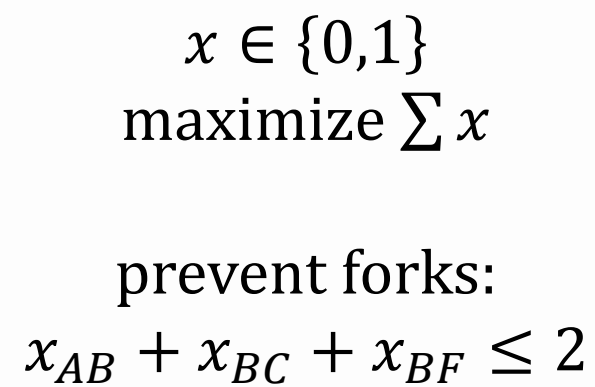
$x \in \{0,1\}$
 maximize $\sum x$

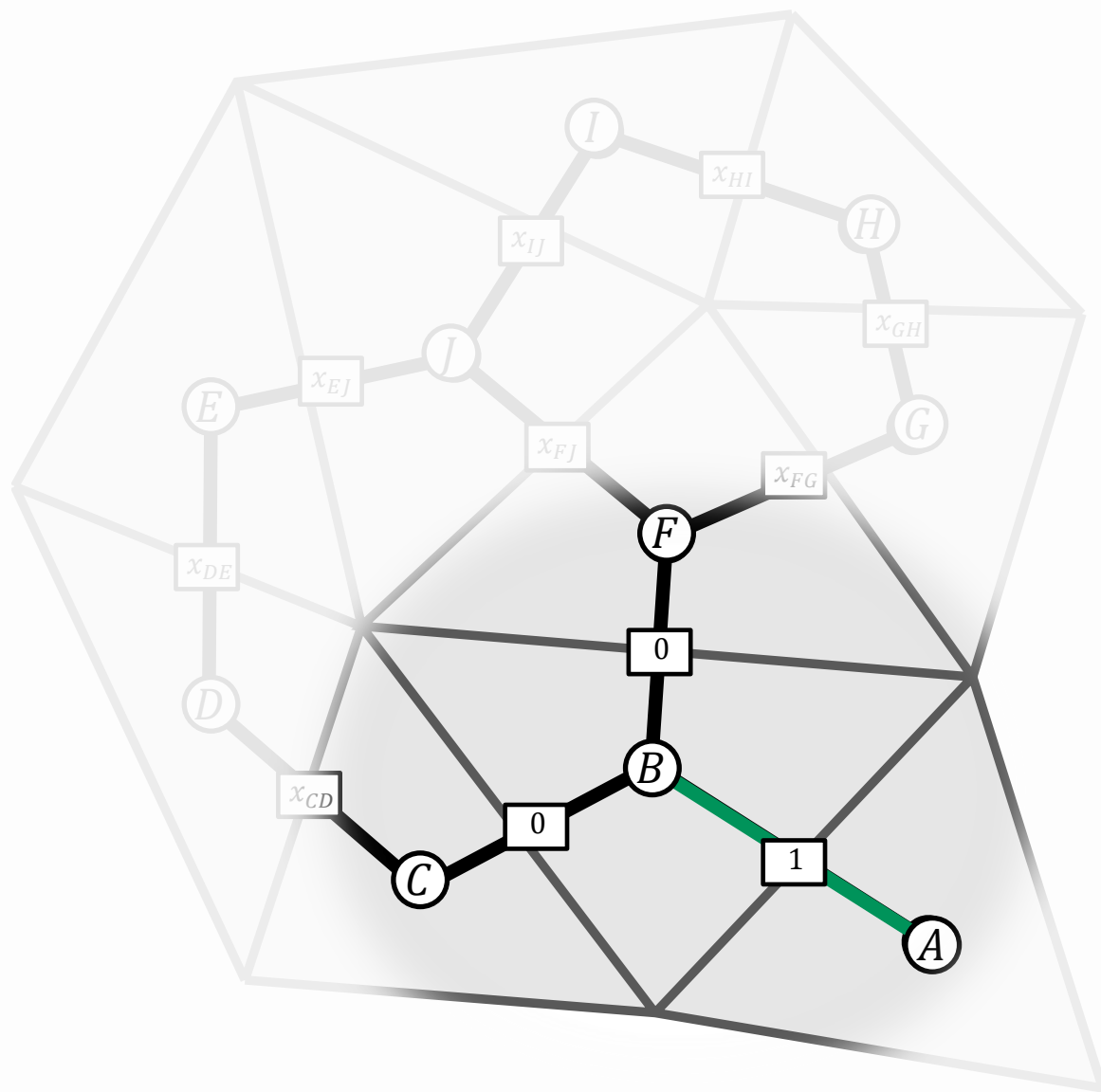
prevent forks:
 $x_{AB} + x_{BC} + x_{BF} \leq 2$



$x \in \{0,1\}$
 maximize $\sum x$

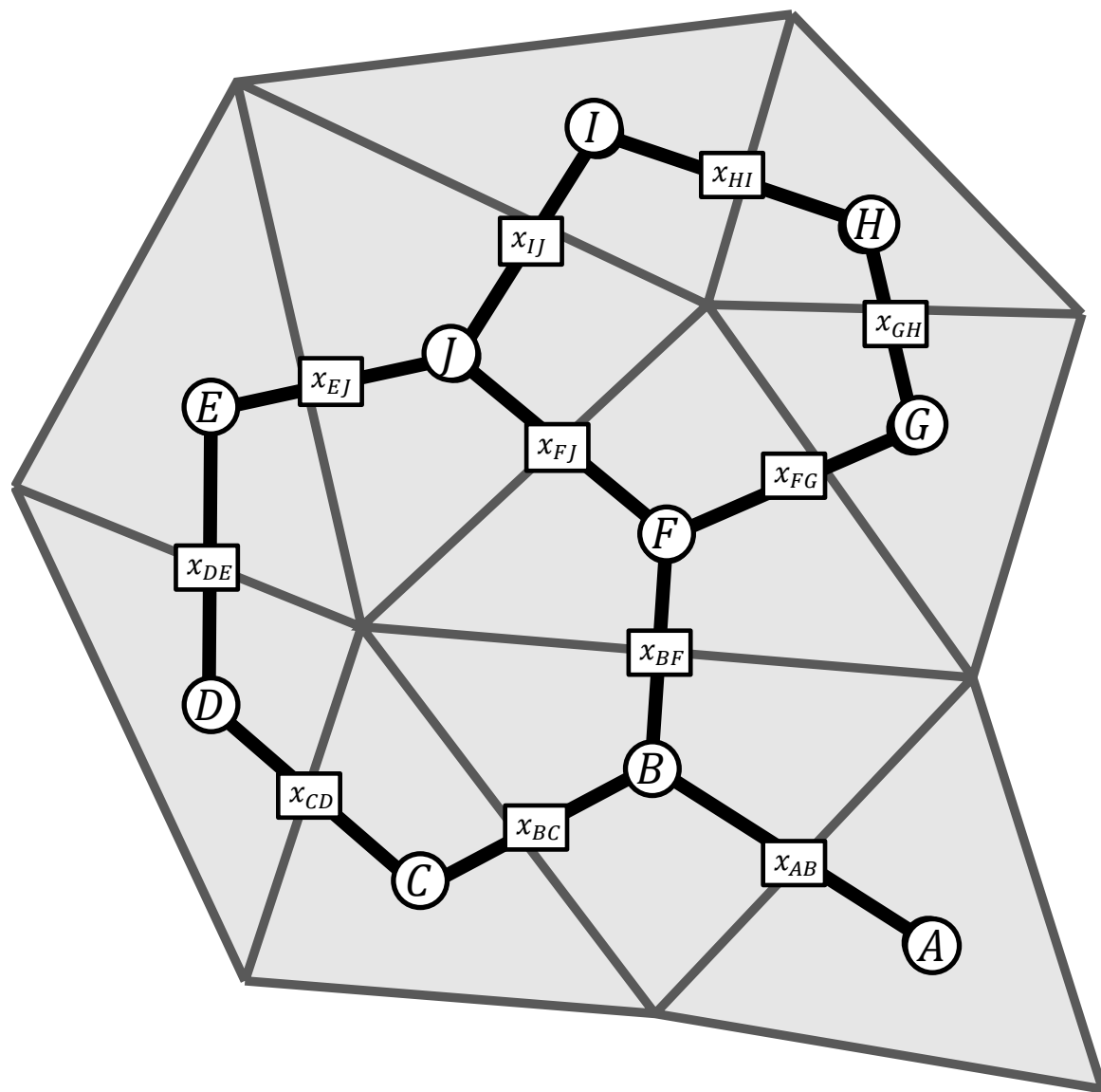
prevent forks:
 $x_{AB} + x_{BC} + x_{BF} \leq 2$





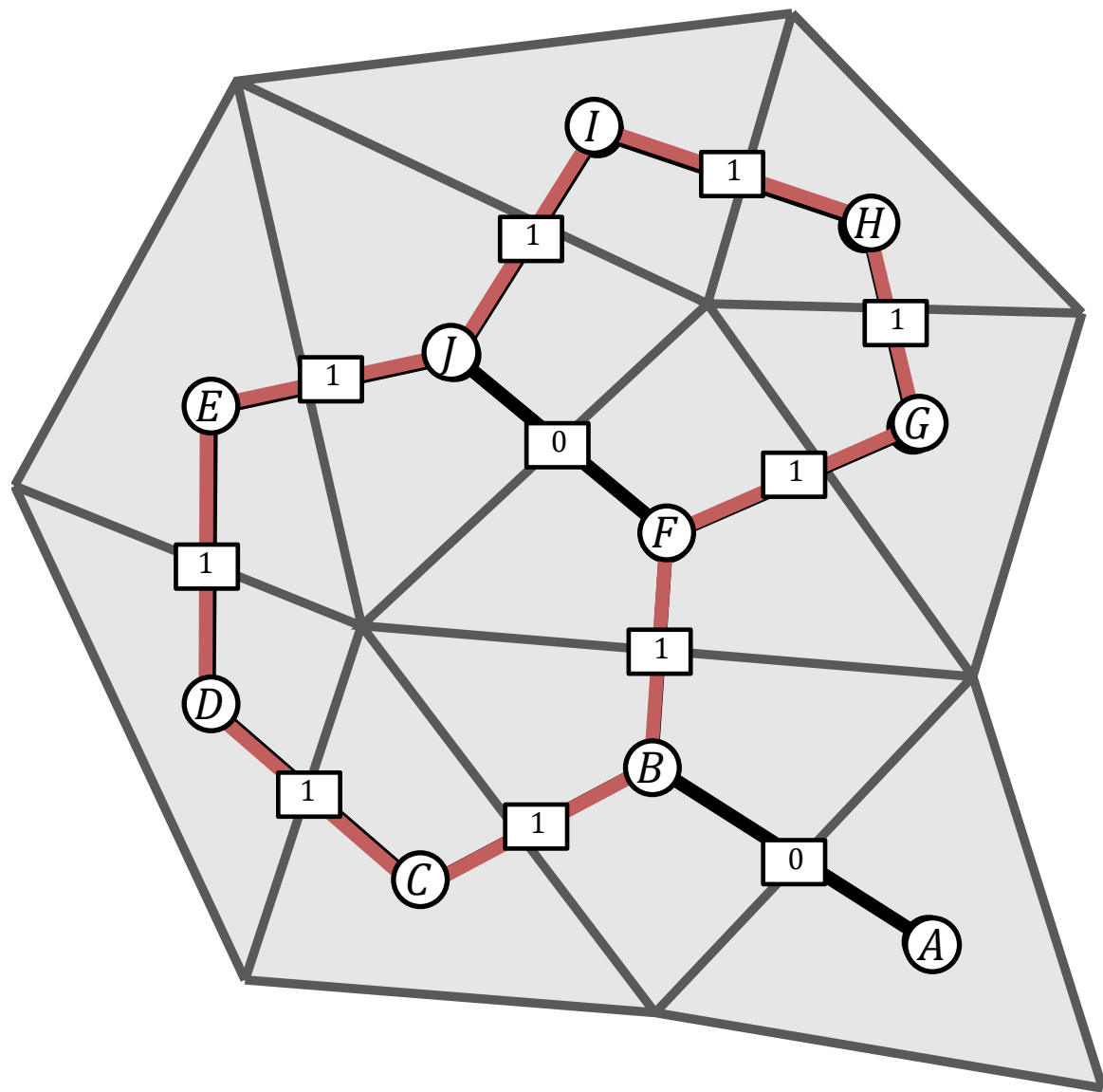
$x \in \{0,1\}$
 maximize $\sum x$

prevent forks:
 $x_{AB} + x_{BC} + x_{BF} \leq 2$

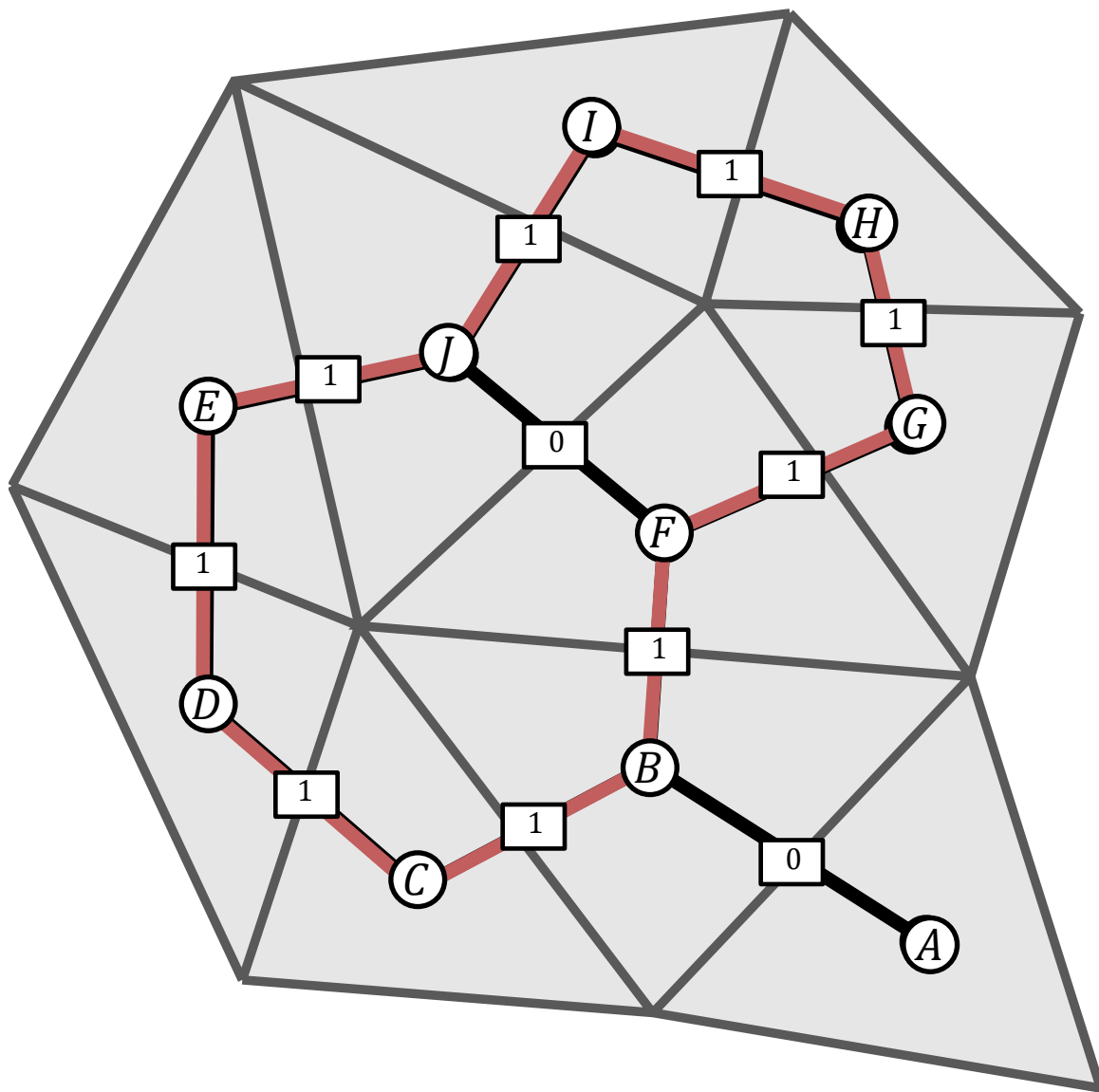


$x \in \{0,1\}$
 maximize $\sum x$

prevent forks:
 $x_{AB} + x_{BC} + x_{BF} \leq 2$



$$\sum x = 9$$

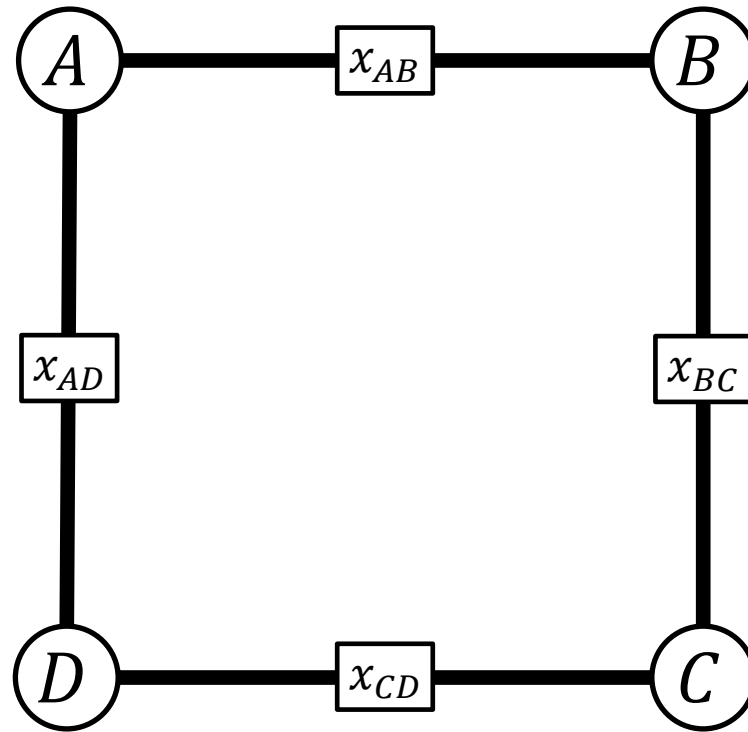


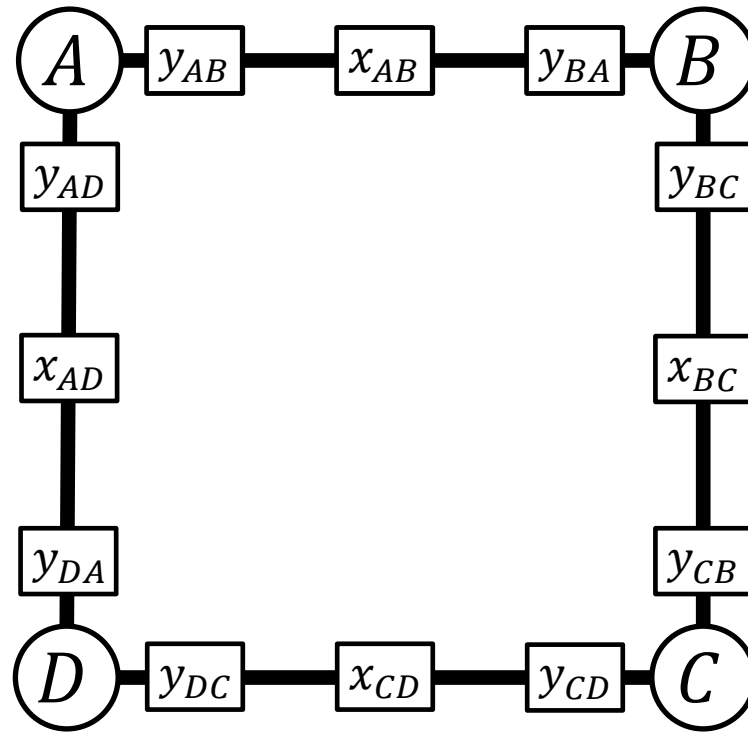
$$\sum x = 9$$

$\Rightarrow$ prevent cycles

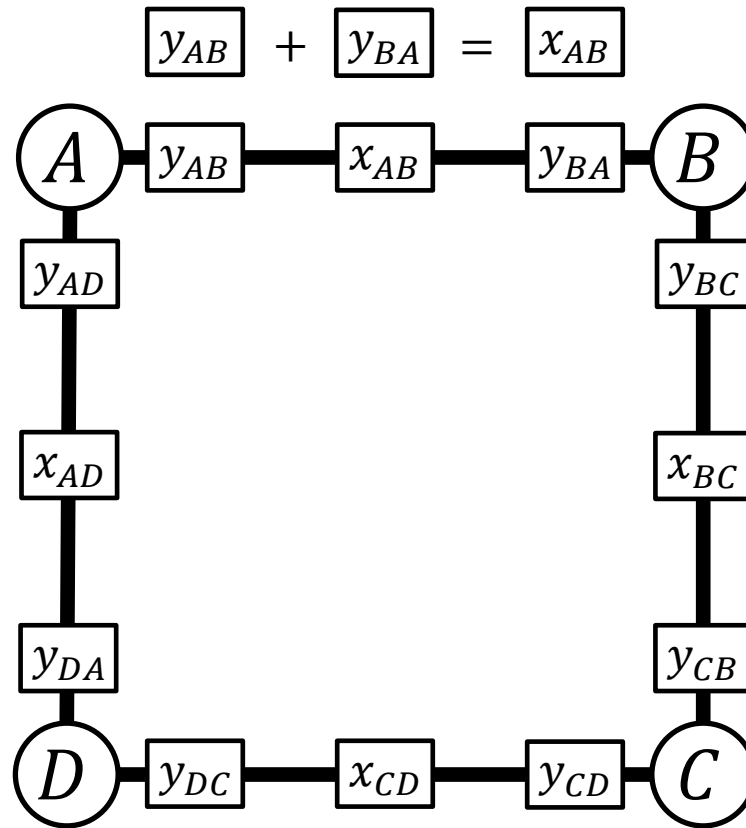
$O(n!)$ constraints ⚡

lazy constraints ⚡

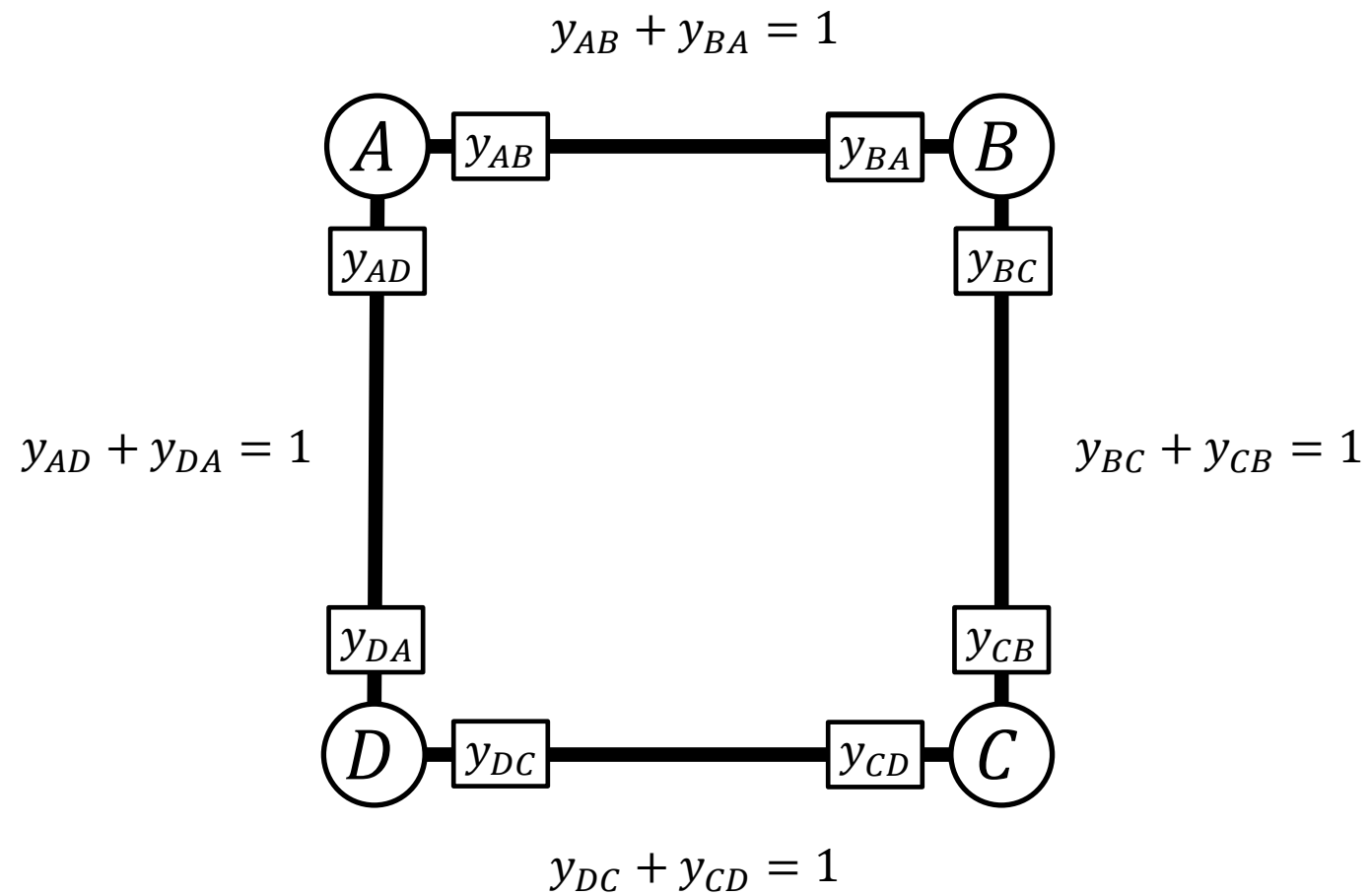




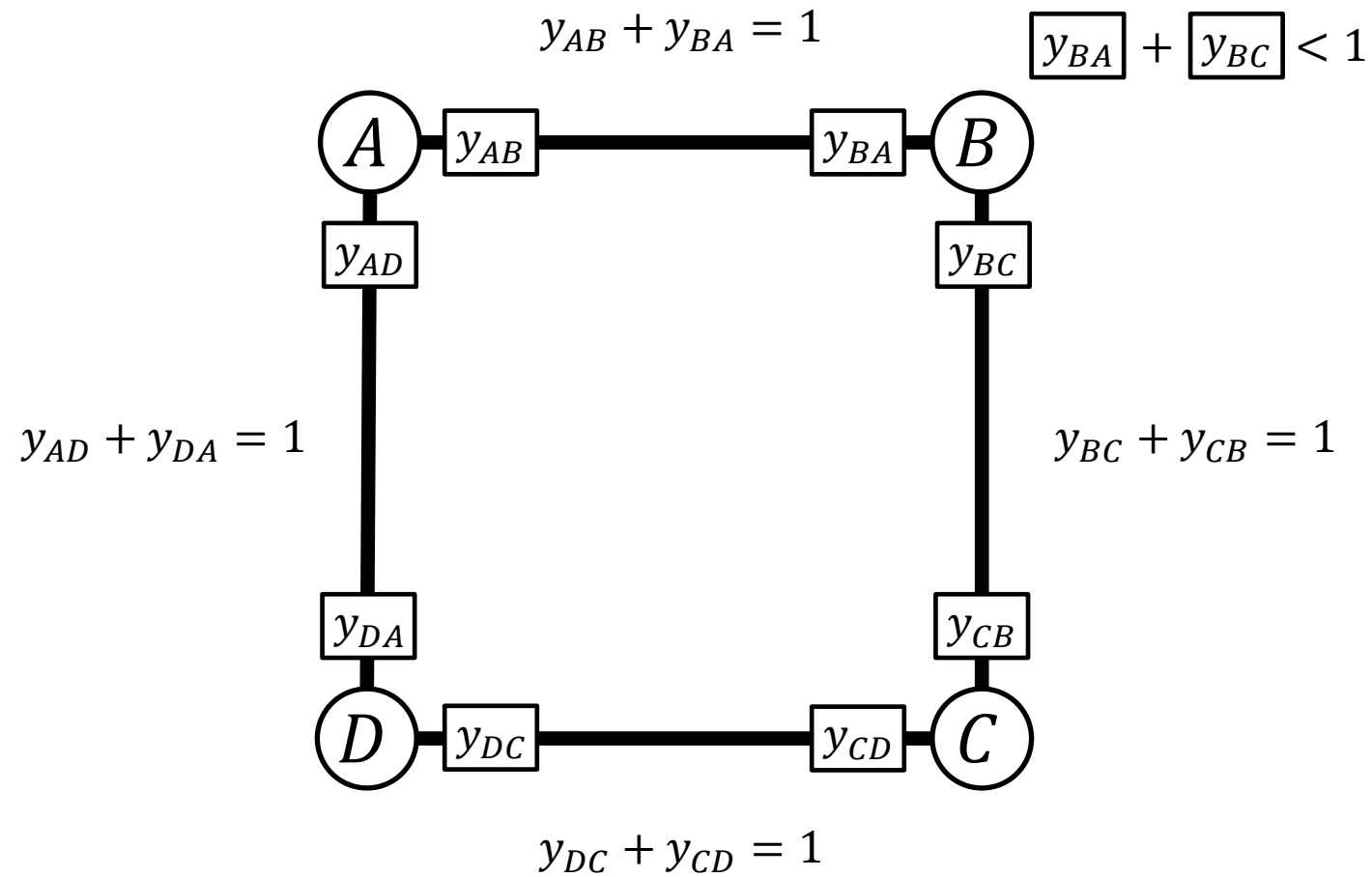
$$y \in \mathbb{R}_{\geq 0}$$



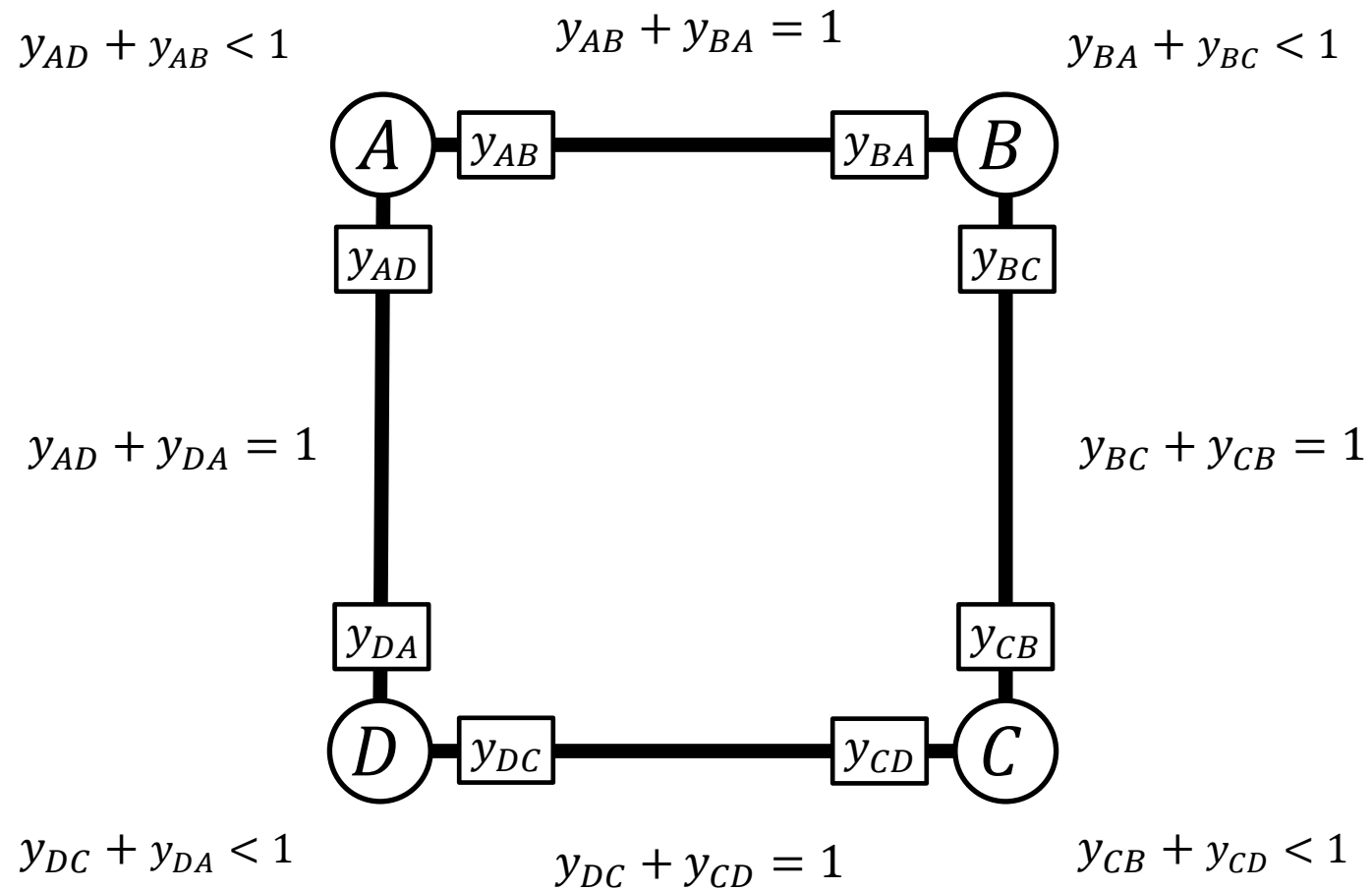
$$y \in \mathbb{R}_{\geq 0}$$



$$y \in \mathbb{R}_{\geq 0}$$

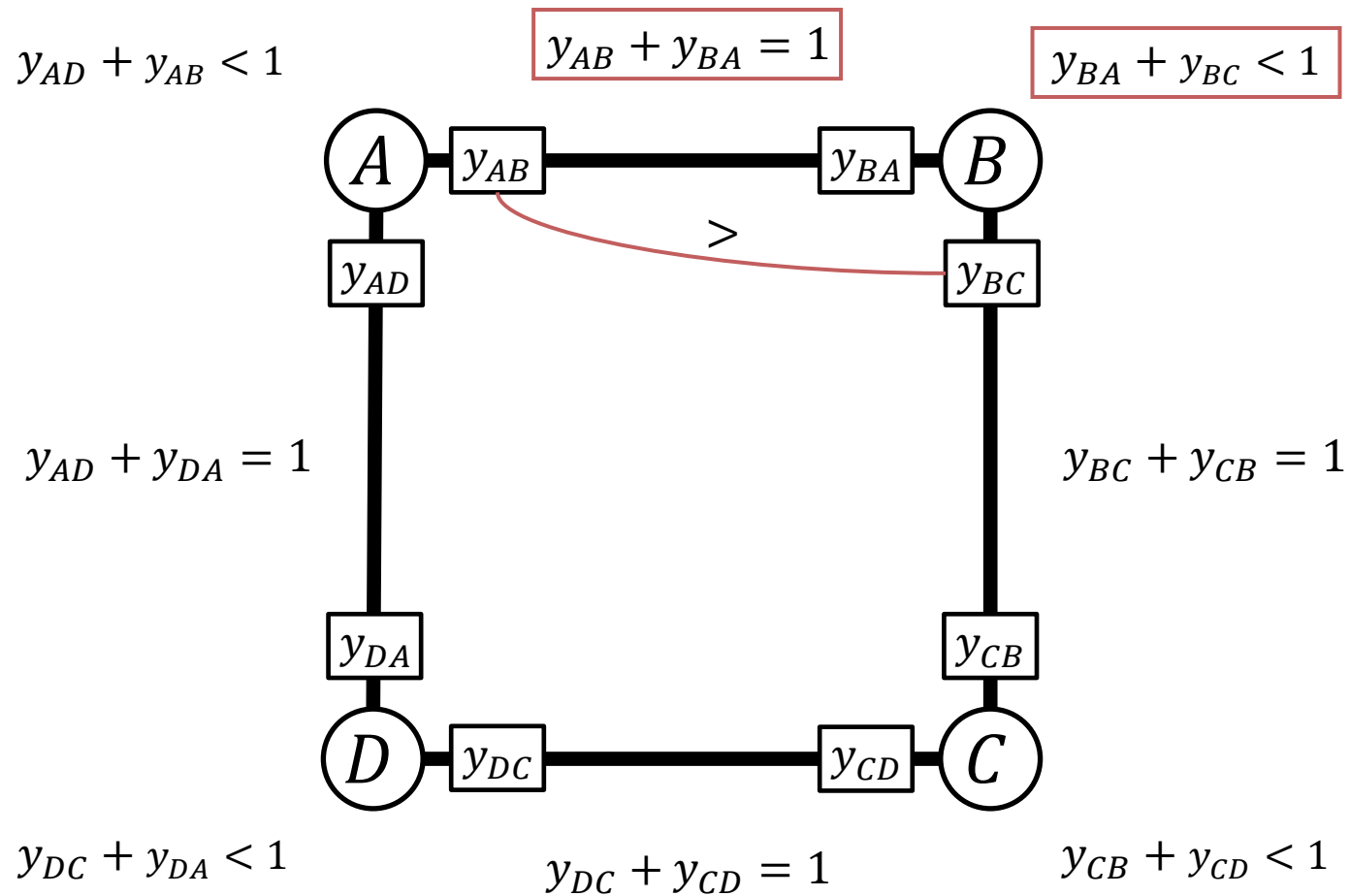


$$y \in \mathbb{R}_{\geq 0}$$

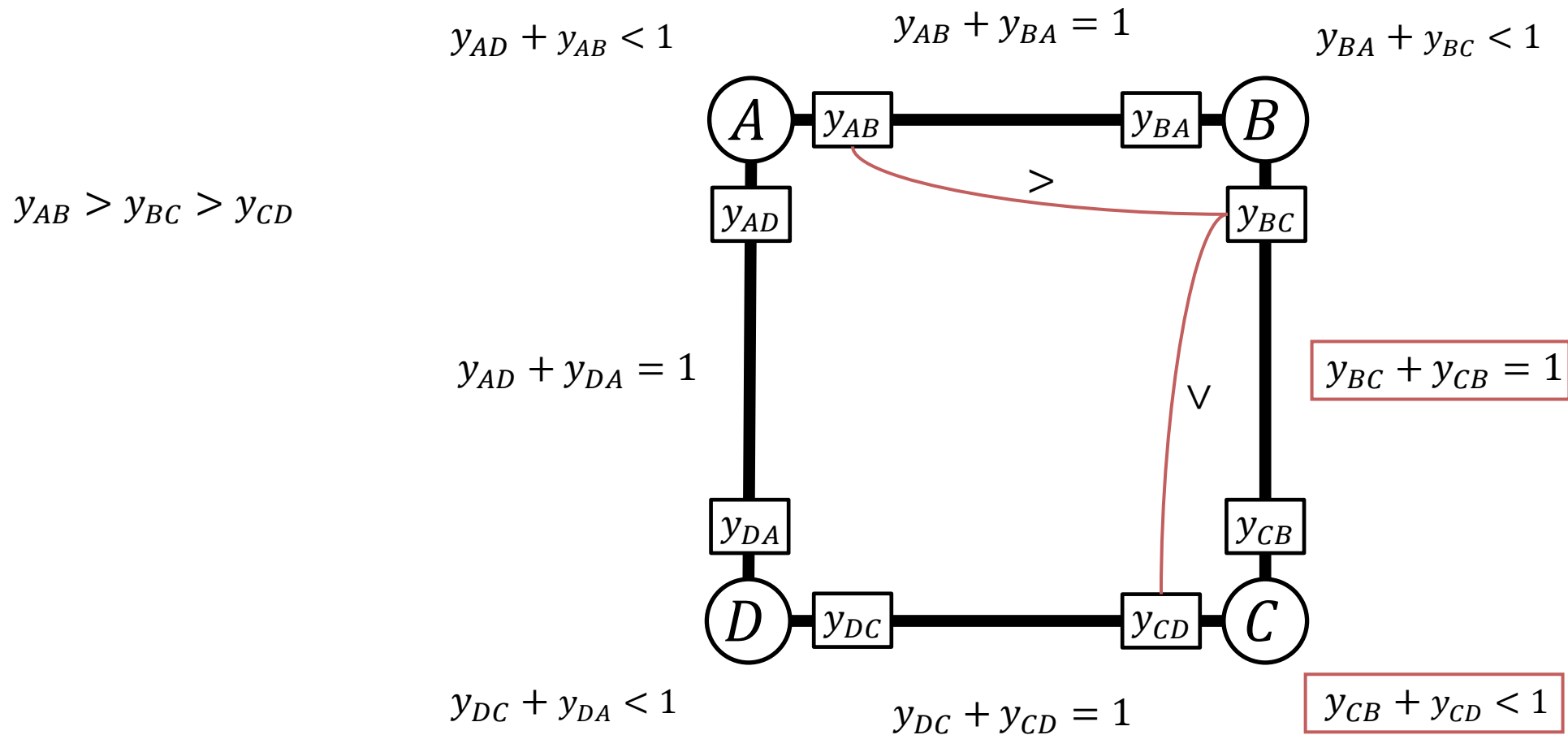


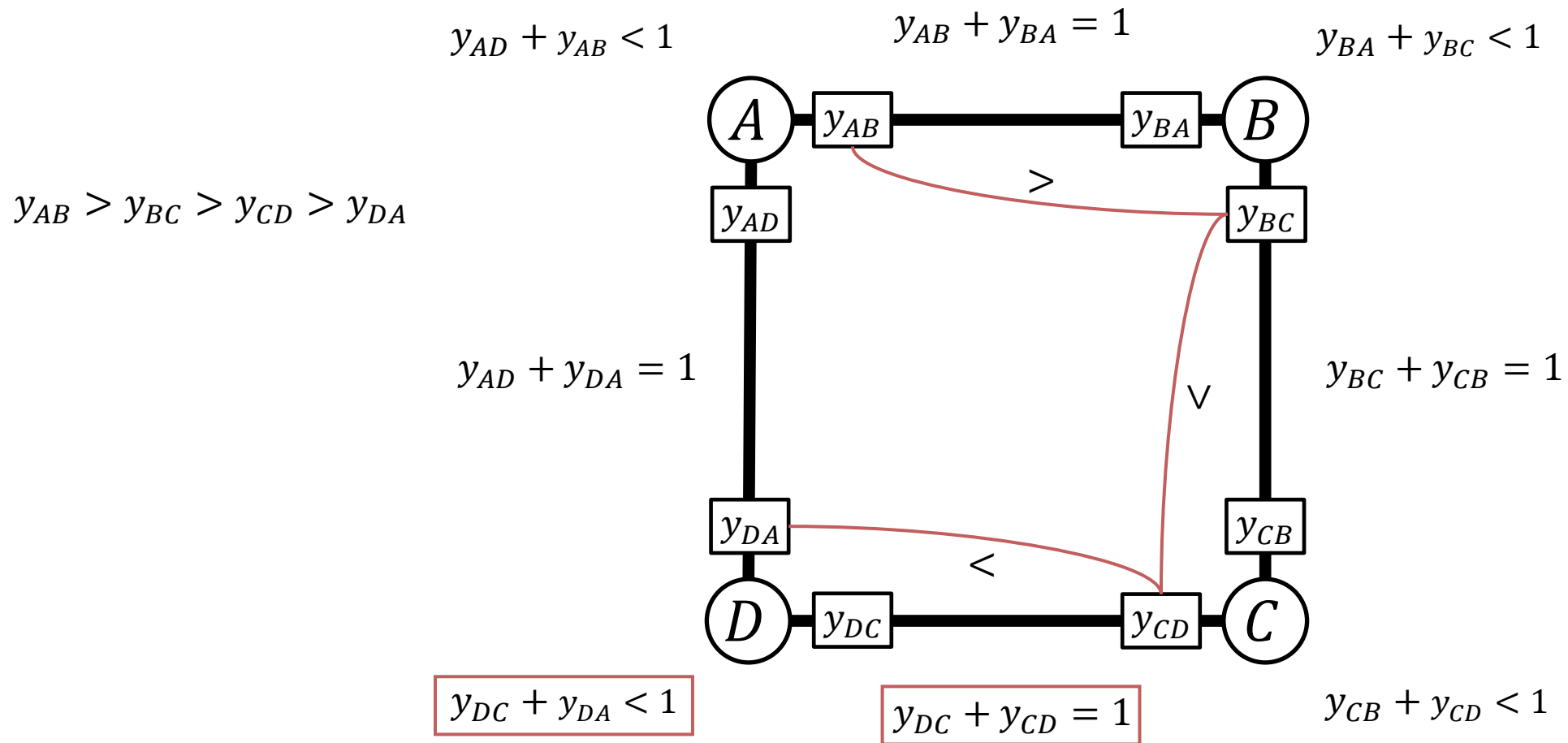
$$y \in \mathbb{R}_{\geq 0}$$

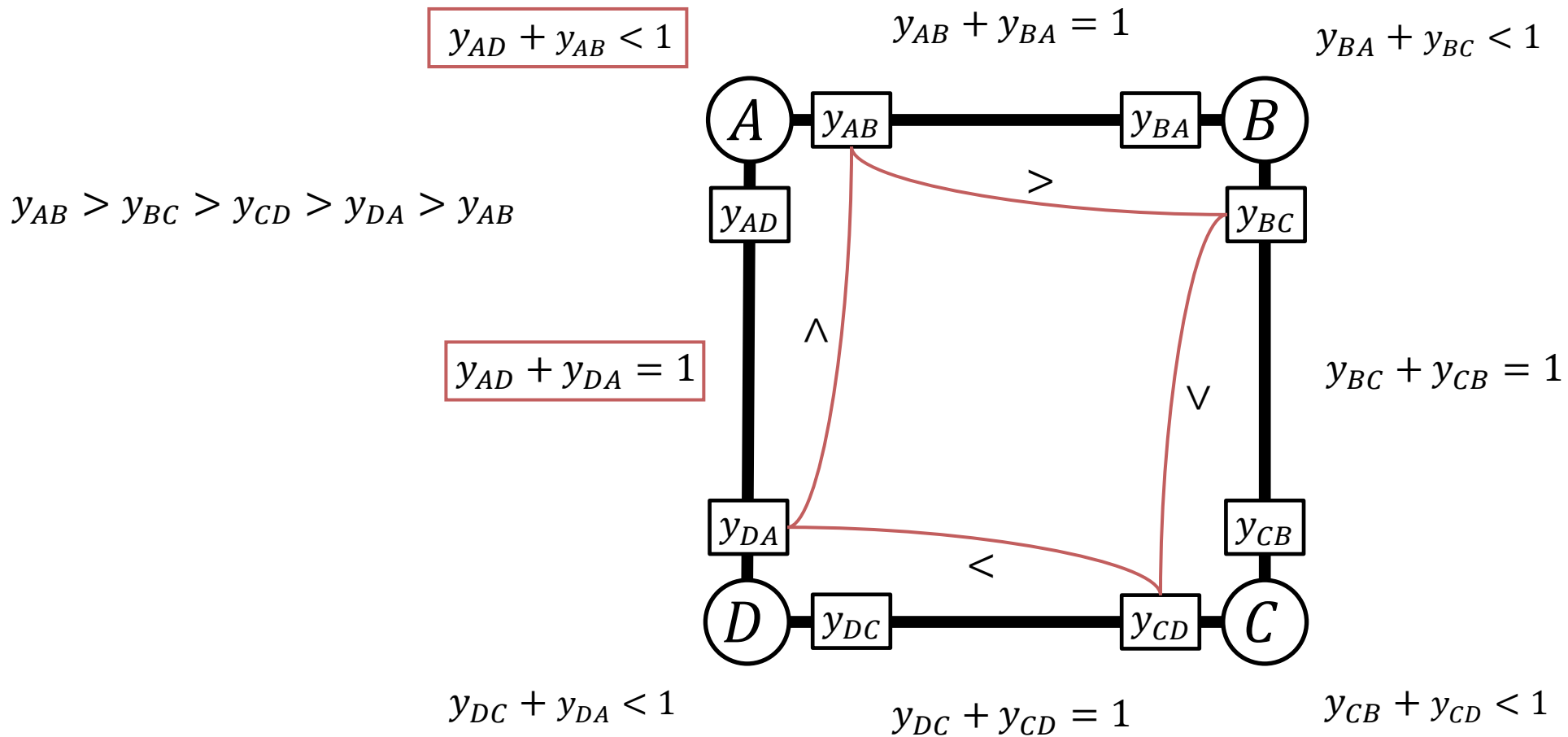
$$y_{AB} > y_{BC}$$



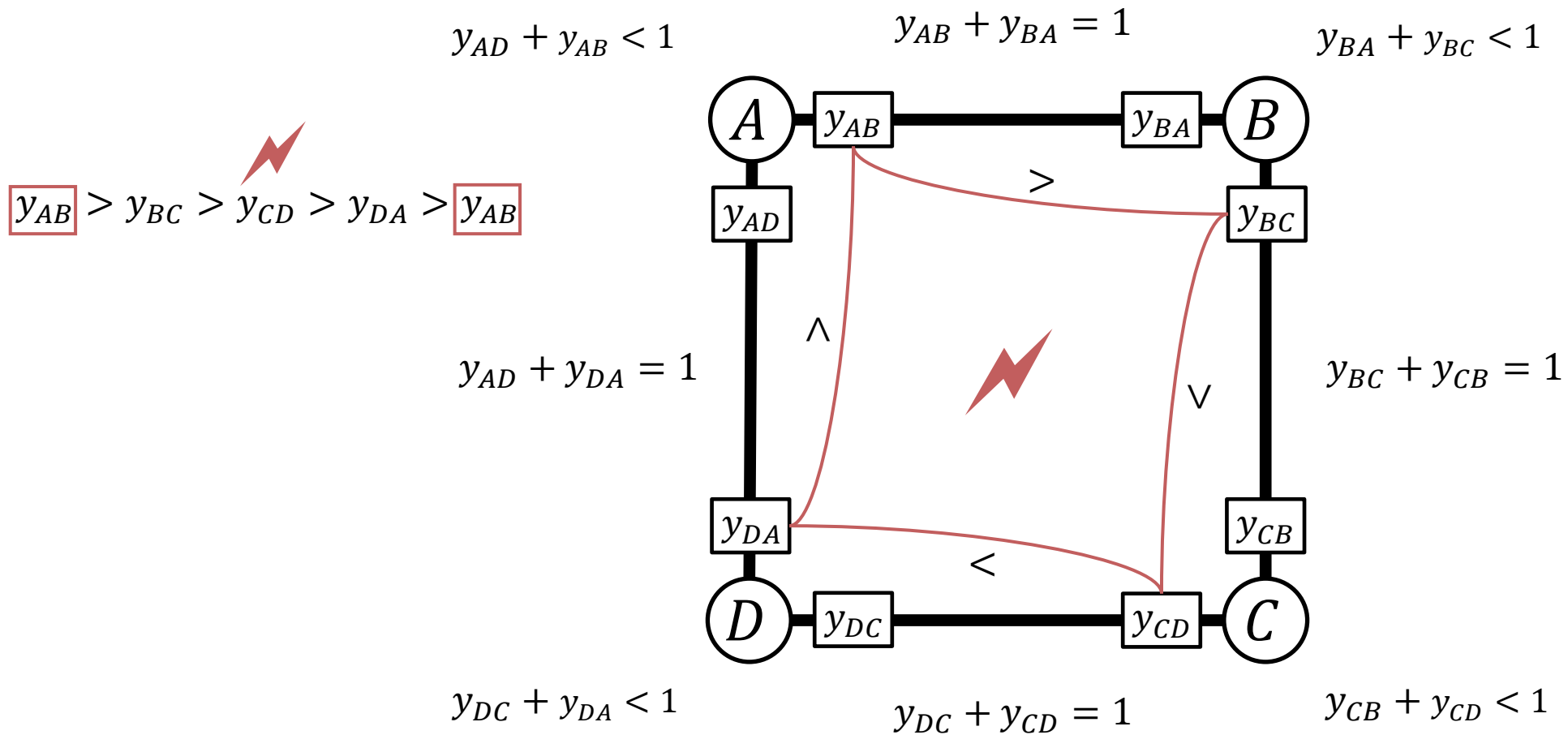
$$y \in \mathbb{R}_{\geq 0}$$

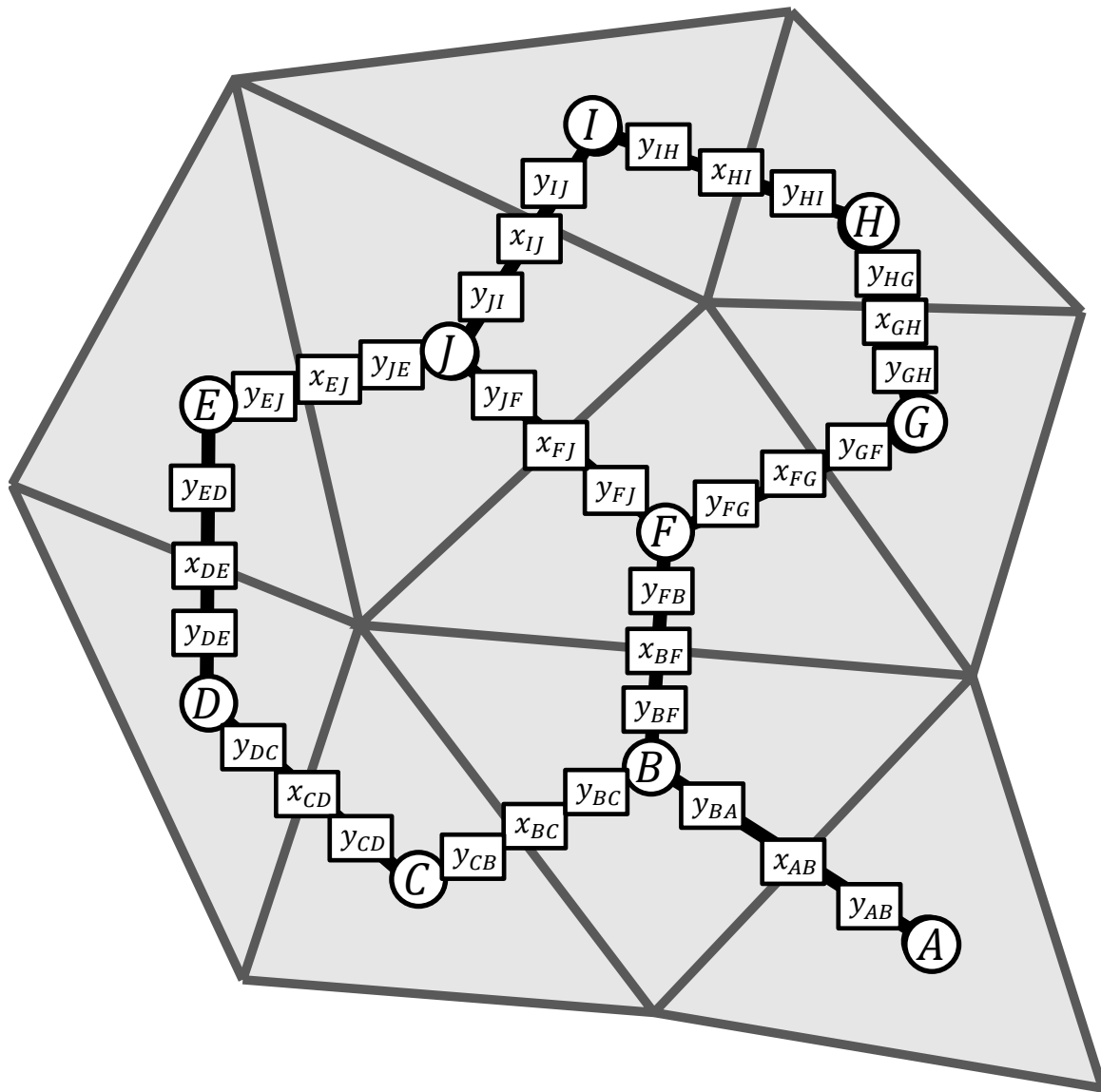






$$y \in \mathbb{R}_{\geq 0}$$





$$x \in \{0,1\}$$

$$\text{maximize } \sum x$$

prevent forks:

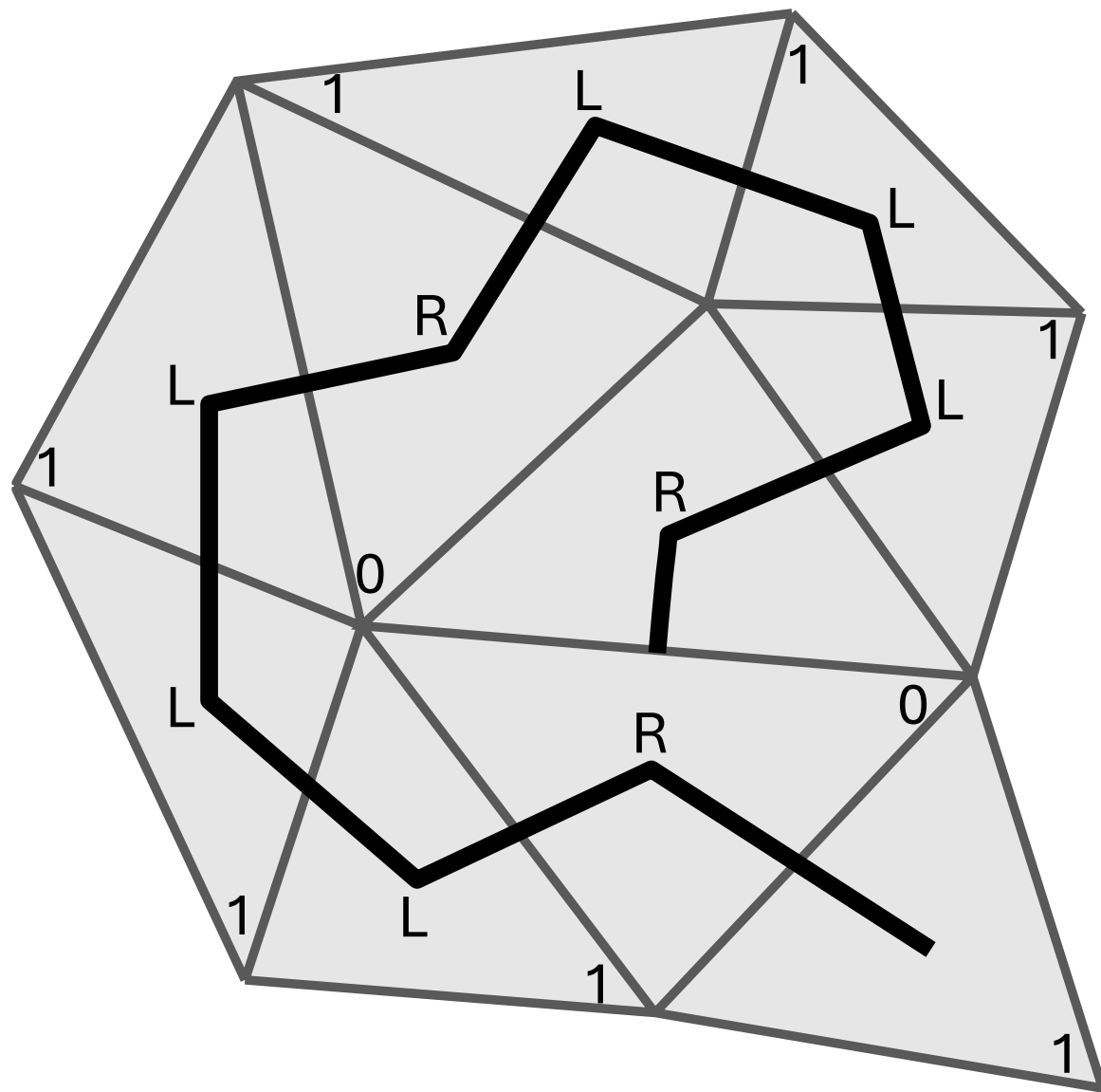
$$x_{AB} + x_{BC} + x_{BF} \leq 2$$

prevent cycles:

$$y \in \mathbb{R}_{\geq 0}$$

$$y_{AB} + y_{BA} = x_{AB}$$

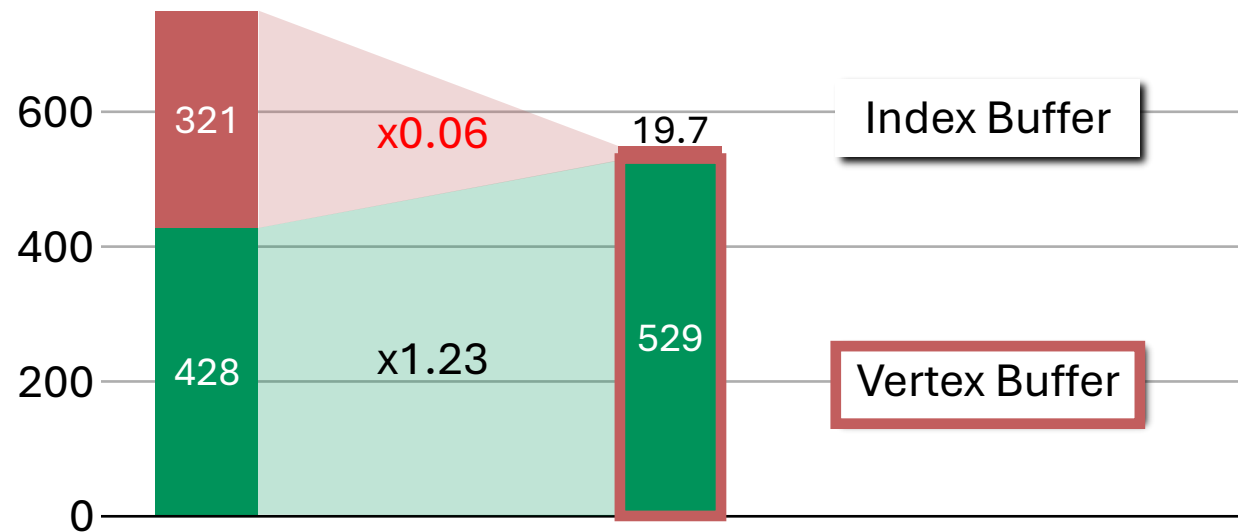
$$y_{BA} + y_{BC} + y_{BF} < 1$$



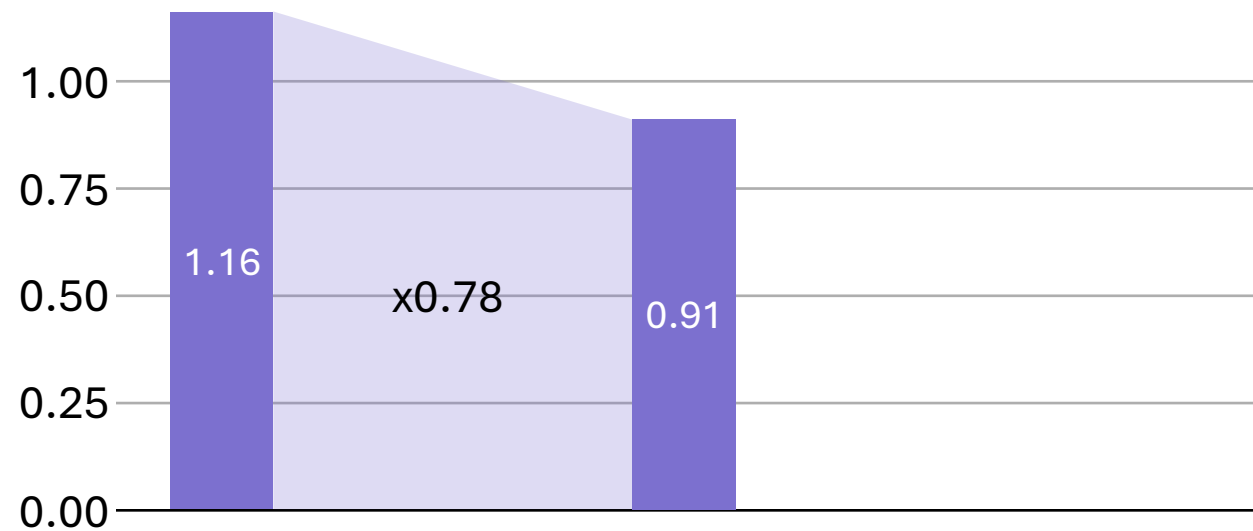
Vertex
Pipeline

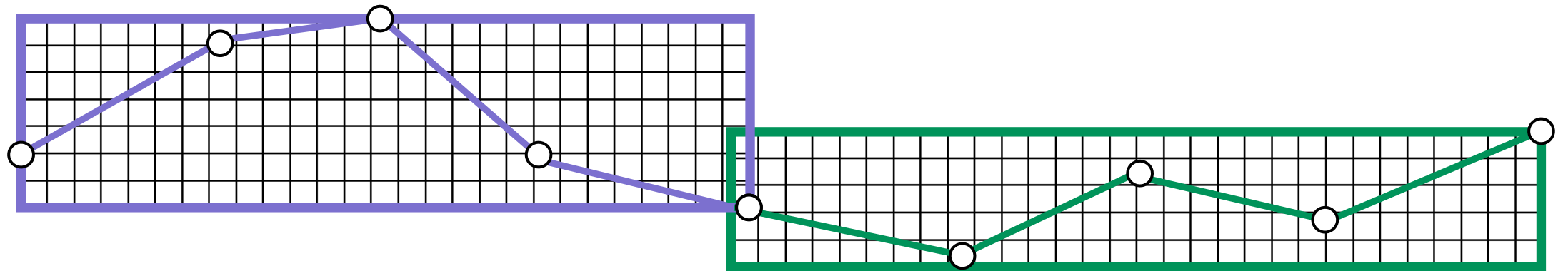
**GTS
Reuse**

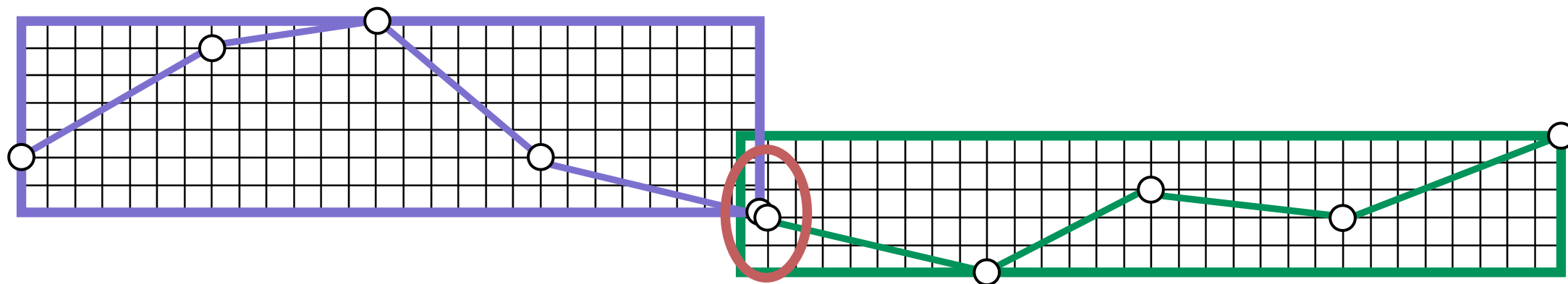
Buffers Size in MiB

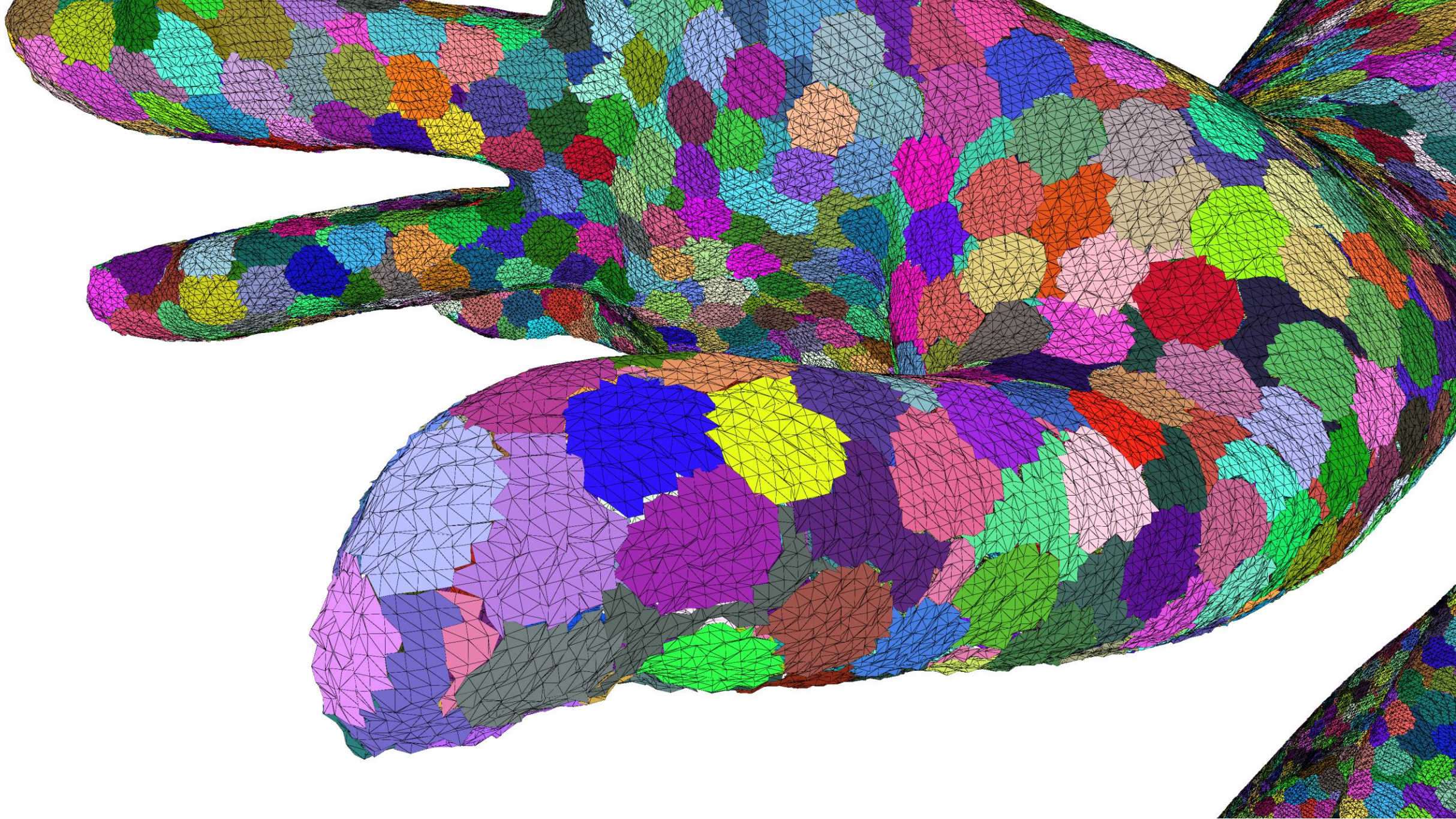


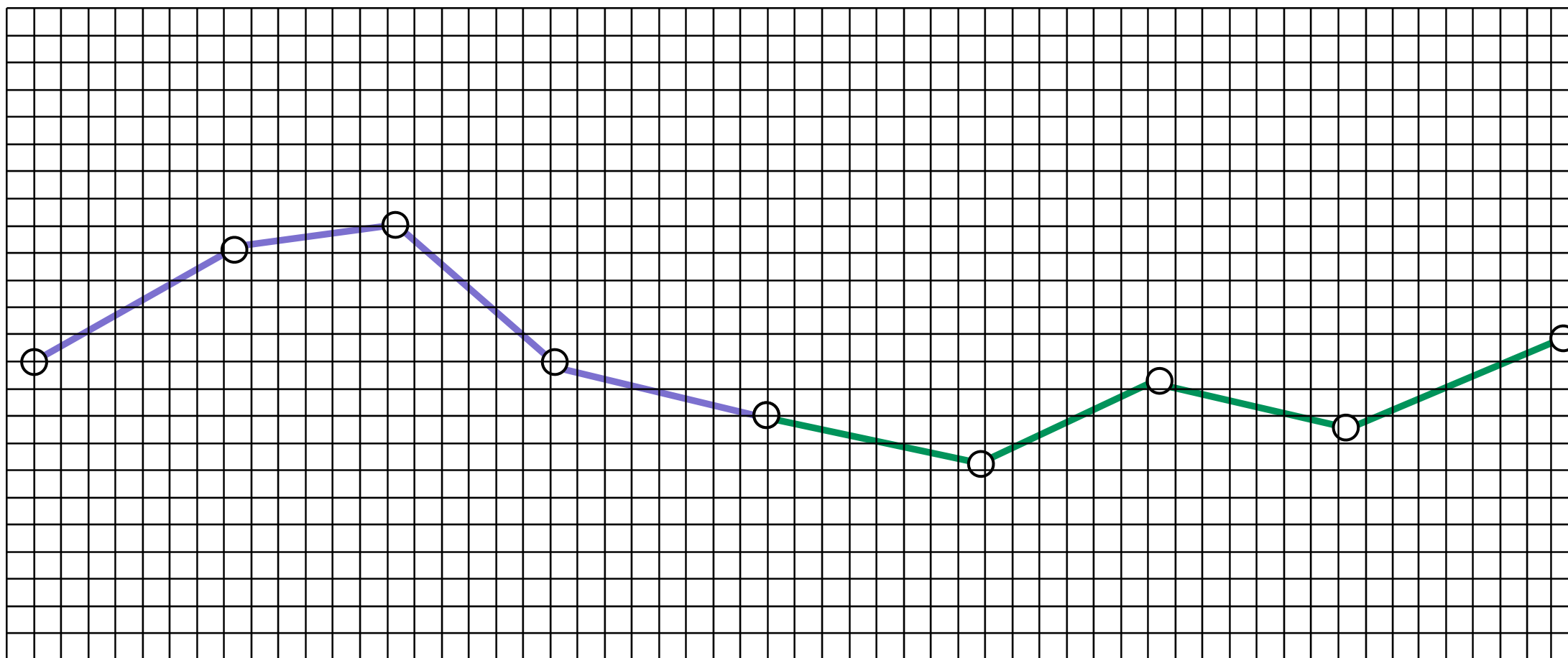
Render Time in ms on RX 7900 XTX

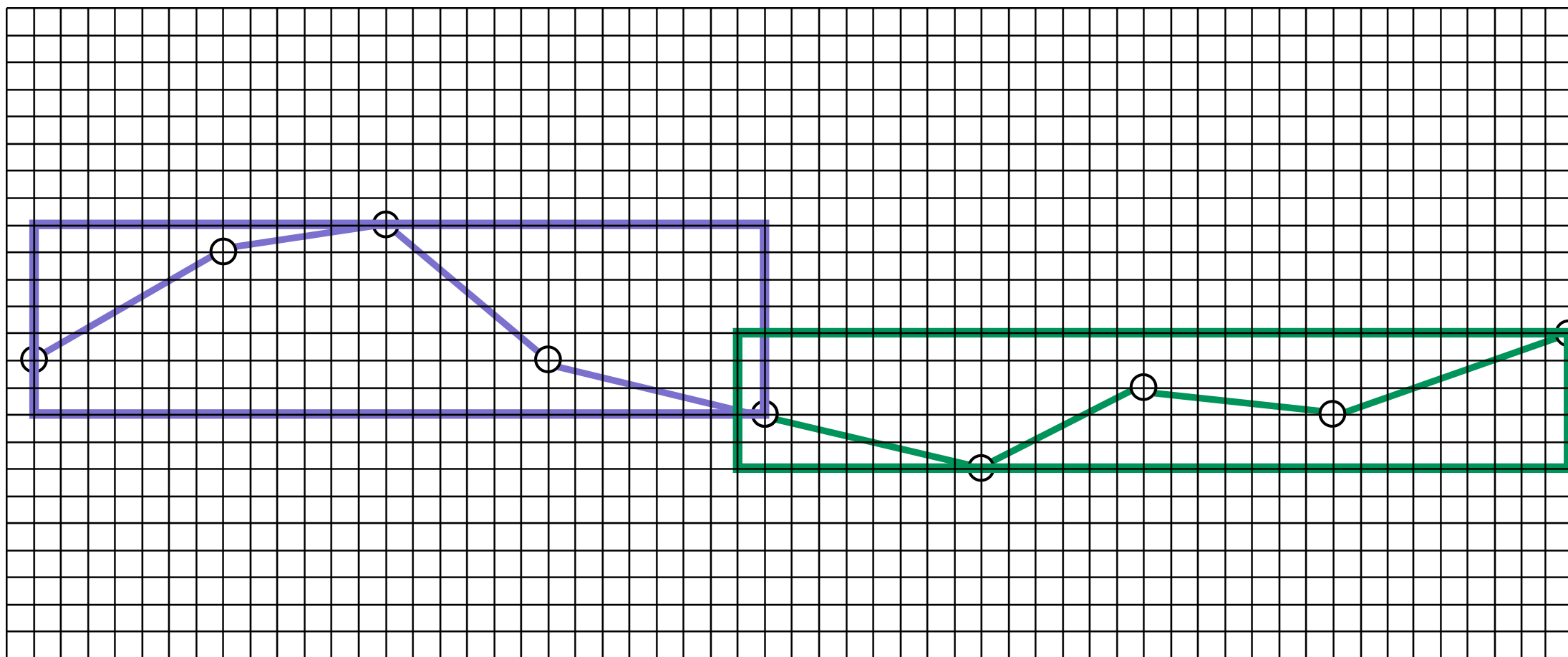


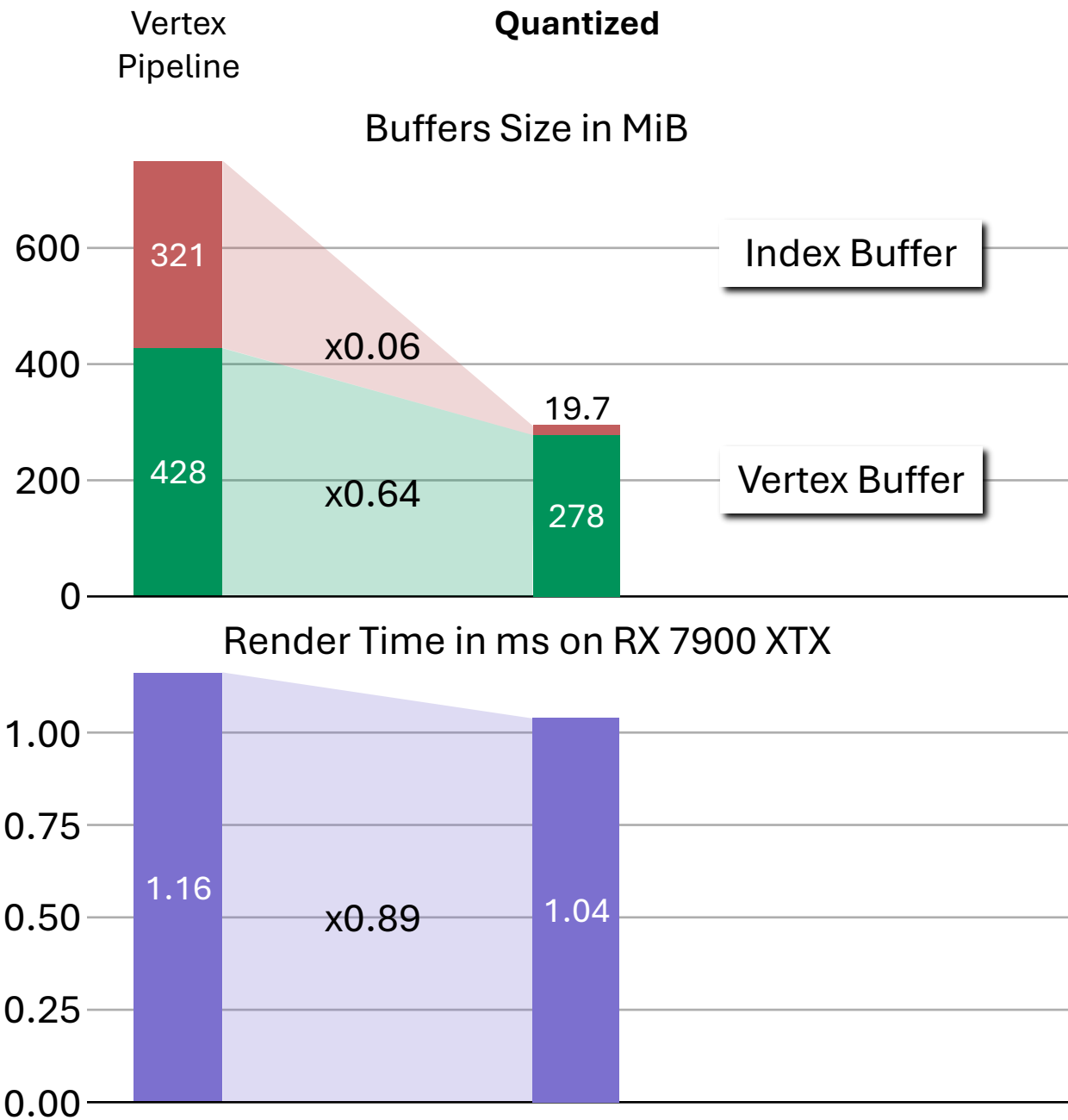
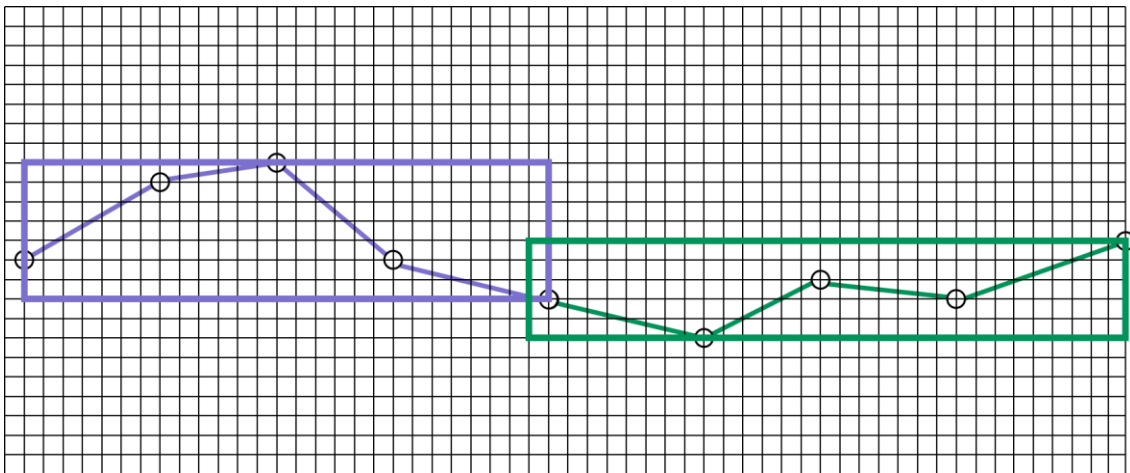


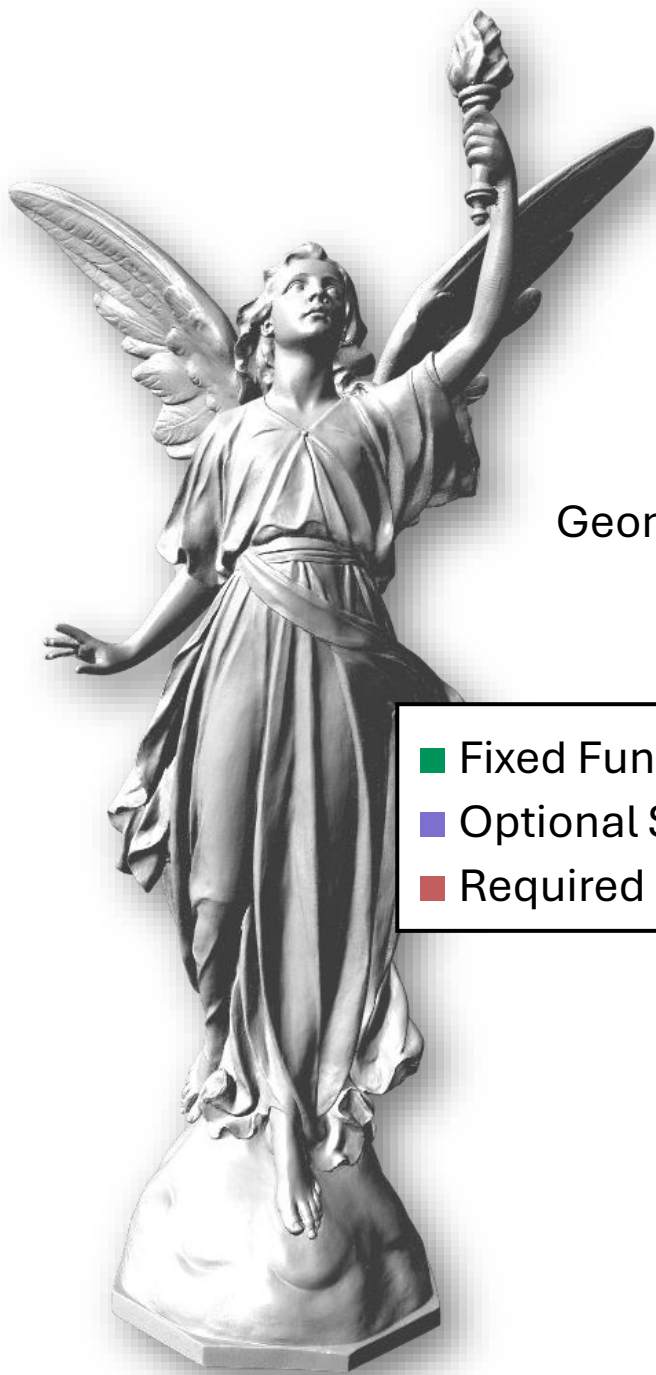










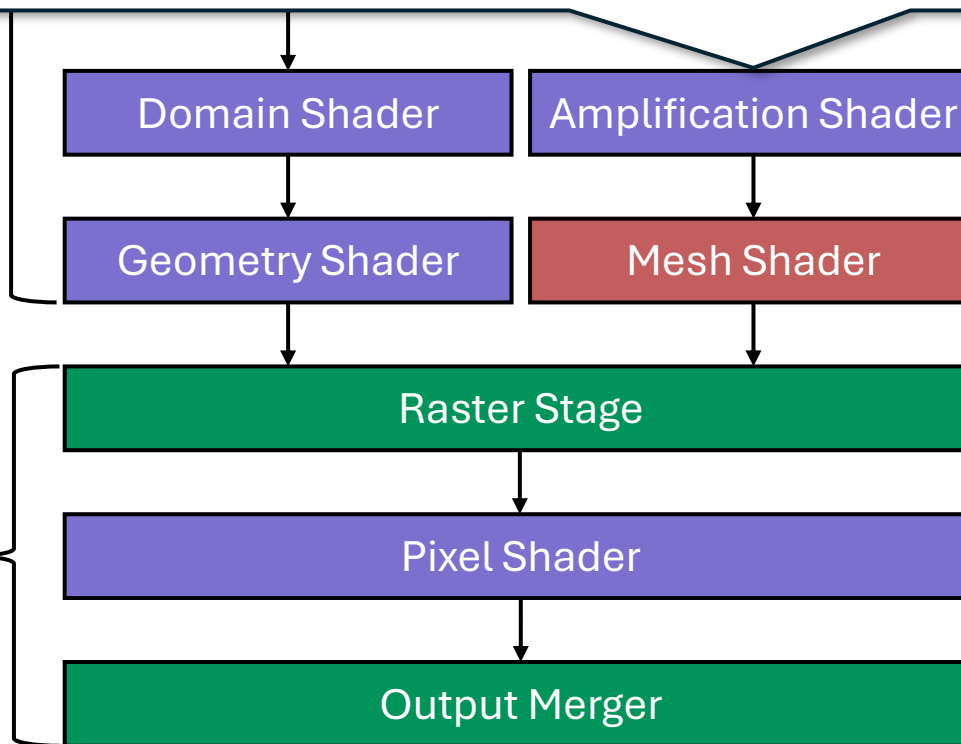


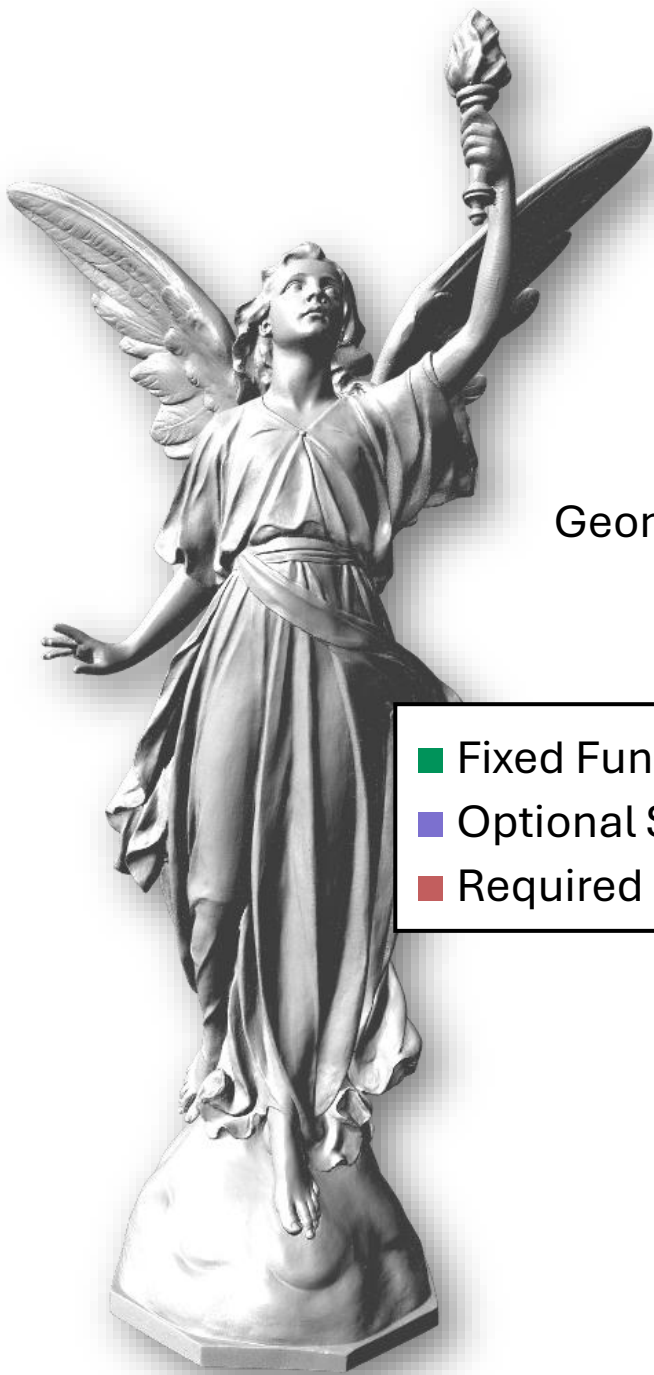
Geometry

```
...
Vertex Mesh-
void AmplificationShader(uint dtid : SV_DispatchThreadID){
    ...
    if visible {
        const uint index = WavePrefixCountBits(visible);
        payload.meshletData[index] = uint2(meshletId, instance);
    }
    const uint meshletCount = WaveActiveCountBits(visible);
    DispatchMesh(meshletCount, 1, 1, payload);
}
```

- Fixed Function Unit
- Optional Stage
- Required Stage

Pixel Stage



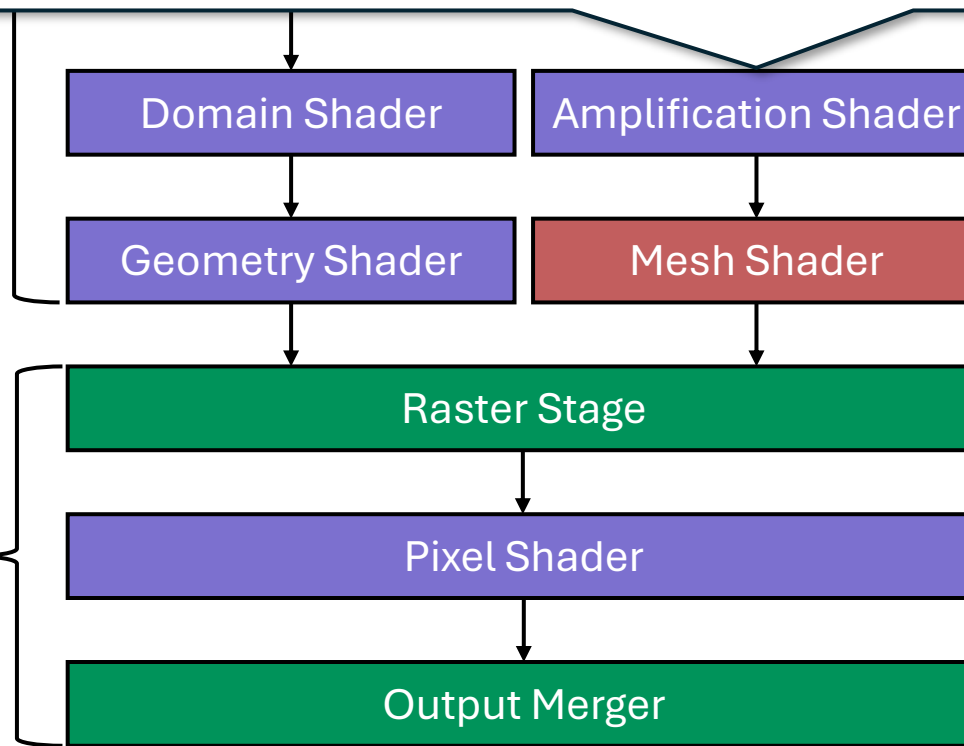


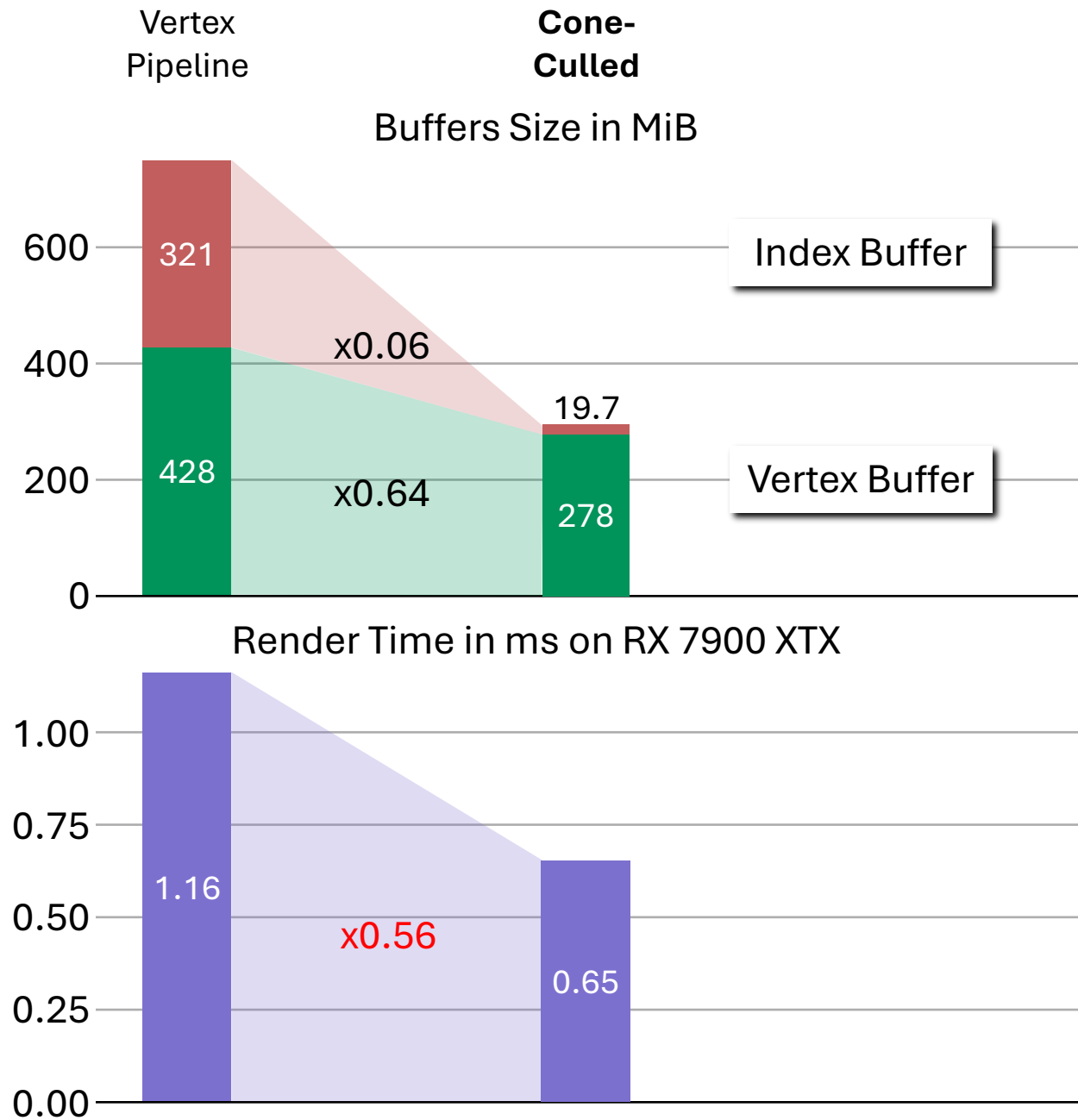
Geometry

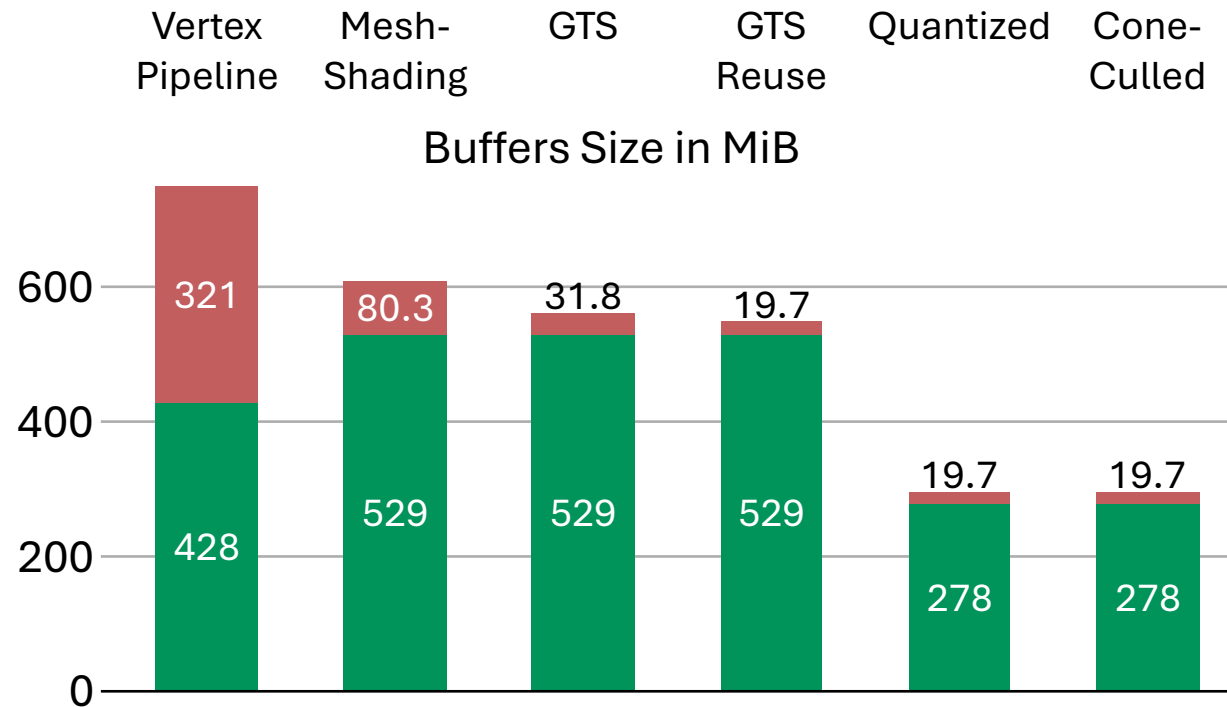
```
...
Vertex Mesh-
void AmplificationShader(uint dtid : SV_DispatchThreadID){
    ...
    if (visible) {
        const uint index = WavePrefixCountBits(visible);
        payload.meshletData[index] = uint2(meshletId, instance);
    }
    const uint meshletCount = WaveActiveCountBits(visible);
    DispatchMesh(meshletCount, 1, 1, payload);
}
```

- Fixed Function Unit
- Optional Stage
- Required Stage

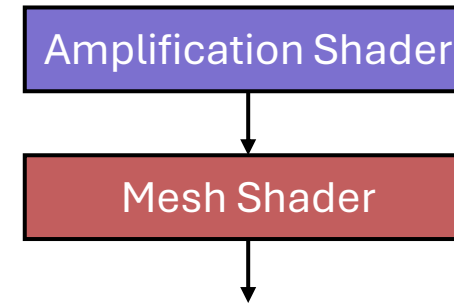
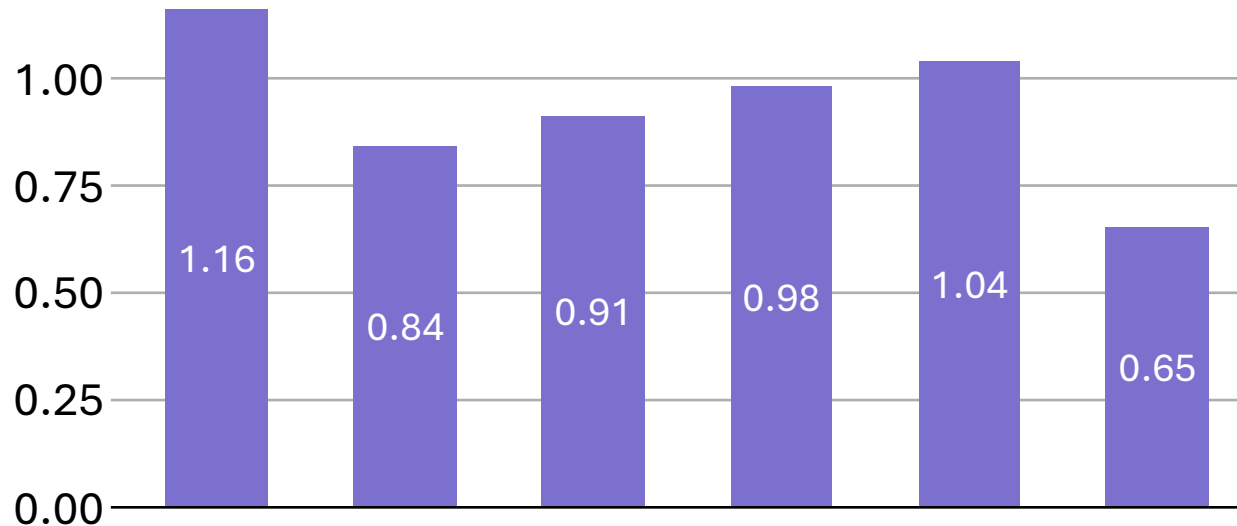
Pixel Stage







Render Time in ms on RX 7900XTX



Compress Indices ~16:1

Compress Attributes 2:1

Speed Rendering 1.8x